



BACHELOR OF COMPUTER APPLICATIONS

SEMESTER 1

ENVIRONMENTAL SCIENCE

Unit 3

Water and Energy Resources

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1. INTRODUCTION

The term “renewable” means something that can be used repeatedly or can be replaced. When we use the term “renewable” with reference to natural resources, it denotes those natural resources around us which never run out, since they are naturally replaced. Any renewable natural resource has an endless supply. For example, wind energy, solar energy, hydro energy, etc.

In more scientific terms, we can say that the rate of depletion or usage of any natural resource which is less than the rate at which it gets replenished is a renewable resource. Some of the natural resources (like sun, air, wind, tides) are also named perpetual resources as their quantity is not affected by human consumption. Other natural resources (like land, soil, water, plants, and animals) need careful management with a focus on how their management affects the quality of our lives (both present and future generations).

Some other resources also come under the category of renewable resources like wood, fish, and oxygen, but they require some time and effort to renew. Many of the precious metals are also renewable though not naturally replaced. But precious metals can be recycled since their extraction and usage do not destroy them.

STUDY NOTE

Kyoto Protocol was the first major international accord that helped in curbing carbon dioxide emission and reduction in global warming.

Natural resources (raw materials or land) are also referred to as natural capital as they are a form of equity. Natural resources are mainly of two types:

1. **Biotic resources:** These are living beings obtained from the biosphere (fish, marine organisms, animals, birds, etc.) Other resources like mineral fuels also come under natural biotic resources since they are formed through the decay of organic matter.
2. **Abiotic resources:** These are non-living resources like land, air, water, ores, etc.

Renewable resources are also a source of clean energy, as they cause less pollution and emit fewer greenhouse gases. In recent years, renewable organic products are being used to make biofuel which has gained prevalence over non-renewable resources like coal, natural gas, etc. Some of the biofuels in use are biodiesel (as an alternative to oil), green diesel (made from algae and a few other plants). The most common renewable resources are:

- Solar energy

- Hydro energy
- Wind energy
- Biomass energy
- Geothermal power
- Tidal energy

The current environmental movement has its focus on renewable resources, both economically and politically. Using renewable resources brings down cost and puts much less restraint on resources in limited supply like fossil fuels. Going green (i.e., adopting sustainable energy) positively impacts our environment, and resources like fossil fuels cause global warming and damage our environment.

The other category of resources, i.e., non-renewable resources such as fossil fuels, minerals, etc., are formed through long geological processes. Due to their extremely slow rate of formation, they cannot be replenished at the same rate as they are consumed; hence they are non-renewable. Non-renewable resources are also called dirty energy due to the pollution and other environmental issues caused by them.

Our energy sources have evolved over time. We used woods before the nineteenth century and later adopted non-renewable resources (coal, petroleum) for the same purposes. These non-renewable resources have been dominant energy sources to date, but the earth has a limited supply of these; hence, there has been a growing need to use renewable resources as energy sources.

Governments of all countries are taking measures to promote renewable resources by designing many incentives for using these alternate energy sources. For example:

- An energy tax is placed as a surcharge on fossil fuels, which makes the price of renewable resources competitive or less than that of non-renewable resources. A survey was done by IRENA (International renewable energy Agency) in 2019 which revealed that the energy produced by renewable resources is quite comparable to the price of fossil fuels.

STUDY NOTE

11% of the energy consumption in the U.S. comes from renewable resources. (As per the report of U.S. Environmental Protection Agency 2017).

- The Green Fund is a mutual fund that invests only in companies that produce goods with more responsibility towards environmental health, and their business dealings are socially conscious.

Renewable resources do pose some challenges in their usage:

- Renewable resources are less reliable than non-renewable resources as seasonal changes can impact their production.
- Renewable resources are not always perfectly renewable. If human consumption increases to a point where the rate of production of renewable resources goes down, then these resources might not be able to replenish themselves completely.
- Urban sprawl, irrigation, cultivation, deforestation, grazing, fishing, hunting could destroy renewable resources which are otherwise abundant in nature.
- If we want to use renewable resources on a large scale, then it can prove costly, and we still need more research to make renewable resources more cost-effective.

1.1 Learning Objectives

After studying this chapter, you will be able to:

- ❖ *Explain what are renewable natural resources and also be able to differentiate them from non-renewable resources*
- ❖ *Define the different types of renewable energy sources*
- ❖ *Explain the scope and importance of environment studies*
- ❖ *Explain the growing need for energy and how alternate energy resources are being used*
- ❖ *Explain about Minerals, Metals and Non-Renewable Energy Sources*
- ❖ *Recognise the challenges and consequences due to overuse of non-renewable natural resources*

2. WATER

The hydrologic cycle or water cycle is a natural cycle that is powered by sunlight and gravity. When water evaporates from the land surfaces and oceans, it becomes vapor and is temporarily held in the atmosphere before it comes down as precipitation. Water is available mainly in three forms in nature:

- **Surface water:** This is the residue of melted snow and precipitation. It is also called runoff. Bodies of water like rivers, streams, lakes form when the average rate of precipitation exceeds the rate of evaporation, absorption by vegetation, or seeping of water into the soil.
- **Groundwater:** Water that infiltrates and goes inside the earth's surface becomes groundwater. Water slowly goes down through layers of porous rocks and soils which are called aquifers. This falling down of water happens due to gravity pull.
- **Frozen water:** Water exists in frozen form as glaciers, ice, etc. in cold regions.

Both the surface and groundwater eventually go back to the oceans and get evaporated, replenishing the water vapor in the atmosphere. From the total water reserve on the earth, 97% is saltwater which is not fit for human consumption. A large portion of the other 3% of fresh water is locked up in polar ice caps. Only 0.003% of freshwater is readily available as surface and groundwater.

Water is critical to all kinds of lives on our planet. The water resources are used for multiple purposes that include:

- Drinking
- Irrigation purposes
- Washing needs
- Waste disposal by industries
- As a coolant for thermal power plants

2.1 Overexploitation of Groundwater and Surface Water

Water exploitation comprises the extraction of these water sources for human consumption and irrigation. With the overuse

STUDY NOTE

In 2016, a report from MoWRGR (Ministry of Water Resources, River Development, and Ganga Rejuvenation), GoI, under the chairmanship of Mihir Shah, concluded that a complex issue like water governance needs a multi-disciplinary approach.

of all forms of water resources, water is getting scarcer over the years. With many rivers, groundwater, and streams drying up, it poses serious challenges for most people and regions. Amongst many reasons for the scarcity of water, some of the prominent ones are:

- **Drastic changes in rainfall patterns:** This is happening mainly due to deforestation which causes reduced rainfall and diminished groundwater recharging.
- **Groundwater mining:** Multinational companies play a vital role in overexploiting groundwater resources. Classic examples of such companies are bottled water and soft drinks businesses. With their plants in rural areas, they exploit water resources as well as pollute the groundwater by releasing effluents. This continuous exploitation of underground water has resulted in lower levels of water. All this has negatively impacted the health of rural people.
- **Subsidence:** Ground subsidence occurs if the groundwater is being withdrawn at a rate higher than its recharge rate. Ground subsidence is a phenomenon in which the sediments in the aquifer get compacted and result in the sinking of the land surface over it. This leads to many economic losses and threats to lives like structural damages in buildings, pipe cracks or breakage, a reversal in the flow of sewers, and tidal flooding.
- **Water logging:** With excessive irrigation, the water tables of the area being irrigated can rise. This leads to waterlogging and salinity problems when salt in the rocks comes in contact with this water. The salt then rises up and evaporates in the upper soil layer, making it saline and hampering the fertility of soils.
- **Floods:** Many factors such as deforestation, rapid industrialization, mining, grazing, global warming, etc., have contributed to the significant rise of floods, which earlier used to be a natural calamity. As more water warms up due to global warming, more water evaporates, causing heavy rainfalls. Prolonged downpours can lead rivers and lakes to overflow and cause floods in low-lying coastal areas.
- **Drought:** Though primarily a meteorological phenomenon, drought can also occur because of many anthropogenic causes like deforestation, grazing, mining, etc. Due to these human activities, there could be below normal annual rainfall and less evaporation, causing drought conditions. Few other reasons for drought are erroneous

STUDY NOTE

Even a rise of 1 degree Fahrenheit in the temperature can cause the atmosphere to hold on 4% more of water vapor.

cropping patterns, increased exploitation of water resources (through canal irrigation and water wells), etc. All this has converted many drought-prone areas into deserts.

2.2 Conflicts Over Water

With the world's population more than tripling in the last century, it is quite natural that water resources are facing tremendous pressures all through the globe. All the world governments see the water problem as an ever-increasing challenge. The water problem has been intensified by environmental damage, variability in the hydrological cycle, industrialization, and urbanization.

The water challenges are more daunting for developing countries since the availability of per capita water is already low in these countries (with urbanization and population growing at an exponential rate). Due to the competing water demands for different users and uses, water disputes are increasing at many levels (local, district level, inter-state, national, and international level). The international conflicts are becoming more apparent now, intertwined with local conflicts.

There are more incidents of two or more countries sharing the water resources (both surface and groundwater). There are more than 300 rivers and 8 major river systems, 100 lakes, and a large number of aquifers currently in India. Some of the incidents of water disputes internationally, which were resolved later are:

- In 1960, there was a dispute between India and Pakistan over the Indus River System which was resolved by the mediation of the World Bank. The Indus river treaty divided the 6 Indus rivers between the two countries. The Sutlej, the Beas, and the Ravi went to India while the Indus, the Chenab, and the Jhelum rivers were given under Pakistan's territory.
- In 1959, Egypt and Sudan had a conflict over Nile waters which was resolved with an agreement of full utilization of the Nile waters by both parties.

There are still many international watercourses that have no agreements or attempts to address the existing issues. Some of the unresolved disputes include the conflict between Iraq, Turkey, and Syria over Euphrates and Tigris. Another dispute that still needs to be

addressed is between the Syrians, the Palestinians, the Israelis, and the Lebanese concerning the Jordan River and its tributaries.

In India, there are primarily 3 stressors that have been causing many interstate water conflicts They are:

- India has a federal structure that makes water part of a state. One example of this issue is the Cauvery water dispute between Karnataka and Tamil Nadu. This law has bestowed the primary right of water on where the water source lies and is skewed towards the upstream users, i.e., Karnataka in this case.
- Food security has been wrongly delineated, which has made crops like wheat and rice high in production. These crops consume a lot of water. This issue began with the green revolution in the 1960s, which increased the yield and production of food grains. The green revolution displaced the low water-consuming crops like ragi and sorghum in many areas.
- India's water policy lacks the understanding of the nexus between land, water, and food. There is no integrated ecosystem approach towards designing the water policy. The water governance in India is a piecemeal program and not an integrated basin approach.

Activity 1

You have to create a mini-hydropower wheel to lift an object. You would need the following materials for this activity:

Plastic soda bottle of 2-liter capacity, ruler and marker, a craft knife, scissor, sewing, thread of 16 inches, any small object which you want to lift (for example an eraser, bottle cap or pencil), sink, duct tape, a large funnel, paper clips, 2 corks, a wooden barbeque skewer.

Method:

1. Measure 6 cms from the bottom of the bottle and mark a few dots using markers and connect the dots.
2. Using the craft knife, cut off the bottom of the bottle where you joined the dots. Now from the cut part of the bottle measure 8 cms and cut it off to get a cylindrical plastic section.
3. Using scissors, cut 2 cms strips from the 8 cms cylindrical section obtained above; you would get 4 such strips. Cut these 2 cms strips in half so that you get 8 curved strips which measure 4 X 2 cms.
4. On the cork draw 8 lines lengthwise which are evenly spaced. Using a knife, cut slits on those lines you just drew.
5. Slide the 8 plastic pieces on the slits in the cork, making sure all the plastic pieces curve on the same side.
6. Unfold the paper clips and flex one end of each of them to create a loop.
7. Take the plastic funnel and use duct tape to affix both the paper clips on each side with the loop facing the top.
8. Divide the skewer and insert one half within each side of the wheel cork. Make a circle on your paper clip holder for each side. Ensure the loops on the paper clip are slack enough for the wheel to spin smoothly.
9. Insert one of the skewers into the other cork and tie thread tightly around it. Tie the loose end of the thread to a weight or other small household object.
10. Tie a small object to the loose end of the string.
11. In your washbasin, run a gentle small stream of water over the entire wheel. Carefully pour water well over wheel, allowing the plastic pieces on the cork to collect and convert the pouring water into mechanical energy.

SELF-ASSESSMENT QUESTIONS - 1

1. Which of the following are primary types of water on earth?
 - a) Surface water, Tap water, Bore well water
 - b) Surface water, Groundwater, Frozen water
 - c) Groundwater, Tap water, Frozen water
 - d) Bore well water, frozen water, Surface-water
2. One of the reasons for water scarcity is groundwater mining. [True/False]
3. Which one of the following choices is not a reason for floods?
 - a) Deforestation
 - b) Global warming
 - c) Industrialization
 - d) Water logging
4. The main issue with water logging is _____ and damage to soil fertility.
5. India's federal structure is one of the reasons for interstate water disputes. [True/False]
6. The water cycle is powered by _____ and _____.



3. RENEWABLE ENERGY SOURCES

Non-renewable energy sources can be hazardous to our environments and human health. They contribute to global warming and some of the methodologies applied to extract these energy sources can endanger life on our planet. For example, the technology used to carry out fracking (liquid at high pressures is injected into boreholes, rocks, etc., to open the existing pores forcibly and extract gas or oil) can pollute water and cause earthquakes.

For these reasons, the use of renewable energy sources to meet the energy demands has been increasing. There are a number of renewable energy sources, each with its own set of ecological and economic benefits.

The most common types of renewable energy resources being used are:

- **Solar or Light Energy:** Solar energy has been harnessed for a long time to grow crops, dry foods, and stay warm. The NREL (National Renewable Energy Laboratory) states that the amount of solar energy falling on earth in one hour is much more than the electricity that the world uses in one year. The sun's radiation can be used to make Photovoltaic cells (made from silicon or few other materials), transforming solar energy into electric energy.

Solar systems are being used to provide electricity to homes and businesses. A single Photovoltaic cell can produce enough energy to power just a calculator. But when they are combined to form solar farms or solar panels, their power is increased manifold. Using solar farms, power can be generated to suffice the needs of thousands of homes.

- **Wind energy:** From the old-fashioned windmills to turbines that are as tall as skyscrapers, there has been tremendous growth in utilizing wind energy. Wind energy works by converting mechanical energy into electricity. Wind energy is generated when the wind turns the blades of the turbines, which feed an electric generator to produce electricity. The energy produced is then used to supply power to homes, buildings, etc. Wind energy can also be stored in the power grid.

STUDY NOTE

Large hydroelectric power plants (or mega-dams) fall under the non-renewable energy category. Since mega-dams tend to reduce and divert the natural flow of water, they restrict access for humans and animals who rely on the rivers for many purposes.

- **Hydroelectric power:** Hydropower comes from fast-moving water like a river or water that is descending rapidly from a high point. Hydropower has been used for thousands of years and is the largest renewable energy resource in the U.S (10% of total electricity usage comes from hydroelectric power). Water dammed in reservoirs or lakes is released through generators and turbines to generate electricity.
- **Biomass energy:** The organic materials from plants and animals constitute biomass like wood, crops, ethanol (from plants like corn), sewage, etc. Biomass can create energy since it has absorbed the energy from the sun. This stored energy is released as heat energy when biomass is burnt. The burning of biomass releases less amount of carbon than fossil fuels.
- **Geothermal power:** A lot of energy is generated deep inside the Earth's core. If you have used a hot spring, you must know that it is geothermal energy. The heat within Earth's core is comparable to the heat on the Sun's surface. There are radioactive particles at the core of our planet that decay and release heat. When deep wells are bored inside the earth's surface, the hot water inside comes out as a hydrothermal source.

STUDY NOTE

Nuclear fusion is another method for obtaining nuclear energy, but it has not been perfected as yet. Nuclear fusion is a process in which light elements undergo nuclear reactions to form heavier elements releasing substantial amounts of energy.

This hydrothermal source can be used to create electricity or heat by pumping it through a turbine. This power source causes very little disruption to land and has immense potential. But it has a slower adoption rate since geothermal plants involve a high upfront cost.

- **Tidal energy:** Water again proves to be a vital source of alternate energy in the form of tides. Though still in the developmental phase, tidal energy has enormous potential. Much like hydroelectric power, tidal energy is also based on water power, though it also uses wind turbines. Just like wind turbines, tidal turbines also use blades to power a generator. These tidal turbines are placed at the seafloor where very strong tidal flows. Tidal stream generators are the most popular use of tidal energy. It uses the kinetic energy of the ocean to supply power to turbines.

SELF-ASSESSMENT QUESTIONS - 2

7. The process of fracking can cause earthquakes. [True/False]
8. _____ cells are used to convert solar energy into electricity.
9. Wind energy works by converting _____ energy into electricity.
10. How is hydroelectric power generated?
 - a) By utilising the solar energy falling on oceans
 - b) By utilising the power of wind on rivers and waterfalls.
 - c) By utilising rapidly flowing water from a high point or fast-moving rivers.
 - d) By utilising tides or waves in oceans.
11. Where is the geothermal energy coming from?
 - a) Within the Earth's deep core
 - b) From the heat of sunlight
 - c) From the tides on oceans
 - d) From rapid falling water

4. USE OF ALTERNATE ENERGY SOURCES

One of the biggest reasons for industrial air pollution is the generation of electricity. Most of the electricity comes from non-renewable power plants like coal, nuclear, etc. Using these resources to produce electricity is extremely damaging for the environment we live in, and pollutes our air, water, and land. For these reasons, methods to tap alternative energy sources are making progress worldwide.

These alternative energy sources are not derived from fossil fuels; hence they produce zero or very little greenhouse gases such as carbon dioxide. If the effect of the greenhouse is lessened by utilizing alternative energy, then there would be fewer causes for extreme climatic changes. Since these energy sources are alternatives to traditional or more common energy sources like coal, natural gas, oil, etc. They are termed as “alternative energy”.

Alternative energy is different from renewable energy sources through many renewable energy sources also fall under alternative energy sources. Solar power, for example, is both a renewable and alternative energy source. Renewable resources are abundant in nature and are constantly replenished, whereas alternative energy like nuclear power uses thorium or

uranium, which are finite resources. The main aim of alternative energy sources is to avoid fossil fuels.

The alternative energy source which is not renewable energy is nuclear energy. Nuclear energy comes from the core of an atom which undergoes a process called fission (or splitting) to gain energy. One of the methods of creating nuclear energy is using reactors to perform controlled nuclear fission. The energy released during fission is harnessed in power plants.

4.1 Growing Energy Needs

With the rising population, there is a steep rise in energy demands as more houses, businesses, and communities need to be powered. Here is some data as projected by IEA (International Energy Agency):

- Between 2005 and 2030, there will be a 55% increase in energy needs. The demand for oil would grow from 11.7 billion tons to 17.7 billion.
- Between 2005 and 2030, there will be a 50% increase in energy consumption. Most of this demand would come from developing countries.

The main reasons for growing energy demand are population and wealth increase in developing countries. With more wealth, technological developments, and a better standard of living, the demand for energy sources would increase. 70% of the increase in energy demands comes from developing countries which India and China lead.

STUDY NOTE

The UK consumes much less energy now than it did in 1970 despite an increase in population by 6.5 million.

To maintain a sustainable energy level and prevent further degradation of ecosystems, it is vital to expand the usage of renewable sources of energy. Today around 26% of the world's electricity comes from renewable energy sources. As per the IEA, this share of renewable energy sources would grow to 30% by 2024. The UK is one of the most efficient countries in both producing and consuming energy. The UK hit a fantastic milestone in 2020 by running only on renewable energy sources for 2 months on 10th June 2020.

SELF-ASSESSMENT QUESTIONS - 3

12. Why is nuclear energy not classified as a renewable source of energy?
- a) It uses methods that produce more carbon dioxide.
 - b) It uses materials like uranium which are finite and not abundant.
 - c) It has a low capacity to produce energy.
 - d) None of the above.
13. Which of the following is a reason for the growing demand for energy?
- a) Growth in population
 - b) Better standard of living
 - c) Affordability
 - d) All of the above

5. MINERALS, METALS AND NON-RENEWABLE ENERGY SOURCES

Minerals and metal ores, fossil fuels and groundwater are the examples of non-renewable resources. The metals themselves are available in abundant quantity in the Earth's crust, and the extraction takes place where they are concentrated by natural geological processes like heat, pressure, organic activity, and other processes sufficient for sustainable extraction.

These processes of formation usually take huge timeframe through plate tectonics and crustal recycling. We usually choose non-renewable natural resources assuming the fact that they are affordable and easy to process and use. It is easy to fill the vehicle with petroleum products, use gas stoves for cooking and use an uninterrupted supply of electricity for illuminating the world. A little bit of energy is required to produce large amount of power required for the above-mentioned activities. Apparently, renewable energy sources are not a competition to non-renewable sources. The advancement in the technology coupled with the non-renewable energy products has simplified human life a lot.

STUDY NOTE

The prominent types of non-renewable energy resources are oil, natural gas and nuclear energy and they are collectively named as fossil fuels.

However, the non-renewable natural resources are not everlasting; they will expire at some point in life. The pace at which non-renewable energy resources have been running out is gradually increasing with the increase in population and technological advancements. The speedy usage of non-renewable natural resources causes serious environmental changes

remarkably triggered by toxic gas emissions. The increasing price tag of non-renewable energy is striking the chord of economic hardship.

5.1 Minerals

Minerals from which metals are extracted are categorised as non-renewable natural resources. Bricks, cement, metals, etc., are used in the construction works. Diamonds, gypsum, limestone, potash, salt, sulphur, and peat are considered as minerals. Most metals extracted from minerals possess low commodity value except diamond.

Asbestos is a group that comprises minerals. This is found in nature as bundling of fibres that may be split into thinner, resilient strands or threads that can be used in commercial and industrial purposes such as automotive breaks, wallboard materials, etc. Certain kinds of asbestos cause human health problems, particularly lung diseases. Limestone, a carbonated sedimentary rock, is mainly used to manufacture cement and make plaster for buildings. Potassium-containing fertilisers are derived from potash which is from the mineral potash feldspar. We use salt, or sodium chloride as a food additive and flavouring ingredients, examples of minerals. Sulphur originated from hydrogen sulphide obtained from sour-gas wells, pollution-control scrubbers at metal smelters, and deposits of elemental sulphur. The mineral sulphur derived from hydrogen sulphide is used in the chemical manufacturing industries. Mineral materials like sand, gravel, and crushed stones are collectively referred to as aggregates. Aggregates are relevant and their usage is inexorable in the construction industry.

STUDY NOTE

Metals are found in the combined state due to their reactive nature. Metals and their compounds are present in the mineral ore.

5.2 Metals

Metals carry wide array of beneficial physical and chemical properties. Metals can be used in their basic form, such as alloys and compounds containing metals and non-metals.

Practical uses of metal:

- Making of equipment, machines, and electricity-conducting wires
- In constructing buildings and other structures.

The most important metals that are used in industries are as follows:

- Aluminium (Al)

- Chromium (Cr)
- Cobalt (Co)
- Copper (Cu)
- Iron (Fe)
- Lead (Pb)
- Manganese (Mn)
- Mercury (Hg)
- Nickel (Ni)
- Tin (Sn)
- Uranium (U)
- Zinc (Zn)

The precious metals gold (Au), platinum (Pt), and silver (Ag) are well known for jewellery uses; however, they have some industrial uses like conductors in electronics.

Examples of metal alloys:

- Brass (containing at least 50% Cu, plus Zn)
- Bronze (mostly Cu, plus Sn and sometimes Zn and Pb), and
- Steel (Majorly Fe, a small amount of carbon, Cr, Mn, and/or Ni)

The tailings (the waste materials left after the target mineral is extracted from ore) are usually dumped onto a landfill, into the nearby lake, or into the ocean, depending on the area's geography. Suppose the electronic goods containing metals or alloys expire. In that case, the metals in the product can be recycled to make new products, or discarded as waste. If we look according to geographic presence, high quality mineral, tin and other ores are not available everywhere uncommon to find. The deposits located satisfactorily close to the surface are most economical for mining, and such ores have comparatively high concentrations of metals.

5.3 Other Non-Renewable Energy Sources

Coal:

Coal is a fossil fuel which is solid and a carbon-heavy rock formed as a result of transformation of decayed plants and animals that lived in the early age. It is considered as the most carbon-intensive fossil fuel that is used for burning in generating energy.

Features:

- Coal is a non-renewable energy, and it is categorised into non-renewable energy source owing to the fact that the formation of coal takes millions of years.
- Coal contains plethora of carbon and hydrocarbons.
- The energy stockpiled by plants lived in the early age usually in the wetlands are retained in the coal.

According to 'Press Information Bureau, Government of India', India is ranked as the second-largest coal-producing country, producing approximately 729 million tonnes of coal as per December 2020 Reports. The states where coal reserves are found in huge quantities in India are Jharkhand, Odisha, West Bengal, Chhattisgarh, and many more.

Types of coal:

Coal is classified into four main varieties based on the type, amount of carbon content and capacity to produce heat energy. The amount of pressure and temperature that forces on the plants over the period of time determines the rank of a coal deposit.

- Lignite
- Sub-bituminous
- Bituminous
- Anthracite

Anthracite:

- The carbon content in Anthracite accounts for 86%–97%.
- It possesses the highest heating value and is considered as high-quality coal.
- In India, Anthracite is generally found in Jammu and Kashmir.
- According to business research report 2020, there is huge rise in demand for Anthracite for domestic purposes and electricity production units in developing countries like

India and China. The demand is higher in these countries due to rapid economic growth, population and other structural developments.

Bituminous:

- The carbon content in Bituminous accounts for 45%–86%
- It is used to generate electricity, utilised in the iron and steel industry and used as a raw material for producing cooking coal
- In India, it is found in states like West Bengal, Jharkhand, Odisha and Madhya Pradesh. The increased usage of bitumen products across different sectors would push the Asia Pacific market to approach USD 35 billion in 2027.

Subbituminous:

- The carbon content in Subbituminous accounts for 35%–45%
- It has a lower heating element than bituminous coal
- In the United States, during the year 2019, 44% of the coal production attributed to sub-bituminous variety
- In the U.S., it is mostly extracted from Wyoming and Montana. Moreover, it is also found in Alaska, Colorado and New Mexico.

Lignite:

- The carbon content in Lignite accounts for 25%–35%
- Lowest energy content among other variants
- Lignite deposits are young compared to others and are not subject to extreme temperature
- Lignite is fragile in nature and holds high moisture
- Furthermore, it is also used to produce synthetic natural gas
- In India, Lignite has presence in states like Rajasthan, Assam, and Tamil Nadu.
- According to world data atlas report 2020, the lignite coal production in India took place around 48,339 thousand short tonnes in 2020. India's lignite or brown coal generation surged at approx. more than 3.10 percent annualised rate from approx. 27,000 thousand short tonnes in 2001 to 48,339 thousand short tonnes in 2020.

Methods of coal extraction:

The widely used methods of coal extraction involve:

- Underground mining: Underground mining utilises heavy machinery to extract the coal from mines under the earth.
- Surface mining: Surface mining is also termed as strip mining. strip mining is the commonly used method to extract coal in the United States. The most catastrophic, uprooting and polluting mode of mining is strip mining compared to surface mining.

Coal in India:

Coal mining in India started from the year 1774. The fourth largest coal reserves in the world are situated in India. In India, Dhanbad city holds the record of largest coal producing city. More than 90% of the coal deposits in India attributes to the states Jharkhand, Odisha, Chhattisgarh, West Bengal, Madhya Pradesh, Telangana and Maharashtra. In India Energy derived from coal in India outpaces the energy derived from oil. This is quite in contrast to the world scenario where the energy derived from the oil outpaces the energy derived from coal.

Oil:

Oil is an example of liquid fossil fuel. Crude oil is mainly composed of comparatively volatile liquid hydrocarbons. Besides, it also contains the elements nitrogen, sulphur, and oxygen. All these form numerous varieties of complex molecular structures.

Crude oil is usually extracted from:

- Underground reservoirs
- Cracks, crevices, and pores of sedimentary rock
- Tar sands near the earth's surface

Extraction of crude oil:

- Drilling on land or sea
- Strip mining of tar sands oil and oil shale

The extracted oil is then converted to usable or consumable fuels like gasoline, kerosene, jet fuel, etc. and other products like plastics and paint.

There is a scarcity of new oil discoveries in India. Coupled with the fact that the Covid-19 pandemic hit the oil production, leading to a 7.1% fall in crude oil production in May 2020 compared to the previous year, it substantiated the decline in India's crude oil production. All these factors increased the reasons for India's dependence on oil imports.

Natural Gases

Natural gas is also referred to as fossil gas or gas. It is a hydrocarbon gas mixture mainly containing methane along with a small proportion of higher alkanes, carbon dioxide, nitrogen, hydrogen sulphide, or helium.

Features:

- It is a naturally occurring gas.
- It is a non-renewable energy source.
- Natural gas formation takes millions of years. Natural gas is a resultant of the long-term process in which numerous layers of decomposing plants and animals under the surface of the Earth are exposed to intense temperature over a period of time. The chemical energy stored by the plants as a result of the conversion of energy obtained from the sun is transferred to natural gas in the form of chemical bonds.
- Uses include heating, cooking, electricity generation, fuel for the vehicle. In addition, it is also used in the production of plastics and organic chemicals for commercial purposes as a chemical feedstock.
- It is either conventional or non-conventional, based on its underground location.

Conventional natural gas:

- Found in porous and permeable or mixed into oil reservoirs
- Extracted by standard drilling

Unconventional natural gas:

- A gas that involves a long process to extract by regular drilling is classified into unconventional natural gases.
- Extracted via special stimulation techniques, like fracking.

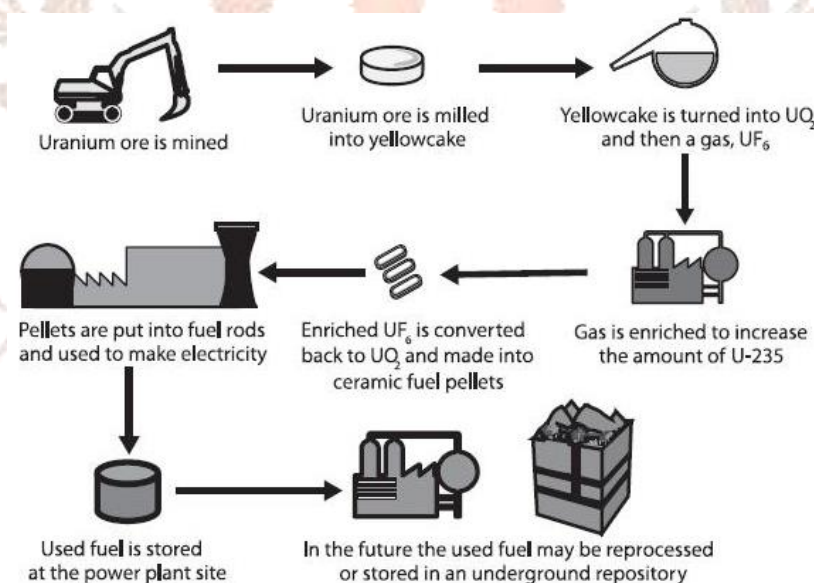
Uranium

Nuclear fuels are non-renewable energy resources. Nuclear fuel Uranium is extracted from Uranium ore. Uranium is a radioactive element that is naturally generated below the earth's surface or sea water. Nuclear plants use uranium for nuclear fission. Uranium is treated as a non-renewable energy source, though found in rocks worldwide. Nuclear power plants use U-235 for fuel because their atoms are easily split into parts. Although uranium is much more common than silver, U-235 is comparatively rare.

The process of mining uranium ore, the stages it goes through in a nuclear reactor, and disposal are collectively called the uranium fuel cycle.

Features:

- It is denoted using the symbol U
- It's atomic number is 92
- Silver grey metal



Source- <https://www.e-education.psu.edu/eme444/node/238>

Fig 1: Uranium Fuel Cycle

Uses:

- Depleted uranium is used in high-density penetrators as a shielding material for containers storing and transporting radioactive materials
- It is used as counterweights for aircraft control surfaces
- Ballast is a material that provides stability for missile re-entry vehicles
- It fuels the nuclear power plants

Thorium:

Thorium is an example of a weak radioactive metallic chemical element denoted by the symbol Th and has an atomic number 90. Thorium turns silver and tarnishes black in colour when exposed to the air due to the transformation into thorium dioxide. Also, it is hard enough, malleable in nature and has a high melting point. It is quite reactive and can catch flame in air when finely divided.

Thorium is more beneficial than uranium. Thorium results in less radioactive waste compared to Uranium. Besides, thorium excels on chemical stability and resistance to radioactivity compared to uranium. Thorium is more common than uranium; however, the deposits are not utilised. In addition, moving to thorium-based fuel production invites a lot of technical challenges.

Downsides:

Thorium cannot directly power a nuclear reactor unlike uranium. This is because thorium doesn't contain sufficient brittle material to trigger a nuclear chain reaction. The nuclear industry lacks in operational experience with thorium and being conservative at the same time because of insufficient experience using thorium. The preparation of thorium fuel is a cumbersome process, which creates difficulty in processing the thorium. It takes 550 degrees more for thorium dioxide than traditional uranium dioxide to melt at high temperature. Hence, high temperatures should be provided to generate high-quality solid fuel. Also, Thorium is quite immobile, causing difficulty in the chemical processing.

Irradiated Thorium is more effectively radioactive and hazardous in the short term. The thorium uranium cycle invariably produces gamma rays. These gamma rays are very hard to block, admitting the need for more expensive fuel handling and/or reprocessing. Above all, the commercial scale efficiency of thorium is not proven practically.

SELF-ASSESSMENT QUESTIONS – 4

14. Which of the following is an example of category of coal?
- a) Lignite
 - b) Anthracite
 - c) Bituminous
 - d) All of the above
15. Which of the following describes the features of uranium element?
- a) Used as fuel in nuclear power plants
 - b) Low density penetrators
 - c) All of the above
 - d) None of the above
16. Pinpoint the incorrect statement:
- a) Thorium cannot directly power a nuclear reactor unlike uranium
 - b) Chemical stability and resistance to radioactivity of thorium is higher than uranium
 - c) Thorium is directly used in nuclear reactors, whereas uranium is indirectly used in the nuclear power reactors
 - d) Atomic number of Uranium is 92
17. Coal is the most carbon-intensive fossil fuel that humans burn for energy generation. [True/False]
18. Oil is an example of liquid _____.

5.4. Use and Overuse of Non-Renewable Natural Resources**Overconsumption of Natural Resources:**

The industrial revolution coupled with the generation of technology, development, and research in the modern era made huge mineral and oil exploration-exploitation an easy process. The increased exploitation led to a hike in natural oil, gas, and mineral depletion and many mines entered a production decline. This is exemplified by the fact that minerals such as Gasoline, Copper, and Zinc production will be reduced in the next 20 years. The up-surging demand for petroleum products triggers more and more oil mining.

The release of toxins and chemical by-products from industries increase with progress in technological advancement in the industries. These harmful wastages are deposited in the

living world and possess the potential to alter the natural habitat of the living species. This is exemplified by the formation of acidic lakes, dead zones, and the death of wildlife.

Factors steering the overuse of non-renewable natural resources:

- Incessant emergence of technological advancement allowing the quick and efficient extraction of natural resources. This is exemplified by the increased rate of oil and mineral extraction, rate of deforestation triggered by urbanisation and urban development.
- Another predominant factor is the increasing population. According to the UN reports, the world population is 7.8 billion in 2020, and is expected to reach 9.9 billion by the year 2050. The demand for food and non-food products is directly proportional to the increase in population.
- The culture of consumerism increased production and stimulated economic growth worldwide. However, the downside of increased production and consumption is marked by the overuse and depletion of renewable and non-renewable natural resources in the earth.
- Materialistic views and consumerism lead to the extraction of renewable and non-renewable natural resources to form both necessary and unnecessary products in excess quantities. Overuse and excessive production results in the accumulation of wastage and by-products that often contaminate the environment and other natural resources in the nature.
- Due to the technological advancement and research, effort, time and money invested by the people increased the rate of extraction, availability and consistent supply of non-renewable natural resources. This is corroborated by the fact that non-renewable resources like fossil fuels are not easy to find around the globe.
- The efficiency of non-renewable resources like fossil fuels are undeniable. A lot of energy can be derived from a minimal quantity of fuel.
- Ease of transportation: Fossil fuels can be easily transported by using underground pipes for oil and gas.
- Easy set-up: A fuel plant can be set up at any location, irrespective of a large quantity of fuel to generate power.

STUDY NOTE

Overuse of natural resources results in resource depletion, pollution, subsequently making the earth unfit for human life.

Resource depletion:

Resource depletion persists when the consumption of resources outpaces its replenishment in nature. This indicates the fact that the depletion of a particular resource increases its value.

Common types of resource depletion:

- Aquifer depletion
- Deforestation
- Mining for fossil fuels and minerals
- Pollution or contamination of resources
- Slash-and-burn agricultural practices
- Soil erosion and
- Overconsumption, excessive or unnecessary use of resources

Consumption of stones, sand, and gravel in the construction industry is an example of the overuse of minerals. The production of minerals like Gasoline, Copper, Zinc, Aluminium, Coal, Iron is estimated to decline during the coming years.

The oil industry will experience a continuing decline in production subsequent to the maximum global petroleum extraction, and the period is referred to as peak oil. Furthermore, the decrease in supply and increase in demand of petroleum products raises the price of these products.

Water

It is a widely known fact that only 2.5% of the earth is fresh water (mostly takes the form of ice or snow); the remaining is salt water and not edible for humans. Reports from UN organisations forecast that by 2025, 1.8 billion people will face scarcity of drinking water.

Groundwater is utilised through wells and lakes for private, agricultural, and public use. Around the globe, the availability of groundwater is 22.6 million cubic kilometres and the renewable portion accounts for only .35 million. Around the globe, the ground water replenished is scarcer compared to its consumption and its renewal takes around 50 years on a human time scale. Groundwater extraction has doubled from the year 1960 and caused an increase in groundwater depletion.

On the other hand, the increase in population drives deforestation at a faster pace. Increasing population demands expansion of farming, agriculture and housing activities. Deforestation has resulted in the depletion of one half of the forest that once shielded the earth.

Resources in wetlands are overused or converted for

- Large-scale agriculture
- Real estate activities
- Urbanisation
- Recreational lakes
- Hydro-power generation
- Hydrologic alteration
- Drainage
- Dredging
- Stream channelisation
- Ditching
- Levees
- Deposition of fill material
- Stream diversion
- Groundwater drainage
- Impoundment
- Marinas/boats
- Industrialisation and industrial development
- Silviculture/Timber harvest
- Mining
- Atmospheric deposition

Coal

It is forecast that the maximum coal will be extracted between the year 2025 and 2048. Furthermore, various estimations indicate that there exists sufficient coal to satisfy the demand for 188 years. However, if the demands increase at a fast pace, the estimated years will show a decrease.

Phosphorous

Phosphorous is extracted from phosphorous rock and guano. Phosphorous is widely used as fertilisers in agriculture. As per the research outcome of the Global Phosphorous Research initiative, the expected **global peak** in **phosphorus** production is predicted to occur around **2030**.

SELF-ASSESSMENT QUESTIONS – 5

19. The depletion of water resources are considered as serious issue than the current oil depletion because:
 - a) Replacement for oil is possible but not for drinking water
 - b) Only 2.5% of the world's total water storage is fresh water
 - c) Proper access to hygienic drinking water is a prime criterion for global development.
 - d) All of the above
20. Which of the following triggered overuse of non-renewable natural resources?
 - a) Technological advancement
 - b) Culture of consumerism
 - c) Materialistic view
 - d) All of the above
21. Urbanisation is not a reason behind resource overuse. [True/False]
22. Increase in population and urban development drives deforestation but in a slow pace universally. [True/False]
23. Overuse of non-renewable natural resources results in the resource depletion. [True/False]
24. Energy efficient equipment reduces the carbon dioxide _____.

6. SUMMARY

- Renewable energy resources are those natural resources that never run out i.e., the rate at which they are consumed is less than the rate at which they are replenished.
- Some of the most common renewable natural resources are sunlight, water, and wind.
- Amongst many functions of land, some of the prominent ones are:
 - It is a store of wealth for communities, groups, or individuals.
 - It is a storehouse of raw materials and minerals for human needs.
 - It is a biological habitat for animals, plants, and other microorganisms.
- Land use is defined by the purpose a certain land cover fulfills like recreation, sports, agriculture, etc.
- The primary reason for the land degradation is the unprecedented rate at which the population is growing globally.
- Some of the other major reasons for land degradation are:
 - Water logging
 - The advent of many socio-economic changes like trade liberalization, privatization, decentralization of decision making, and globalization of markets has made it difficult for humans to manage resources efficiently.
 - Changes in climate and natural (or human-induced) disasters also degrade the quality of the land.
- Some of the significant effects of land degradation are:
 - Soil erosion
 - Desertification
 - Land conflicts
 - Food shortage for consumption
- Forest is a biotic (living) renewable resource that provides us clean air, water, wood products, stable soil, habitat for wildlife, a source of recreation, and a beautiful environment that is rich in biodiversity.
- Forests are being cut down to make room for something other than forests to meet different needs. Deforestation happens when forests are removed permanently.
- The main reasons for deforestation are:
 - Agriculture

- New constructions
- Mining
- Some of the major adverse effects of deforestation are:
 - Deforestation affects biodiversity: Many rare and fragile species, including plants, animals, mammals, birds, amphibians, and insects, have their homes in forests. If forests are destroyed, it creates a natural imbalance and puts life at threat.
 - It hampers the livelihood of the local or tribal population: Globally, 1.6 billion people depend on healthy forests for their livelihoods. Amongst them, around 1 billion are the poorest in the world. They cannot continue their small-scale agricultural programs by hunting and gathering raw materials from forests.
 - Reduced rainfall: Deforestation affects the way water vapor is produced over canopies in the tropical rainforests. This causes a reduction in rainfall.
 - Soil erosion: The weakening and degradation of soil is another massive downside of deforestation.
- Water is available mainly in three forms in nature:
 - Surface water: This is the residue of melted snow and precipitation. It is also called runoff. Bodies of water like rivers, streams, lakes form when the average rate of precipitation exceeds the rate of evaporation, absorption by vegetation, or seeping of water into the soil.
 - Groundwater: Water that infiltrates and goes inside the earth's surface becomes groundwater. Water slowly goes down through layers of porous rocks and soils which are called aquifers.
- Frozen water: Water exists in frozen form as glaciers, ice, etc., in cold regions. Amongst many reasons for the scarcity of water, some of the prominent ones are:
 - Drastic changes in rainfall patterns
 - Groundwater mining
 - Subsidence
- Due to the competing demands of water for different users and uses, water disputes are increasing at many levels (local, district, inter-state, national, and international).
- The most common types of renewable energy resources being used are:

- **Solar or light energy:** The sun's radiation can be used to make Photovoltaic cells (made from silicon or few other materials) which can transform solar energy into electric energy.
 - **Wind energy:** Wind energy is generated when the wind turns the blades of the turbines, which feed an electric generator to produce electricity.
 - **Hydroelectric power:** Hydropower comes from fast-moving water like a river or water that is descending rapidly from a high point.
 - **Biomass energy:** The organic materials from plants and animals constitute biomass like wood, crops, ethanol (from plants like corn), sewage, etc. Biomass can create energy since it has absorbed the energy from the sun.
 - **Tidal energy:** Much like hydroelectric power, tidal energy is also based on water power, though it also uses wind turbines.
- Alternative energy is different from renewable energy sources, though many renewable energy sources also fall under alternative energy sources. Solar power, for example, is both a renewable and alternative energy source.
 - Renewable resources are abundant in nature and are constantly replenished, whereas alternative energy like nuclear power uses thorium or uranium, which are finite resources.
 - Nuclear energy is an alternative energy source (not renewable) that comes from the core of an atom which undergoes a process called fission (or splitting) to gain energy.
 - Non-renewable energy sources are neither replenishable on a human timescale nor ensure sustainable production and supply. Fossil fuels, such as coal, natural gas, and oil, are the basic but important sources of non-renewable energy serving the demands of vast number of industries. Large scale usage of non-renewable energy sources multiplies the drawbacks of non-renewable energy, inclusive of negative environmental impact.
 - The fossil fuel which possesses highest carbon content is coal. It is derived from the plants and animals' sediment over the years.
 - Fuels like gasoline, diesel, petrol, jet fuel, etc. are derived from the extract of liquid fossil fuel which is also called as crude oil or petroleum products.

- Natural gas is extracted from the naturally processed remains of tiny sea plants and animals in its early age. The main chemical compound present in natural gas is methane.
- The nucleus of an atom when subjected to either nuclear fission or nuclear fusion, releases nuclear energy.
- Uranium is a naturally formed radioactive element found in different rocks.
- Thorium is fertile rather than fissile and it is indirectly used in nuclear reactors. The availability of thorium is higher than uranium in the world.
- Occurrences like increase in population, rapid modernisation, urbanisation and industrialisation triggers the use and overuse of natural resources.
- Overuse of non-renewable natural resources will have a significant impact in a negative way on our environment, economy and health.
- Conservation strategies restore the quality of nature and promote the use of renewable energy sources.
- Sustainable human development and non-renewable resource development are complementary and multidimensional. The dimensions of sustainability are based on economic, social and ecological factors.
- Sustainable utilisation of resources should be in line with poverty elimination, sustainable livelihood, women empowerment, environment protection and regeneration, efficient governance.

7. GLOSSARY

Anthropogenic: It refers to human activities that cause pollution and environmental degradation.

Effluents: It is the discharge of waste sewage (water and gas) to natural bodies of water like rivers or the sea.

Environmental economists: Those who study the economics of natural resources are called environmental economists. They study the extraction and use of natural resources as well as how economic incentives can be used to help the environment.

Marshes: A wetland that is dominated by herbaceous plant species. They can be found at the edges of rivers, streams, lakes, etc., where there is a transition from terrestrial to aquatic ecosystems.

Subsistence farming: When farmers grow crops and raise livestock sufficient only for their own use (no surplus for trade), it is referred to as subsistence farming.

Solar farms: These are large-scale installations of solar panels. They are also called solar parks and operate as power plants similar to natural gas power plants.

Terrestrial: This refers to things related to land like forests, deserts, grasslands, etc.

By-products: An incidental product produced during a production process.

Biomass: Plant or animal substance possessing the potential to generate heat or electricity.

Contamination: Contamination is defined as the presence of an undesired substance in the resources and making it unfit for further usage.

Fertiliser: Natural or synthetic material possessing the potential to nourish the soil and plants by providing essential nutrients.

Greenhouse effect: The process which increases the temperature of earth atmosphere. This occurs when greenhouse gases attract the heat emitted by the surface of the earth.

Greenhouse gas: Gas which absorbs, store and reradiates the infrared radiation emitted by Earth's surface.

Nuclear fission: Nuclear reaction in which the nucleus of an atom is separated into multiple lighter nuclei.

Oil spills: leakage of liquid petroleum hydrocarbon into the earth surface.

Ore: Naturally available substance from the ground usually in the form of rock or sediments, from which minerals or metals are extracted.

Recycling: It is the process of converting unwanted material into useful products.

Renewable energy: Energy derived from nature will replenish naturally on a human timescale.

Sustainability: Ability to provide and retain the resources incessantly.

Solar power: Conversion of sunlight into electricity.

Urbanisation: Process leading to the increase in urban area and decrease in rural area.

8. CASE STUDY

Starbucks

On January 21, 2020, Starbucks announced the decision to cut down the greenhouse gases and other waste it releases to half over the next decade. Furthermore, by the year 2030, they plan to replenish or conserve 50% of the water used for business operations and coffee production.

The company proposed various long-term strategies for attaining the concept of getting greener. The tiny footsteps towards getting greener are achieved by reusing packaging and taking more healthy cuisine in their menu. By 2021 on its 50th anniversary, Starbucks plans to disclose more data on its environmental goals.

The pertinacious nature of Starbucks on achieving its sustainability goals is questionable. Starbucks accomplished the goal of purchasing enough renewable energy to power the operations of entities in the US and Canada within the deadline. Whereas in 2008, Starbucks set the initiatives to serve in reusable containers (25% of drinks served by 2015), but few years later the goal was reduced to 5% and even by 2018, Starbucks was able to serve only 1.3% of its drinks in recyclable containers.

Starbucks must indulge in the determined attempt to reduce its environmental footprint owing to the fact that the entity's greenhouse gas emission is equivalent to pollution caused by nearly fourteen power plants using coal as fuel or on par with giants like Microsoft. Moreover, the water consumption of Starbucks is equivalent to the water sufficient to fill 4 lakhs Olympic-sized swimming pools.

Reports from Starbucks revealed the fact that 80% of their carbon footprint attributes to the consumption of energy in the stores. Hence, as an initiative to reduce the carbon footprint, Starbucks aims to reduce energy consumption in its retail outlets by 25%.

Starbucks has been working in various directions of saving energy. By 2013, they attained 50% of their goal and since then they incessantly challenged themselves to attain 100% of the goal set.

In tune with reducing the energy consumption, Starbucks replaced incandescent and halogen lighting with LED lights in more than 7,000 stores across the United States, Canada, Europe, and Asia to reduce the energy consumption by 80%. As per department of energy reports, it is recorded that using LED lights save \$30 in annual energy consumption costs. This indicates that for a 1,000 square foot store, the average savings will be \$600 per annum and the equivalent CO₂ elimination will be of 10 barrels of oil.

In the year 2016, Starbucks implemented an Energy Management System. Starbucks also focused on pushing suppliers to come up with efficient energy saving equipment. However, Starbucks had to offset the savings from replacing equipment with the cost associated with changes in their menu. This is exemplified by the requirement and increased usage of oven and refrigerator for heating stored food.

Renewable Energy

Despite the setbacks faced, Starbucks sets initiatives to invest in renewable energy. As per EPA's Green Power List, Starbucks is one among the pioneer buyers of organic electricity in the whole coffee industry. They aim to stay with this by procuring Green-E certified Renewable Energy Certificates (RECs).

One of the prominent initiatives of Starbucks was to drive energy sustainability. This was done by necessitating a few building standards, utility efficiency, and collaborating with greener suppliers. Starbucks initiated this by creating a verification program called Greener Retail which they want to implement in 10,000 new stores by 2025.

Another important initiative by Starbucks is the effort to make coffee 100% sustainable. Starbucks established the group "The Sustainable Coffee Challenge" and partnered with other entities to make coffee sustainable. The major goals of the initiative include:

- Ensuring a profitable coffee production and subsequent improvement in the income for associated stakeholders like producers, workers, and their families.
- Implementing sustainable agricultural practices and tripling the productivity as well.

Restricting the activities leading to deforestation of forests and reducing number of trees.

Starbucks gained competitive advantage over other entities in their sector through these conservation methods for energy conservation.. With proper funding and plans, Starbucks is well on the path and pace to becoming a leading entity in energy conservation.

Source- www.starbucks.in

Discussion Questions:

1. According to you, did Starbucks take the right step to protect the environment? How will implementing 'Green Retail' practice help them?
2. List out ways through which companies can reduce consumption of any non-renewable resources. What are the substitutes available to them?

9. TERMINAL QUESTIONS

SHORT ANSWER QUESTIONS

- Q1. Define the term "renewable" in reference to natural resources.
- Q2. Give a brief introduction to minerals and coals.
- Q3. Briefly explain major fossil fuels utilised for energy generation.
- Q4. List out the triggers that entail the usage of non-renewable energy resources.

LONG ANSWER QUESTIONS

- Q1 Write a short note on non-renewable natural resources.
- Q2. What are the three main water source types?
- Q3. Explain any three adverse effects of overexploitation of water resources.
- Q4. Explain any four types of renewable energy sources.

10. ANSWERS

SELF ASSESSMENT QUESTIONS

1. B. Surface water, Groundwater, Frozen water
2. True
3. D. Waterlogging
4. Salinity
5. True
6. Sunlight, gravity
7. True
8. Photovoltaic
9. mechanical
10. C. By utilising rapidly flowing water from a high point or fast-moving rivers.
11. A. Within the Earth's deep core.
12. B. It uses materials like uranium which are finite and not abundant.
13. D. All of the above.
14. D. All of the above
15. A. Used as fuel in nuclear power plants
16. C. Thorium is directly used in nuclear reactors, whereas uranium is indirectly used in the nuclear power reactors
17. True
18. Fossil fuel
19. D. All of the above
20. D. All of the above
21. False
22. False
23. True
24. Emission

TERMINAL QUESTIONS**SHORT ANSWER QUESTIONS**

Answer 1: The term “renewable” means something that can be used repeatedly or can be replaced. When we use the term “renewable” with reference to natural resources, it denotes those natural resources around us that never run out since they are naturally replaced.

Answer 2: Minerals used to obtain metals are classified as non-renewable natural resources. Minerals include diamonds, gypsum, limestone, potash, salt, sulphur, and peat. Except for diamonds, several metals derived from minerals have a low financial value.

Coal

Coal is a mineral, a black stone that may be obtained and consumed as a source of energy. Coal combustion is responsible for the majority of the country's electrical generation. While most coals are found in layer deposits, igneous outbursts can cause the reserves to be exposed to increased pressure and temperature.

Answer 3: Fossil fuel plays a superior role in the worldwide energy system. Petroleum, coal and natural gas attribute to the major share of fossil fuels used for energy production and consumption.

- A. Coal:** Coal is an important substance in the worldwide electricity generation. Coal fired power plants satisfy 37% of global electricity demand.
- B. Natural gas:** Natural gas is composed of hydrocarbon, and is categorised as fossil fuel. It is utilised for heating, cooking, electricity generation, producing plastics and organic chemicals and as vehicle fuel..
- C. Petroleum or oil:** Petroleum or crude oil is treated and refined to produce fuels for consumption. The examples of petroleum products are gasoline, diesel fuel, heating oil, jet fuel, petrochemical feedstock, waxes, lubricating oils, and asphalt.

Answer 4: The increase in population, technological innovations and advancements, industrialisation, urbanisation and urban development are some of the triggers that cause use and overuse of non-renewable natural resources.

LONG ANSWER QUESTIONS

Answer 1: Non-renewable natural resources are those resources that cannot be replenished by the nature within a normal meaningful timescale after its utilisation. Most carbon-based fossil fuels are examples of non-renewable energy resources. The main type of non-renewable energy sources are coal, oil and natural gas.

Most non-renewable natural resources are formed from the remains of plant and animals that lived millions of years ago. Replenishment of non-renewable natural resources will take millions of years owing to the fact that the formation of most non-renewable energy resources takes millions of years.

Non-renewable resources, whether in the form of gas, liquid, or solids, are exposed to processing in order to purify them into edible forms. Non-renewable resources such as coal and oil are the country's principal form of energy.

Answer 2: Water is found mainly in three forms in our environment:

- **Surface water:** This is the residue of melted snow and precipitation. It is also called runoff. Bodies of water like rivers, streams, lakes form when the average rate of precipitation exceeds the rate of evaporation, absorption by vegetation, or seeping of water into the soil.
- **Groundwater:** Water that infiltrates and goes inside the earth's surface becomes groundwater. Water slowly goes down through layers of porous rocks and soils which are called aquifers. This falling down of water happens due to gravity pull.
- **Frozen water:** Water exists in frozen form as glaciers, ice in cold regions.

Answer 3: Adverse effects of overexploitation of water resources are:

- **Groundwater mining:** Multinational companies play a vital role in overexploiting groundwater resources. Classic examples of such companies are bottled water and soft drinks businesses. Their plants in rural areas exploit water resources and pollute the groundwater by releasing effluents. This continuous exploitation of underground water has resulted in lower levels of water.
- **Subsidence:** Ground subsidence occurs if the groundwater is being withdrawn at a rate higher than its recharge rate. Ground subsidence is a phenomenon in which the sediments in the aquifer get compacted and result in the sinking of the land surface

over it. This leads to many economic losses and threats to lives like structural damages in buildings, pipe fractures, a reversal in the flow of sewers, and tidal flooding.

- **Water logging:** With excessive irrigation, the water tables of the area being irrigated can rise. This leads to waterlogging and salinity problems when salt in the rocks comes in contact with this water. The salt then rises up and evaporates in the upper soil layer, making it saline and hampering the fertility of soils.

Answer 4: The four types of renewable energy sources are as follows:

- **Solar or light energy:** Solar energy has been harnessed for a long time to grow crops, dry foods, and stay warm. Solar systems are being used to provide electricity to homes and businesses. Using solar farms, power can be generated to suffice the needs of thousands of homes.
- **Wind energy:** Wind energy works by converting mechanical energy into electricity. Wind energy is generated when the wind turns the blades of the turbines, which feed an electric generator to produce electricity. The energy produced is then used to supply power to homes, buildings, etc. Wind energy can also be stored in the power grid.
- **Hydroelectric power:** Hydropower comes from fast-moving water like a river or water that is descending rapidly from a high point. Hydropower has been used for thousands of years and is the largest renewable energy resource in the U.S (10% of total electricity usage comes from hydroelectric power).
- **Biomass energy:** The organic materials from plants and animals constitute biomass like wood, crops, ethanol (from plants like corn), sewage, etc. Biomass can create energy since it has absorbed the energy from the sun.

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BACHELOR OF COMPUTER APPLICATIONS

SEMESTER 1

ENVIRONMENTAL SCIENCE

Unit 4

Biodiversity and its Conservation

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1. INTRODUCTION

In the previous chapter, we learned about minerals, metals and different natural resources such as coal, oil, natural gases, uranium etc. We also explained how these sources of energy are used and how the overuse of these resources has brought many challenges and consequences to our environment. The different conservation strategies and sustainable utilisation of resources have been discussed in detail previously.

The next concept that we need to concentrate on is biodiversity, which measures the variation of different living organisms from multiple perspectives such as terrestrial, marine, and aquatic organisms, as well as the various ecological complexes to which they relate, and includes diversity among species and ecosystems. Biodiversity has several ecological and economic importance. It provides us with housing, fuel, nourishment, clothing and several other resources. Monetary benefits can be extracted from it through tourism. Thus, it is crucial to have a good knowledge regarding biodiversity to experience a sustainable livelihood.

Biodiversity is the existence of variability among living organisms on the earth, including the variability within and between species, and within and between ecosystems. There are three types or levels of biodiversity: species diversity, genetic diversity, and ecosystem diversity. Biogeographic regions or biogeographic zones are geographical areas that are defined based on the species found there.

It provides crucial information to ecologists and managers of natural resources to understand large-scale processes that affect species and ecosystems. The understanding of the fact that the number of species found on Earth varies globally, locally as well as time is known as the pattern of biodiversity.

Biodiversity zones are bio-geographic places with large global biodiversity and sustainability that are found along with an uncommon concentration of endemic species. Geographically, India is situated at the confluence of three realms: Afro-tropical, Indo-Malayan, and Paleo-Arctic. As a consequence, it has attributes from each of them. Endangered plants and endangered animals are on the verge of extinction.

Those species which are found in some unique areas and nowhere in the world are known as endemic species. Biodiversity is an essential factor for the survival of living beings in this world. The major threats to biodiversity include habitat loss, wild-life poaching and man-wildlife conflict. Biodiversity provides us with various benefits such as biological resources, ecosystem services, social and aesthetic benefits. There are two main methods that are used for the preservation of biodiversity - In-situ Conservation and Ex-situ Conservation.

STUDY NOTE

70% of the World's species are found in mainly 12 countries which are Brazil, Australia, India, China, Columbia, Madagascar, Mexico, Costa Rica, Ecuador, Indonesia, Peru and Republic of Congo.

In this chapter, we will study the meaning and definition of biodiversity, its significance, the different levels of biodiversity and different biogeographic zones. We will also discuss biodiversity patterns and hotspots. We will focus on how India is a mega-biodiversity nation, endangered and endemic species, different threats to biodiversity, methods of conserving biodiversity, and different values and services provided by biodiversity.

1.1 Learning Objectives

After studying this chapter, you will be able to:

- ❖ *Explain the meaning of Biodiversity, its significance and its different types or levels*
- ❖ *Explain the meaning of Biogeographic Zones and different Biogeographic Zones of India*
- ❖ *Describe Biodiversity Patterns and Hotspots*
- ❖ *Explain India as a Mega-diversity nation, and list Endangered and Endemic Species*
- ❖ *Describe the Threats to Biodiversity and Methods of Conservation of Biodiversity*
- ❖ *Illustrate the Biodiversity Values and Services*

2. MEANING, LEVELS AND TYPES OF BIODIVERSITY

The expression "biodiversity" differentiates between two species of plants, animals, and microbial organisms that exist in nature and the ability to perform everyday activities. It incorporates the genetic information contained in these organisms. The richness and variety of life on earth are described by biodiversity. Life would not be able to sustain without biodiversity.

It forms the most complicated, yet significant feature of our planet. The term biodiversity was coined in 1985. It forms a crucial part both in natural and artificial ecosystems. The variability of nature and the variability among plants, animals and micro-organisms is dealt with here.

Thus, biodiversity can be characterised as the variation among different living organisms from multiple sources such as terrestrial, marine, and aquatic organisms, and the various ecological complexes to which they contribute, which includes diversity among plants, species, and habitats.

Significance of Biodiversity:

Biodiversity is significant in maintaining water cycles, oxygen production, carbon dioxide reduction, soil protection, etc. It helps to preserve ecological processes, support global life, fix and recycle nutrients, etc. Healthy biodiversity offers many benefits which are as follows:

- It helps in the regulation of different cycles such as a means of regulating the level of carbon dioxide in the atmosphere by forests by the release of oxygen as a by-product during photosynthesis. It controls rainfall and soil erosion.
- It prevents depletion, contamination, and pollution of water supplies.
- It aids in the development and preservation of the soil.
- It helps in the storage and recycling of nutrients.
- It aids an ecosystem's recovery from unforeseen incidents.
- It offers biological services such as food, medicine, wood products, prescription medications, breeding stock, and so on.
- It offers recreational and tourism services.

STUDY NOTE

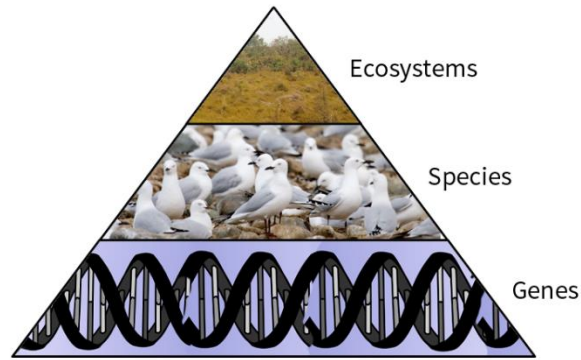
The highest tree diversity is reportedly found in Lambir hills National Park, Sarawak.

Levels and Types of biodiversity:

The presence of heterogeneity among living organisms on the planet, including variability within and between species, as well as within and between habitats, is known as biodiversity.

There are three types or levels of biodiversity, which are the following:

- **Species Diversity:** Species diversity refers to the various types of plants, animals, organisms, and fungi that can be found in a given area. More than 30 million species are estimated to be there on earth. Species diversity varies from one ecosystem to another. Conservation biologists have described and classified about 1.8 million organisms to date. Hotspots of diversity are areas of a high diversity of biodiversity. A tropical biome, for example, has greater diversity than a temperate ecosystem. Invertebrates refer to those animals that don't have the backbone and these species are the most diverse in nature.
- **Genetic Diversity:** Inside a population, genetic variability refers to the variety of genes that exist. Genetic variation refers to the various forms of genes present in plants, organisms, and microbes. It can happen both within a species and between species. This genetic diversity is vital for the successful reproduction of a species' population. For example, Poodles, German shepherds, and golden retrievers, , are all puppies, but they have different appearances, colours, and skills.
- **Ecosystem Diversity:** The diversity in species, natural populations, and environments is referred to as ecosystem diversity. It applies to the different interactions that animals have with one another and their surroundings. Tropical and temperate forests, mountains, grasslands, rivers, hot and cold deserts, lakes, and coral reefs are examples of habitat diversity. Each ecosystem is made up of a complex web of interactions between biotic and abiotic elements.
 - High-temperature discharges
 - Nutrients (like phosphorus and nitrogen)
 - Microorganisms (like bacteria, viruses, and Giardia)



Source- es.govt.nz

Fig 1: Levels of Biodiversity

SELF-ASSESSMENT QUESTIONS – 1

1. Biodiversity doesn't contribute to the stability of climate. [True/False]
2. The term biodiversity was coined in the year _____.
3. Which of the following is/are type/s of biodiversity?
 - a) Species Diversity
 - b) Generic Diversity
 - c) Ecosystem Diversity
 - d) All of the above

3. BIOGEOGRAPHIC ZONES

Biogeographic regions or biogeographic zones are geographical areas that are defined based on the species found there. It provides crucial information to ecologists and managers of natural resources to understand large-scale processes that affect species and ecosystems.

It is a biogeographic classification process that divides a large area into distinct geographical zones. Each zone contains a combination of species, habitats and physical features which are unique or different from others.

Biogeographic zones of India: With 2.4 percent of the world's overall land area, India contributes to 8% of the world's biological diversity. In terms of biogeography, India is divided into ten distinct biogeographic areas, as follows:

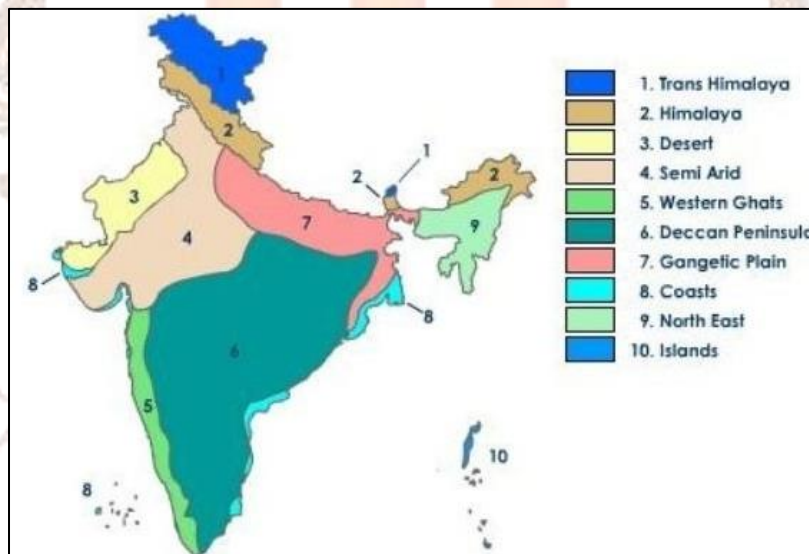
STUDY NOTE

Drawing its objectives from Convention of Biodiversity (CBD), National Biodiversity Act aims at conserving biodiversity, sustainable use and sharing of the benefits.

- **Trans-Himalayan Region:** It encompasses the high mountain regions of Ladakh, Jammu & Kashmir, North Sikkim, and parts of Himachal Pradesh, accounting for 5.6 percent of the total geographical area. Having alpine steppe vegetation, this zone has various endemic species. It is the most suitable habitat for the world's major populations of wild sheep and goats and other rare fauna, including Snow Leopard and the migratory Black-necked Crane. The cold desert of this zone represents an extremely fragile ecosystem.
- **Himalayan Zone:** It covers 6.4 percent of the overall land area and contains some of the world's highest peaks. The Himalayan zone is one of the richest zones in terms of habitats and species. The grassy meadows, alpine forests and mixed deciduous forests provide diverse habitats for endangered species such as Himalayan Tahr, Bharal, Takin, Markhor and Ibex. Other such endangered species found in this region include Hangul and Musk Deer.
- **Indian Desert Zone:** It accounts for 6.6 percent of the overall land area. It covers the Thar and Kutch deserts. Grasslands host a variety of endangered mammals, including the Wolf, Desert Cat, and others.
- **Semi-arid Region:** It contributes 16.6 percent of the total geographical region. It is a zone of transition between the desert and the Western Ghats. Many artificial and

natural lakes, marshy lands are present in this zone. The Lion, an endangered carnivore species, Caracal, Jackal and Wolf are some of the endangered species that are characteristic of this region.

- **Western Ghats:** It makes up 4% of the nation's total land area. It is one of India's largest tropical evergreen forest areas. The Western Ghats are home to most of the vertebrate species found in peninsular India. The significant species found in this region include Nilgiri Langur, Lion Tailed Macaque, Grizzled Giant Squirrel, Malabar Civet, Nilgiri, Tahr and Malabar Grey Hornbill.
- **Deccan Plateau:** It is India's largest biogeographic region, occupying 42% of the country's total land area. This semi-arid region falls in the rain shadow area of the Western Ghats. The zone comprises deciduous forests and thorn forests. Species found in this region are Chital, Barking deer, Sambar, Nilgai, Chou Singha, Gaur, etc.
- **Gangetic plain:** It accounts for 10.8% of the overall geographical area. The characteristic fauna of this region includes Rhino, Elephant, Buffalo, Hog-Deer, etc.



Source- alchetron.com

Fig 2: Biogeographic Zones of India

- **North East Region:** It accounts for 5.2 percent of the overall land area. This region is a transition zone between the Indian, Indo-Malayan and Indo-Chinese bio-geographical regions. This zone is the biogeographical 'gateway' to most of India's fauna and flora.

- **Coastal region:** It is characterised by beaches, mangroves, coral reefs, and marine angiosperm pastures, making them the wealth and health zones of India. The coastline from Bengal to Gujarat is about 5,425 km long. 25 islets of coral origin constitute the Lakshadweep and have a reef lagoon system that is rich in biodiversity. However, it has no natural vegetation.
- **Andaman and Nicobar Islands:** In India, it is one of three tropical moist evergreen forest areas. Flora and fauna present on this island are special. These islands, which are rich in endemism, are home to some of India's finest evergreen forests and a diverse collection of corals. In India, Andaman and Nicobar Islands is the only place where endemic island biodiversity is found. The endemic fauna includes Narcondam hornbill, South Andaman krait, etc.

SELF-ASSESSMENT QUESTIONS – 2

4. Indian Desert Zone includes the deserts of Thar and Kutch. [True/False]
5. Semi-arid Region constitutes 42% of the total geographical area of India. [True/False]
6. _____ are geographical areas which are defined on the basis of the species found there.
7. Which of these endangered species is found in Himalayan Zone?
 - a) Ibex
 - b) Chital
 - c) Desert Cat
 - d) Narcondam Hornbill

4. BIODIVERSITY PATTERNS AND HOTSPOTS

Patterns of Biodiversity: The understanding of the fact that the number of species found on Earth varies globally, locally as well as with time is known as the pattern of biodiversity.

- **Species varying globally:** Species that are distantly related and living in habitats of similar nature in different parts of the world generally act similarly. Example: Emus of Australia, Ostriches of Africa are all flightless birds.
- **Species varying locally:** Galapagos Islands consist of a group of islands, located close to one another but have different atmospheres and climates. Species on these islands adapt to the atmosphere they live in.
- **Species varying over time:** Armadillo of modern times resembles the fossil remains of glyptodont.

Biodiversity Hotspots: Biodiversity hotspots are bio-geographic areas with unusually high levels of biodiversity and abundance and a high concentration of endangered species. These animals, however, are under threat of exploitation and extinction. Listed below are two conditions for a biodiversity to be considered a hotspot,

- There are at least 1,600 vascular plants that are endemic to the region.
- They must be in extreme danger of being lost.

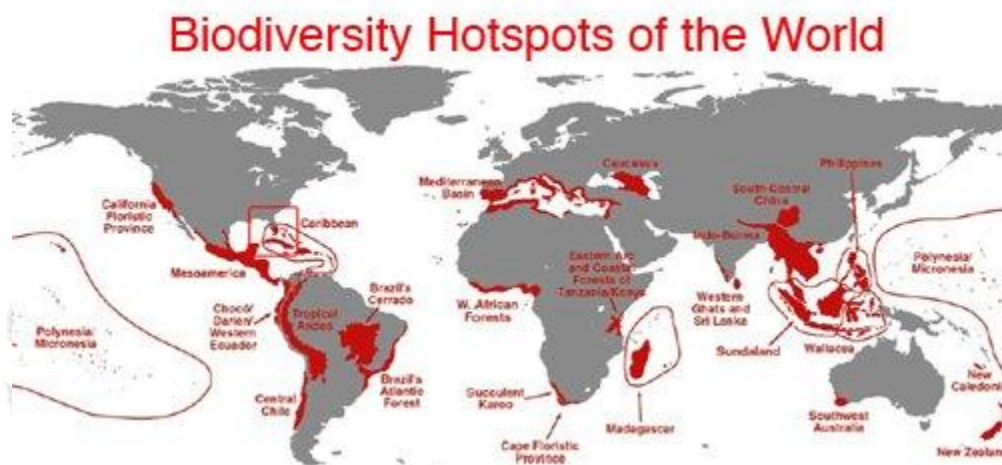
Biodiversity hotspots have been designated in 35 locations around the world. The hotspots cover 2.3 percent of the land area and host a variety of endemic plants and animals. Here we have a few examples of different regions and countries falling in those regions:

1. **North and Central America:** California Floristic Province, Mesoamerica and Madrean pine-oak woodlands
2. **The Caribbean:** Caribbean Islands
3. **South America:** Atlantic Forest, Cerrado, Chilean Winter Rainfall-Valdivian Forests, Tumbes-Chocó-Magdalena, Tropical Andes
4. **Europe:** Mediterranean Basin
5. **Africa:** Cape Floristic Region, Coastal Forests of Eastern Africa, Eastern Afromontane, Guinean Forests of West Africa, Horn of Africa,

STUDY NOTE

44% of all plant species are found in only 25 biodiversity hotspots. Just 1.4 percent of the earth's surface area contains 35 percent of all terrestrial vertebrate animals.

6. **Central Asia:** Mountains of Central Asia
7. **South Asia:** Eastern Himalaya, Nepal, Indo-Burma, India, Myanmar, Western Ghats, India, Sri Lanka
8. **South-East Asia and Asia-Pacific:** New Caledonia, New Zealand, Philippines, Polynesia-Micronesia, Sundaland, Wallacea,
9. **East Asia:** Japan, Mountains of Southwest China,
10. **West Asia:** Caucasus, Irano-Anatolian region,



Source-conservationbytes.com

Fig 3: Biodiversity Hotspots of the world

4.1 India as a Mega-Biodiversity Nation

India is one of the 12 super biodiversity countries on the planet. Ten biogeographic zones have been developed in the region. The country's physical characteristics and climatic environments have resulted in the creation of varied natural landscapes such as trees, deserts, coasts, grasslands, wetlands, and coastal ecosystems, all of which support vast biodiversity.

- India is geographically located at the crossroads of three realms: Afro-tropical, Indo-Malayan, and Paleo-Arctic. As a result, it has traits from each of them. The country's biological diversity is abundant and distinctive due to the combination of traits from three distinct domains.

STUDY NOTE

Asiatic Lion, also known as the Indian Lion is only found in and around Gir Forest National Park of Gujarat. These are listed as endangered species. These are one of the five big cats found in India, the others being Indian Leopards and Bengal Tigers.

- India is one of the 12 main centres of origin for domesticated plants and animals. It is thought to be the source of 169 important plant species and 116 domesticated animal breeds.
- Around 5,000 species of flower plants are native to the region. North-East India, the Western Ghats, the North-West Himalayas, and the Andaman and Nicobar Islands are all home to these creatures.
- About 62 percent of amphibian species are native to India, with the Western Ghats accounting for the rest. India is home to over 1,300 bird species, 460 reptile species, and 46, 000 plant species.
- The country has numerous alternative medicines prepared by plant-based raw materials. Such medicines are Ayurveda, Unani, Siddha and Homeopathic system. Herbal formulations for medicinal and cosmetic uses account for a significant portion of India's conventional biodiversity.
- National parks, wildlife sanctuaries, and biosphere reserves have been developed to conserve biodiversity. There are nearly 18 biosphere reserves in India.
- Approximately 4.3 percent of the country's total geographical area has been set aside for biodiversity protection in situ. It has been created by way of a protected area of 86 national parks and 450 wildlife sanctuaries.

4.2 Endangered and Endemic Species

Endangered Species: The species that are at risk or threatened of extinction are known as Endangered Species. These species are native to a particular region with a population strength reduced from 50% to 5%.

Various factors such as hunting, climate changes, pollution, habitat loss and other natural calamities risk the survival of animals and plants. Species having limited population strength are more prone to become endangered. The extinction rate is higher than before because of human activities like deforestation, industrialisation, etc. Around 460 plant species, 120 mammals and 160 types of birds are considered endangered in India. More species are becoming endangered every year due to reasons such as overexploitation of resources and habitat destruction.

STUDY NOTE

The Tura Range in Garo Hills of Meghalaya is a gene sanctuary for preserving the rich native diversity of wild citrus and musa species.

Some of the critically endangered animals are:

Amur Leopard, Hawksbill Turtle, Leatherback Turtle, Javan Rhino, Sumatran Tiger, Cross River Gorilla, etc.

Some of the endangered animals are:

Asiatic Black Bear, Great Indian Rhinoceros, Indian Elephant or Asian Elephant, Red Panda, Blue Whale, Indus River Dolphin, Ganges River Dolphin, Capped Leaf Monkey, etc.

Endemic species: Endangered animals are those that are threatened by extinction. As a result, the 'endemic location,' 'national endemic,' or 'geographical area endemic' refers to the place where the species is endemic. Kangaroos, for example, are native to Australia and are found nowhere else on the planet. Humans introduced them outside their natural environment while they were in captivity, which is why they have been seen in areas other than their natural habitat. Plants and species that are native to a single geographical area are more vulnerable to extinction. As a result, extra work is needed to save them. The following are some of India's endangered species:

1. Asiatic Lion, Gir Forest
2. Lion-tailed Macaque, Western Ghats
3. Kashmir Stag, Kashmir Valley
4. Sangai Deer, Loktak Lake
5. Nilgiri Tahr, Nilgiri Hills



Source-slideserve.com

Fig 4: Critically Endangered Species

SELF-ASSESSMENT QUESTIONS – 3

8. Leatherback Turtle, Javan Rhino, Sumatran Tiger, are examples of endangered animals. [True/False]
9. _____ is a bio-geographic area with unusually high levels of biodiversity and abundance, as well as a high concentration of endangered plants.
10. India has alternative forms of medicines such as:
 - a) Ayurveda
 - b) Unani
 - c) Siddha
 - d) All of the above
11. Which of the following is/are critically endangered species?
 - a) Salim Ali's Fruit Bat
 - b) Sumatran Rhinoceros
 - c) Blue Whale
 - d) Both a and b

5. THREATS TO BIODIVERSITY

Biodiversity is essential for the future of all living things on this planet. There would be a decline in the number of humans as the number of plants and animals reduces. The following are some of the main threats to biodiversity:

- **Habitat Loss:** Misuse of natural resources has led to a major loss of biodiversity. Urbanisation, rapid industrialisation and an increase in population has led to massive deforestation and habitat loss. Productive forests have turned into deserts. Mangroves have been cleared to make way for fuelwood and prawn farming, which has harmed aquatic fish breeding.
- **Poaching of wildlife:** Poaching of animals for trade and economic purposes has been a major cause of extinction of many animals and the endangerment of others, such as whales and Asian tigers. Overharvesting for food is one of the major causes of extinction followed by fashion and business. Elephants are poached for ivory, and leopards are poached for their meat, resulting in significant habitat loss.

STUDY NOTE

Mass extinction is a phenomenon where large number of species is permanently lost over a short period of time. This mainly occurs due to changes in Earth's environment and atmosphere.

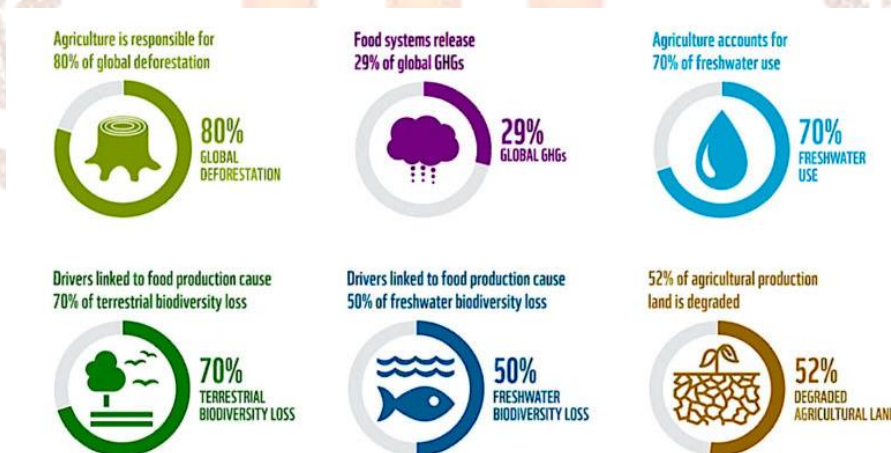
- **Man-wildlife conflict:** It applies to the relationship between wild animals and humans, as well as the negative consequences for both. Recently, the degree of conflict has increased due to a gradual increase in the human population. Natural wildlife territory has been displaced due to human race encroaching on wild animal habitats. The decrease in the availability of natural prey has led these wild animals to seek alternate sources. The major consequences of man-wildlife conflict are:

- It results in the loss of animal habitat
- It results in human and wildlife injury and death
- It results in crop and livestock depredation
- It results in damage to human property
- It results in a decline in wildlife population

Apart from the above threats, biodiversity is also threatened by climate change, non-native species invasion, and other factors.

STUDY NOTE

Scientists predict that human activity would obliterate 10 million animals (roughly) by the year 2050. At the present rate of extinction, almost 25% of the world's biodiversity will be extinct within a few decades.



Sources: scroll.in

Fig 5: Threats to Biodiversity

5.1 Conservation of Biodiversity

Biodiversity is of vital importance for living beings. Biodiversity provides us with various benefits such as biological resources, ecosystem services, social and aesthetic benefits. are Two main techniques used for the conservation of biodiversity are:

In-situ conservation: The conservation of species within their natural ecosystem of the environment is known as In-situ or on-site conservation. It is considered the most practical

method of biodiversity conservation. Here, the genetic resources are conserved through their maintenance within the environment of their origin.

Examples: National Parks, Biosphere Reserves, Gene Sanctuaries, Wild Life sanctuaries, etc. In-situ conservation is advantageous in the following ways:

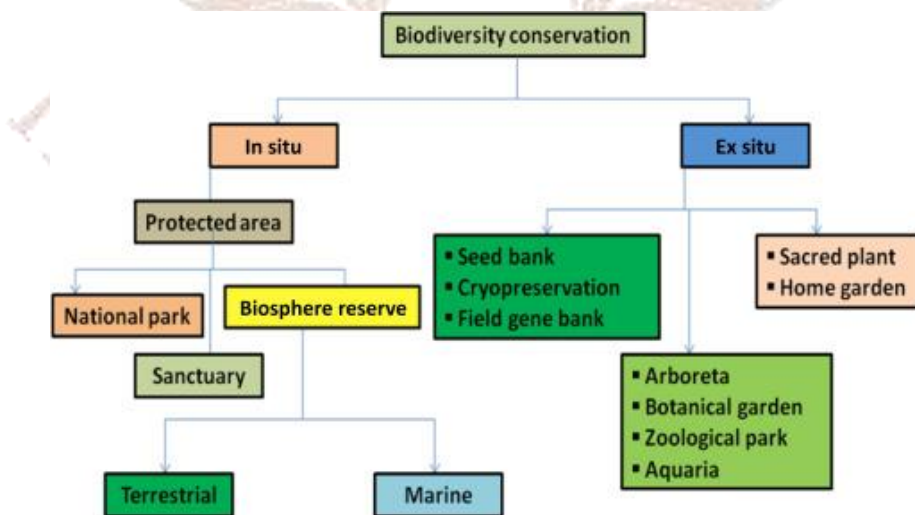
1. It is an economical and convenient method
2. It helps in preserving numerous organisms
3. Living organisms are allowed to adapt to environmental conditions in a natural habitat.

STUDY NOTE

The illicit wildlife trade is expected to produce between \$8 billion and \$25 billion in revenue per year. In 2015, the United Nations General Assembly adopted a resolution to combat wildlife trafficking.

Ex-situ conservation: Ex-situ conservation refers to the preservation of different elements of ecological diversity outside of their natural environments. This approach involves removing threatened or endangered animals and plants from their natural habitat and placing them in a special care facility where they will be safe and given the opportunity to mature naturally. Plants and animals are cared for in this environment, which has been created artificially. Captive breeding, seed banks, botanical gardens, in vitro fertilisation, cryopreservation, tissue culture, and zoos are also examples of captive breeding. This approach can be carried out in a variety of ways:

1. Creating gene banks, where sperm, ova, and seeds are preserved at temperatures that are manually regulated.
2. By artificially propagating plants and breeding animals in captivity.



Source: sciencedirect.com

Fig 6: Biodiversity conservation

5.2 Biodiversity Values and Services

Humans contribute basic values to biodiversity because we are so reliant on it for our cultural, physical, and environmental needs. Biodiversity offers the following programmes and values:

- **Environmental Value:** Observing each ecological phase and defining the resulting ecosystem resources can be used to assess the environmental importance of biodiversity. Wetlands, for example, serve as breeding and nursery areas for certain fish and provide a safe haven for animals during droughts.
- **Social Value:** There are significant variations in landscape and biodiversity preferences due to age, social conditions, and cultural factors. Peepal, Banyan, and Tulsi trees are also revered in India.
- **Economic Value:** Biodiversity is a vital opportunity for a number of sectors that fuel the global economy. Biodiversity is the foundation of the world's main fuel supplies, such as wood and fossil fuels. Important chemicals are derived from a wide range of flora and fauna and are used in a variety of medical fields.
- **Consumptive use value:** This category includes natural resources that are specifically used for cooking, such as wood, fuel wood, and so on. Humans use nearly 45,000 plant and animal species on a daily basis.
- **Productive Use Value:** These figures refer to items that are harvested and sold commercially. Biotechnologists use wild plant species to produce higher yielding and disease-resistant varieties.
- **Ethical and Moral Value:** These principles are linked to the preservation of biodiversity. It is founded on the idea that all kinds of life on Earth have the right to live. Morality and ethics teach one to protect all aspects of life and to prevent causing undue damage to other species.
- **Aesthetic Value:** Biodiversity is responsible for the uniqueness of our planet; without it, Earth would mimic most desolate worlds in the world. The uniqueness of an ecosystem is due to biodiversity, which has contributed to eco-tourism.

STUDY NOTE

According to the 2003 United Nations List of Protected Areas, there are 1,02,102 sites under protection covering 18.9 million km² and 1.65 million km² of underwater protected area.

- **Ecosystem Services:** These services manage the human needs and activities by managing production ecosystems. Some of the ecosystem services include:
 - Plants and aquatic algae provide oxygen
 - Vegetation slows runoff, traps sediment, and removes nutrients
 - Soil, microbes break down pollutants
 - Greenhouse gas mitigation by sequestering ambient carbon in biomass, among other roles.

Activity 1

Think of all the activities in which you were engaged in the last week. List the products, commodities or services which were made available to you by the different components of biodiversity. Analyse each of them in detail, and segregate them under different values which have been discussed above.

Repeat the same activity for another week. You will now be more conscious of the different values and services of biodiversity in our daily life. Prepare a summarised report of the values of biodiversity in real life aspects and the effect of its absence or the problems we might face due to its absence.

SELF-ASSESSMENT QUESTIONS – 4

12. Social Value deals with recreational, cultural, aesthetic and spiritual values.
[True/False]
13. In-situ conservation refers to the preservation of different types of ecological diversity outside of their natural environments. [True/False]
14. _____type of conservation is the retention of certain aspects of biological diversity beyond their natural habitats.
15. Which of the following is a biodiversity value?
- a) Ethical and Moral Value
 - b) Aesthetic Value
 - c) Both a and b
 - d) None of the above
16. Which of the following is a threat to habitat?
- a) Habitat Loss
 - b) Wild-life Poaching
 - c) Man-wildlife conflict
 - d) All of the above



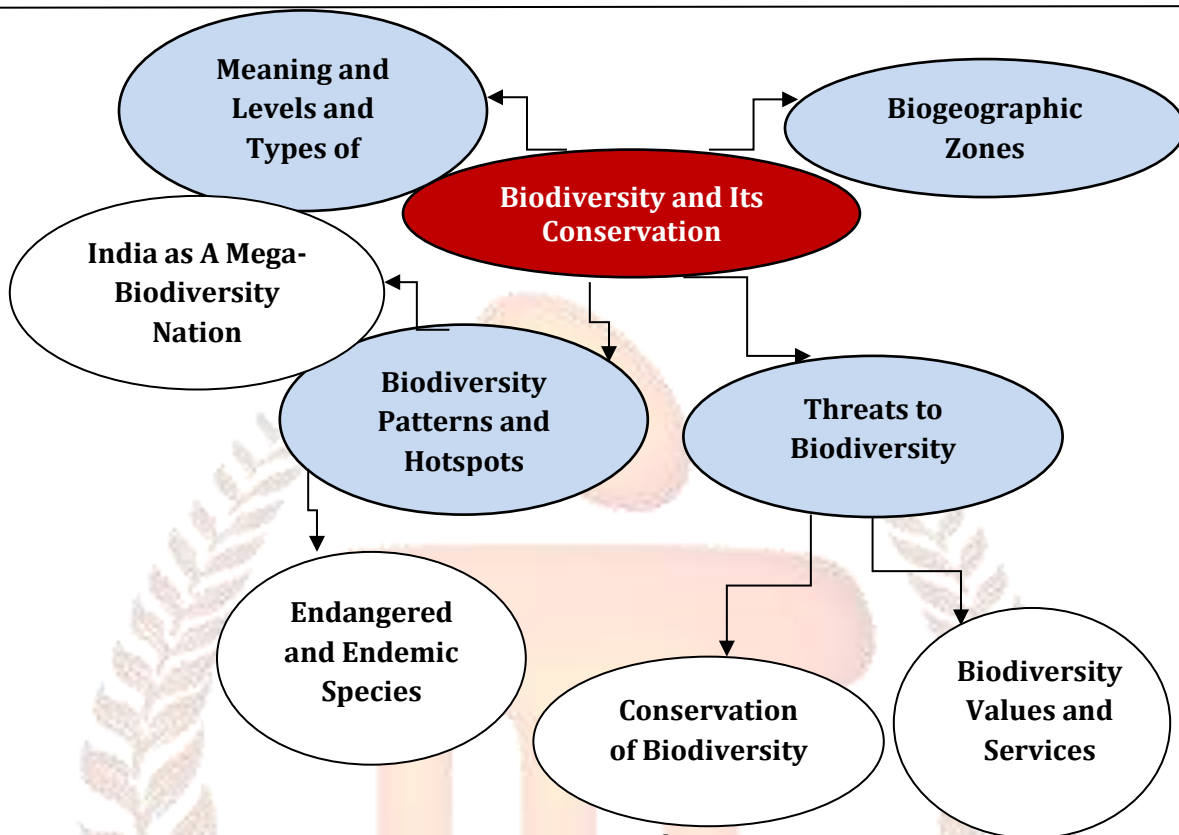


Fig 7: Conceptual Map

6. SUMMARY

- Biodiversity is the number of all the distinct species of plants, animals, and microscopic creatures that exist on Earth, as well as the diverse environments in which they live.
- Biodiversity helps to maintain water cycles, oxygen production, carbon dioxide reduction, soil protection, etc.
- Biodiversity has mainly three levels: Species Diversity, Genetic Diversity and Ecosystem Diversity.
- Biogeographic regions or biogeographic zones are geographical areas that are defined based on the species found there. There are around 10 biogeographic zones in India.
- Biodiversity hotspots are bio-geographic areas with unusually high levels of biodiversity and abundance, as well as a high concentration of endangered species.
- India is one of the 12 super biodiversity countries on the planet. Around 5,000 species of flower plants are endemic to the region. India is home to over 1,300 bird species and 460 reptile species.
- The species that are under risk or threat of being extinct are known as Endangered Species.
- Habitat destruction, animal smuggling, and man-wildlife violence are also significant threats to biodiversity.
- In-situ conservation and ex-situ conservation are the two primary approaches for habitat conservation.
- Biodiversity offers Environmental Value, Social Value, Economic Value, Ecosystem Services, Consumptive Value, Productive Use Value, Ethical and Moral Value and Aesthetic Value.

7. GLOSSARY

Abiotic: The term which is used to describe the absence of life, refers to all the non-physical elements of the ecosystem.

Biotic: The term which is used to describe the presence of life, refers to all the living components of a community.

Depletion: The gradual decrease in the number of something, such as plants, animals, trees, etc.

Ex-situ: A place that is situated away from the natural or the original location.

Habitat: The natural environment of an organism, where it lives and grow.

In-situ: A place where an event takes place, a place of origin.

8. CASE STUDY

MANDAVALLI FOREST RANGE

In the catchment of the river Sharavati, the village of Mandavalli is located 25Km from Sagar Town, in the district of Shimoga in Western Ghats. The river has been dammed for Linganamakki Reservoir. Mainly metamorphic rocks are found in this region with red sandy and lateritic soil type. The forest is covered by deciduous and evergreen trees.

Principal Observations:

- A decade ago, the northern part of the village was covered by dense forests, but due to deforestation only a thin forest remains.
- The water requirements are met by three lakes in the village.
- The overall literacy is 60% and female literacy is low.
- 148 species of medicinal plants are found in the forest, of which 15 are endangered.
- Trees constitute 48% of the plant population, herbs form 12%, shrubs form 32% and climbers form 8%.
- It is a relatively high rainfall area with low development and low population pressures.

Threatening Factors:

- Rainfall has considerably decreased to an extent of 40% resulting in the death of herbaceous species.
- In the last two decades, deforestation to an extent of 15% has been reported.

- Water bodies have been silted up to 32% of their capacity which has resulted in reduction in water-holding capacity.
- Honey Bee colonies have decreased to the extent of 70%.
- The dam has resulted in submersion of 15sq km of the forest area which has led to loss to medicinal plants.
- Expansion of roads and installation of power lines has caused damage to forests.

Source: Article from Economic times newspaper

Discussion Questions:

Q1. What would the decrease in Honey Bee colonies lead to?

Q2. What else do you think can cause damage to the forest?

Q3. What can be done to prevent this ecological imbalance that this village is heading towards?

9. TERMINAL QUESTIONS

SHORT ANSWER QUESTIONS

- Q1. Mention four significant uses of biodiversity.
- Q2. What is genetic diversity?
- Q3. What are the features of Western Ghats? Mention two species of animals found in the Western Ghats.
- Q4. List four endangered animals and endemic animals.
- Q5. List four biodiversity hotspots around the world.

LONG ANSWER QUESTIONS

- Q1. What are the types or levels of biodiversity?
- Q2. What are endangered and endemic species? Give some examples.
- Q3. What are the different threats to biodiversity?
- Q4. What are the methods of conserving biodiversity?
- Q5. What are the different services or values of biodiversity?

9.1 Answers

SELF-ASSESSMENT QUESTIONS:

1. False
2. 1985
3. D. All of the Above
4. True
5. False
6. Biogeographic Zones
7. A. Ibex
8. True
9. Biodiversity Hotspots
10. D. All of the Above.
11. D. Both a and b
12. True
13. False
14. Ex-situ

15. C. Both a and b

16. D. All of the above

TERMINAL QUESTIONS:

SHORT ANSWER QUESTIONS:

Answer 1: Uses of biodiversity are:

- It helps in the storage and recycling of nutrients.
- It helps in checking pollution.
- It contributes to the stability of the climate.
- It helps an ecosystem to recover from unpredictable events.

Answer 2: Inside a population, genetic variability refers to the variety of genes that exist. Genetic variation refers to the various forms of genes present in plants, organisms, and microbes.. This genetic diversity is vital for the successful reproduction of a species' population.

Answer 3: The Western Ghats cover 4.0 percent of India's total land area. It is one of India's most important tropical evergreen forest regions. The Western Ghats are home to the majority of peninsular India's vertebrate animals. Two species of animals found in the Western Ghats are Nilgiri **Langur** and **Lion Tailed Macaque**.

Answer 4: Four endangered animals are:

- Asiatic Black Bear, Great Indian Rhinoceros, Indian Elephant or Asian Elephant, and Red Panda.

Four endemic animals are:

- Asiatic Lion of Gir Forest, Lion-tailed Macaque of Western Ghats, Kashmir Stag of Kashmir Valley and Nilgiri Tahr of Nilgiri Hills.

Answer 5: Four biodiversity hotspots around the world are:

1. **North and Central America :** California Floristic Province, Mesoamerica and Madrean pine-oak woodlands,
2. **The Caribbean:** Caribbean Islands
3. **South America:** Atlantic Forest, Cerrado, Chilean Winter Rainfall-Valdivian Forests, Tumbes-Chocó-Magdalena, Tropical Andes

4. **Europe:** Mediterranean Basin

LONG ANSWER QUESTIONS:

Answer 1: There are three types or levels of biodiversity which are the following:

- **Species Diversity:** Species diversity refers to the broad number of plants, animals, organisms, and fungi that can be found in a given environment.
- **Genetic Diversity:** Genetic diversity refers to the distribution in genes found within a population.
- **Environmental Diversity:** The diversity of species, natural populations, and environments is referred to as ecosystem diversity.

Answer 2: Endemic species: Endangered animals are those that are threatened by extinction. As a result, the 'endemic location,' 'national endemic,' or 'geographical area endemic' refers to the place where the species is endemic. Kangaroos, for example, are native to Australia and are found nowhere else on the planet. Humans introduced them outside their natural environment while they were in captivity, which is why they have been seen in areas other than their natural habitat.

Answer 3: The following are some of the main threats to biodiversity:

- **Habitat Loss:** Misuse of natural resources that has led to a major loss of biodiversity. Urbanisation, rapid industrialisation and an increase in population has led to massive deforestation and habitat loss.
- **Poaching of wildlife:** Poaching of animals for trade and economic purposes has been a major cause of the extinction of many animals and the endangerment of others, such as whales and Asian tigers.
- **Man-wildlife conflict:** It applies to the relationship between wild animals and humans, as well as the negative consequences for both. Recently, the degree of conflict has increased due to a gradual increase in the human population.

Answer 4: There are two main techniques used for the conservation of biodiversity:

- **In-situ conservation:** The conservation of species within their natural ecosystem of the environment is known as In-situ or on-site conservation. It is considered the most practical method of biodiversity conservation. Here, the genetic resources are conserved through their maintenance within the environment of their origin.

- **Ex-situ conservation:** Ex-situ conservation refers to the preservation of different elements of ecological diversity outside of their natural environments. This approach involves removing threatened or endangered animals and plants from their natural habitat and placing them in a special care facility where they will be safe and given the opportunity to mature naturally.

Answer 5: The services and values of biodiversity are the following:

- **Economic Value:** Biodiversity is a vital opportunity for a number of sectors that fuel the global economy. Biodiversity is the foundation of the world's main fuel supplies, such as wood and fossil fuels. Important chemicals are derived from a wide range of flora and fauna and are used in a variety of medical fields.
- **Consumptive use value:** This category includes natural resources that are specifically used for cooking, such as wood, fuel wood, and so on. Humans use nearly 45,000 plant and animal species on a daily basis.
- **Productive Use Value:** These figures refer to items that are harvested and sold commercially. Biotechnologists use wild plant species to produce higher yielding and disease-resistant varieties.
- **Ethical and Moral Value:** These principles are linked to the preservation of biodiversity. It is founded on the idea that all kinds of life on Earth have the right to live. Morality and ethics teach one to protect all aspects of life and to prevent causing undue damage to other species.
- **Aesthetic Value:** Biodiversity is responsible for the uniqueness of our planet; without it, Earth would mimic most desolate worlds in the world. The uniqueness of an ecosystem is due to biodiversity, which has contributed to eco-tourism.

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BACHELOR OF COMPUTER APPLICATIONS

SEMESTER 1

ENVIRONMENTAL SCIENCE

Unit 5

Environmental Pollution & Policies -I

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1. INTRODUCTION

Air pollution is pollution that harms the environment and ejects harmful gases, among other things.

Air Pollution alludes to the condition where the presence of harmful substances in the air, produced by different human activities and natural calamities like volcanic ejections, brings about harmful impacts on the environment. As expressed here, this is an intense issue that is hard to treat because of the idea of airborne particles.

As per the World Health Organization (WHO), outside air contamination is arranged into four principal classes: particulate matter, ozone, nitrogen dioxide, and sulfur dioxide. Also, air contamination is additionally isolated into essential and optional poisons. Essential contaminations are delivered directly into the air from a source, where optional poisons happen because of complex compound reactions occurring from two toxins reacting with one another (**Kibble and Harrison 2005**).

As we all know, One of the hazardous parts of biodiversity is water pollution. The basic and essential need of living beings is water. It is inevitable. A major area of the Earth is occupied with water. Water is present in the atmosphere in various ways, like vapours, in seas, oceans and underground. However, in several countries, the amount of freshwater available is limited. A lot of rainwater is wasted and is not used effectively. Water extracted from the underground area gets replenished by seepage of rainwater.

STUDY NOTE

The shocking fact of the world is that most of the people die from drinking harmful elements that are mixed with water rather than dying on road accidents, wars, etc.

Due to a rise in population and industrial and agricultural activities, a huge amount of water is consumed and contaminated. An enormous amount of water is required for construction work, houses, offices, roads, shops, etc. This is due to the increase in demand for these things by the increased population. Almost every industry utilizes water in its production process. The rise in the number of industries causes depletion of water. A majority of farmers use water for irrigation purposes, which has also led to a decrease in water levels.

Water can be polluted through natural and man-made processes. To reduce water pollution, various measures are required to be taken immediately. The discharge of hazardous substances into the water bodies by the industries should be prohibited.

Pollution is harmful to all living kinds- be it plants, animals, or humans. It equally affects all kinds of living creatures and can sometimes be life-threatening. We should consider it a serious concern rather than adapting to these problems. Punishable actions should be taken against those engaging in polluting, to make our world pollution-free.

we will also study a type of land pollution which is soil pollution, in detail. We should be aware of soil pollution because it indirectly relates us to the cause of our living, i.e., food. The autotrophs get adversely affected by those chemicals, sewage and many other kinds of pollutants, which affect animal's health that consumes it directly (Herbivores), and human beings who consume plants in one or other form indirectly.

1.1 Learning Objectives

After studying this chapter, you will be able to:

- ❖ *Explain the meaning of Air pollution and sources of Air Pollution*
- ❖ *Describe the Effects of Air Pollution on Human Health and Environment*
- ❖ *Describe the effects of water pollution on the environment and human health.*
- ❖ *Elaborate on the various types of water contaminants.*
- ❖ *Explain the importance of soil on various components.*
- ❖ *State the consequences of soil pollution on future generations and suggest control measures.*

2. AIR POLLUTION

When particles suspended in the air are combined with the gases, it is known as air pollution. Particles may be suspended as a result of vehicle discharges, gases from production units, burning fuels, dust, and mold spores. Ozone gas which also forms a protective layer against UV rays, is a substantial source of air pollution.

It is the presence of substances in the air that are destructive to the soundness of health people and other living creatures, or cause harm to the environment or materials. There are various kinds of air poisons, like gases (such as, carbon monoxide, sulphur dioxide, nitrous oxides, methane and chlorofluorocarbons), particulates (both natural and inorganic), and organic particles.

STUDY NOTE

Youngsters living close to contaminated zones are more inclined to pneumonia and asthma. Numerous individuals pass on consistently because of the immediate or aberrant impacts of air contamination.

Air contamination may cause infections, sensitivities and other problems to humans and the environment. It might hurt other living and non-living things in the environment. For example, food crops, animals, plants, humans and climate. Both human action and natural calamities can create air contamination.

A. Negative impact of Air pollution

- It is a critical danger factor for various contamination-related sicknesses, including respiratory diseases, coronary illness, Chronic Pulmonary Obstructive Disorder (COPD), stroke and cellular breakdown in the lungs.
- The sweeping impacts of low air quality on human well-being majorly influence the body's respiratory and cardiovascular frameworks. Singular responses to air poisons rely upon the kind of contamination an individual is presented to, the level of exposure, and the person's well-being status and hereditary qualities.
- Indoor air contamination and poor metropolitan air quality are recorded as two of the world's most exceedingly terrible harmful contamination issues in the 2008 Blacksmith Institute World's Worst Polluted Places report.
- According to WHO reports, Ambient (outdoor air pollution) in both cities and rural areas was estimated to cause 4.2 million premature deaths worldwide in 2016. WHO also stated, "Approx. 7 million people lose their lives due to air pollution, as it

causes serious health problems like heart failure, breathing problem, lung cancer, and many other problems.

- Due to the high death rate because of air pollution, the Institute of Health Metrics and Evaluation and World Bank estimated the total cost they lost in welfare losses is approx. US\$5 trillion. Different pollution control advances and systems are accessible to decrease air contamination.

B. Types of Air pollution

- **Particulate Matter:** It is the combination of two particles that are in the form of solids and liquids which includes sulphate, nitrates, carbon, complex chemicals, and water suspended in the air. These generally vary in size as we can see with the naked eye; this can be in the form of dust, soot, smoke or dirt.
- **Nitrogen dioxide:** This is gas which comes from man-made sources such as vehicles, heating plants, and power stations. Diesel vehicles are responsible for emitting these gases.
- **Sulphur dioxide:** This is a colourless gas that cannot be seen and has a very suffocating smell. This originates from burning oil and coal. The major source of Sulphur dioxide gas is industries, which mostly burn fossil fuels.

STUDY NOTE

Chemical reactions, agricultural activities, and transportation are the primary sources of air contamination. The clustering of certain toxins can be reduced only by dispersion, or breakdown whenever they are released into the environment.

SELF-ASSESSMENT QUESTIONS - 1

1. When particles suspended in the air are combined with the gases it is known as _____.
2. Air Pollution majorly affects the respiratory and cardiovascular framework. [True/False]
3. Sulphate and nitrates are among the particles that make up nitrogen dioxide, which comes in the form of different substances. [True/False]
4. Which of the following result in emission of Nitrogen dioxide?
 - a) Heating plants
 - b) Power stations
 - c) None of the above
 - d) Both A and B
5. Sulphur dioxide is one of the gases that causes air pollution; the colour of sulphur dioxide is:
 - a) Black
 - b) Brown
 - c) Colourless
 - d) Smoky grey

2.1 Sources of Air Pollution

Sources of air pollution refer to means by which air pollution is caused. These sources can be natural or even man-made. Some of these sources are:

- A. Toxic gases
- B. Vehicular pollution
- C. Industrial setups
- D. Wood or other substances burning

Pollutants in the air can be released directly into the environment through basic flows or can take shape as a result of compounding interaction with changing environment. According to

WHO statistics from 2016, India is among top 5 polluted countries where more than 10 cities have highest pollution.

Toxic emissions cause air contamination; however, a decline in toxic outflows do not always result in equal reductions in pollutants. There are astounding connections between air contamination outflows and air quality. Chemical variations, unusual changes in the environment, exploitation of resources and global obligations, the effect of climate and geography, are some examples of reasons that cause toxic emissions. Critical cuts in emanations are fundamental for improving air quality.

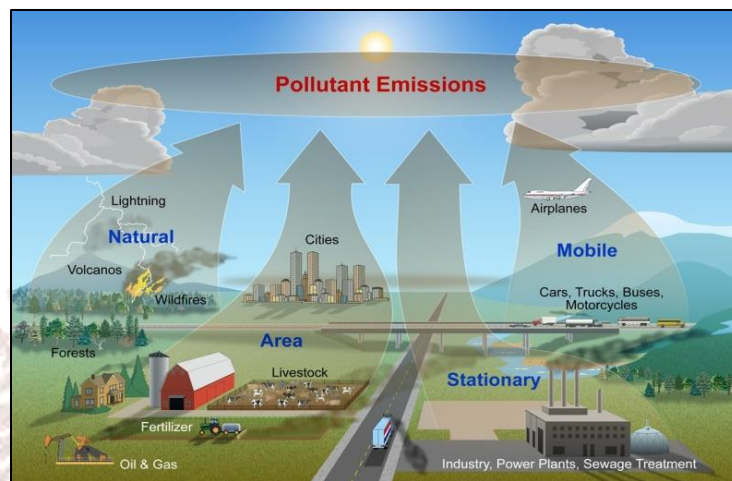
Air poisons are transmitted from a scope of both man-made and common sources, including:

- Consuming of non-renewable energy sources in power age, transport, industry and families
- In mining industries, the physical and mechanical activities take place in which they use insoluble chemicals, materials that mix with air
- Farming and waste treatment
- Characteristic sources include volcanic ejections, windblown residue, ocean salt shower, and outflows of unstable natural mixtures from plants.

There are four principal sources of air pollution:

1. **Portable or mobile sources:** These are sources that can move from one place to another. Portable sources generally consist of vehicles, transports, planes, trucks, etc. These sources create air pollution while running on fuels leading to carbon monoxide being released into the air.
2. **Fixed or stationary sources:** These sources are those that remain stationary at one location like power plants, petroleum processing plants, mechanical offices, and industrial facilities. These are some of the major sources of air pollution all around the world. The government usually mandates these plants to carry out different measures to filter the gases before exposing them to the air.
3. **Zone or area sources:** These sources are usually those which are specific to a certain region in the country like farming territories, urban areas, and wood-consuming chimneys.

4. **Common or natural sources:** These are the sources over which there is no control of humankind and are natural phenomena like the wind-blown residue, fierce blazes, and volcanoes



Source- nps.gov

Fig 1: Sources of Air Pollution

SELF-ASSESSMENT QUESTIONS – 2

6. When _____ sources of air pollution run on fuel, carbon monoxide is discharged into the atmosphere, causing pollution which negatively affect the living beings.. Volcanic eruptions are an example of zone source of Air Pollution. [True/False]
7. The fixed sources of air pollution pollute more than one region [True/False]
8. Which of the group is example of zone source of air pollution?
 - a) Car, plane, truck
 - b) Power plant, car, industrial unit
 - c) Volcanic eruption, wind-blown
 - d) Wood consuming chimneys, farming territories
9. Which of the sources of air pollution remain constant at a particular location?
 - a) Mobile sources
 - b) Natural sources
 - c) Stationary sources
 - d) Area sources

2.2. Effects of Air Pollution

The steady contamination of the air has extremely negative impacts on nature and human well-being. Those long haul and medium-term impacts have become issues that are hard to address. All of those significant impacts can increment if their causes are not quickly followed. The vast majority of them result from excessive contaminating and uncontrolled mechanical activity. Companies around the globe must oversee, treat and control the toxins they radiate into the environment. The negative effects of air pollutants and toxins has a poor impact on human health and environment.

A. EFFECT ON THE ENVIRONMENT

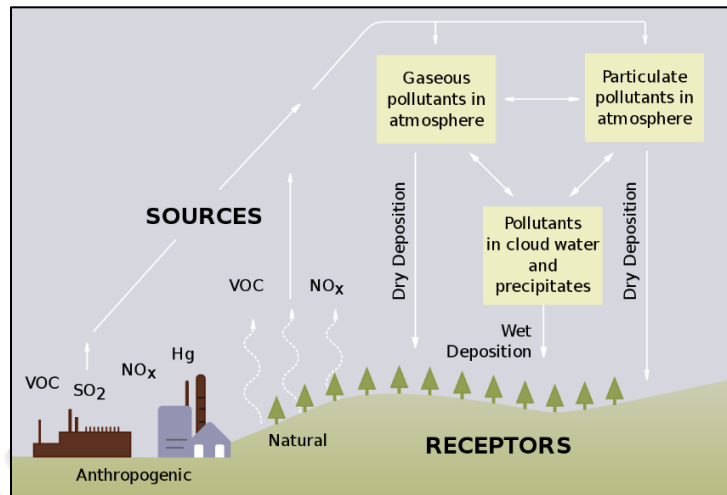
- **A dangerous atmospheric deviation (GLOBAL WARMING):** A dangerous atmospheric deviation is perhaps the most stressful impact for researchers and preservationists. A worldwide temperature alteration is an immediate result of the

primary influence, which is delivered by the high outflow of CO₂ and methane into the air. A major amount of these pollutants comes from the manufacturing industry. Manufacturing plants can either stop or prevent this by limiting activities, or the government can impose restrictions and penalties on the businesses.

- **Environmental Change:** Environmental Change is another result of a dangerous atmospheric deviation. At the point when the temperature of the planet rises, there is an aggravation in the standard climatic cycles, quickening the progressions of these cycles in an apparent manner.
- **Acid Rain:** The gases released by enterprises, power plants, boilers, warming and transport are exceptionally poisonous. Sulfur dioxide (SO₂) and nitrogen oxides (NO_x) are released into the atmosphere due to the consumption of petroleum products such as generators and cars.
- **Smoking Fog (SMOG) Effect:** The smoke occurs when a dull haze is present over the urban areas and fields. That mist is a heap of toxins and can be of 2 sorts: sulfurous exhaust cloud and photochemical brown haze, both hazardous and hurtful to well-being. The two sorts of brown haze are a result of the modern and metropolitan activity. In any case, the sulfurous brown haze has its root fundamentally in the utilization of coal in numerous mechanical cycles. That has been decreased in the most exceptional nations – in any case, there are as yet many non-industrial nations that don't have treatment conventions for toxins.
- **Deterioration of fields:** Corrosive downpour, environmental change and exhaust cloud affect the Earth's surface. Contaminated water and gases permeate the land, altering the composition of the soils. This directly affects agricultural activities, changing production cycles, and the food chain hierarchy.

STUDY NOTE

Contaminations can be regular, like volcanic debris. They can likewise be made by human action, for example, refuse or spill over created by production lines. Contaminations harm the nature of air, water, and land.



Source- https://en.wikipedia.org/wiki/Acid_rain

Fig 2: Effect of Air pollution- Acid Rain

B. EFFECTS ON HUMAN HEALTH

Both, healthy and unhealthy people can suffer from the negative effects of polluted air, such as respiratory discomfort or breathing difficulties during exercise or outdoor activities. The actual risk of negative consequences is determined by the current health status, the surroundings, toxin kind and intensity, and the duration of time an individual is exposed to contaminated air.

Air pollution can have various serious impacts on human health, which may lead to death. Certain air contaminants fall out of the air and end up falling in our meals, drinks, and soils, such as mercury in fish. Climate change and air pollution are somehow linked with each other, which also affects human health. Climate change occurs due to burning trees, fossil fuels and so on, which directly affect the surroundings and create various health issues.

High air pollution levels can cause prompt medical conditions including:

- Disturbed cardiovascular and respiratory ailment
- Added pressure to the heart and lungs, which work towards supplying oxygen to different organs of the body.
- Affected cells in the respiratory framework

Long haul exposure to polluted air can have lasting well-being impacts such as:

- Early-onset of maturing of the lungs
- Limitation of the functioning of lung capacity

- The onset of infections like asthma, bronchitis, emphysema, and perhaps malignant growth
- Life expectancy has been shortened

SELF-ASSESSMENT QUESTIONS – 3

10. Air pollution has adverse effect on human health which causes certain diseases and on environment which causes problem to living and non-living things. [True/False]
11. The gases released by enterprises, power plants, boilers, warming and transport are _____.
12. Which kind of problem to human health can be caused by the Air contamination?
 - a) Respiratory or breathing problems
 - b) Infection
 - c) Lungs problem
 - d) All of the above
13. When the temperature of the planet rises, there is _____ in the standard climatic cycles.
 - a) Aggravation
 - b) No effect
 - c) Less serious
 - d) None of the above

2.3. Methods for Control of Air Pollution

Air contamination control, the procedures utilized to decrease or dispose of the emanation into the climate of substances that can hurt the climate or human well-being. Air contamination control is one of the foremost zones of contamination control, alongside wastewater treatment, strong waste administration, and perilous waste administration. These remember antagonistic impacts on human well-being, property, and environmental perceivability. The air is defenceless to contamination from natural sources, just as from human exercises. Some normal marvels, like volcanic emissions and timberland fires,

may have neighbourhood and local impacts as well as enduring worldwide ones. By the by, just contamination brought about by human exercises, like industry and transportation, is dependent upon alleviation and control.

The necessary steps needed to reduce pollution:

- 1. Usage of Public Transport:** Since vehicular emission is one of the major sources of air pollution, usage of public transport will reduce the said emissions.
- 2. Development of alternative fuel:** When fossil fuels are burnt, they generate harmful toxins. To reduce these, the usage of alternative fuels such as electricity and solar power is the most effective way.
- 3. Eco-friendly packaging and products:** When plastic is burned it releases harmful toxins which can combine with the air and cause air pollution. If we start using eco-friendly packaging it can be more environmentally friendly.
- 4. Avoid fireworks:** The burning of fireworks releases huge amounts of carbon monoxide and other toxins. This results in high pollution in the region where the fireworks are being burnt. To reduce air pollution, avoiding fireworks is the best method.
- 5. Afforestation:** When we plant more trees, more oxygen is produced hence improving the AQI of the region.

SELF-ASSESSMENT QUESTIONS – 4

15. Planting of more and more trees is known as the _____.
16. Usage of public transport reduces carbon emission. [True/ False]
17. One of the ways to reduce air pollution what we can choose as best practise?
 - a) Use ecofriendly products and packaging
 - b) Increase bursting crackers
 - c) Increase use of personal vehicles
 - d) All of the above

3. WATER POLLUTION

Water pollution occurs when there is an existence of harmful chemicals and biological components that are mixed with freshwater. It also includes the injection of chemicals into water bodies through various human activities. These toxins have the potential to harm the human and aquatic ecosystem. The temperature of the water bodies is worse affected due to discharge of hot water from some specific industries, which uses water for cooling in different operations, and due to the release of water from dams.

Due to the effect on temperature, life-forms inside water bodies are in great danger. It can encourage the growth of some species while harming some other species. It affects the balance between the different organisms in the water bodies.

The eggs and larvae of different animals are specifically susceptible to temperature changes—water pollution results in the removal of necessary substances from water bodies. Plants and animals use dissolved oxygen for their survival in aquatic systems, and anything that reduces its level, would badly affect the animals and the plants residing there.

3.1 Point & Non-Point Sources of Water Pollution

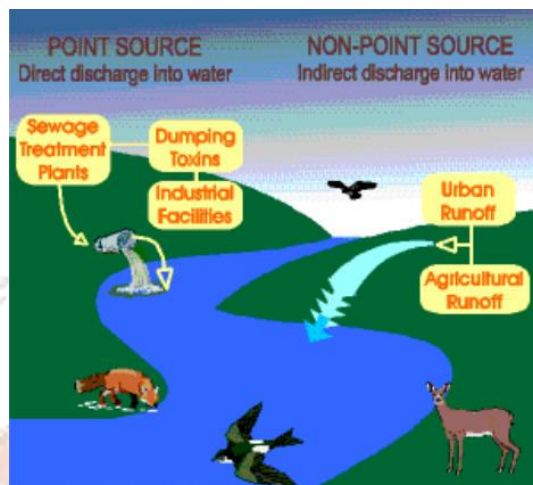
Every process on earth, whether it is man-made processes or natural processes, generate a certain amount of by-product from that particular process. Generally, these by-products are returned to the environment and are known as pollutants. The environment can “self-clean” these pollutants if given enough time. But, the consistent and heavy flow of by-products into the environment makes it extremely difficult for the environment to do its cleaning all by itself. The sources of these pollutants can be classified into point sources and non-point sources.

❖ Point sources:

Pollution arising from a sole, recognizable source, like a drainpipe from a sewage plant, tunnel, well, vessel or factory, is known as point-source pollution. These pollutants in groundwater and surface water are generally found on the top with the highest group of the pollutant nearest to the origin and the declining group far away from the origin.

Point sources of pollution arise from agriculture, including animal feeding operations, animal waste treatment lagoons, mixing and handling, storage, cleaning areas for pesticides,

petroleum and fertilisers. Municipal pollution sources comprise landfills, motor pools, utility stations, fleet maintenance facilities and wastewater treatment plants.



Source: savelakewinnipeg.com

Fig 3: Point Source & Nonpoint Source

These activities involve the usage of dangerous materials as raw materials used in the process. Harmful substances are also discharged through these processes, which, if not treated properly, can get mixed up with the surface water and cause pollution.

1. **Groundwater:** One of the most constant point-source pollutants in groundwater are organic compounds, which encompass toxic substances like oils, paint, solvents and fuel products. In general, these pollutants cause severe health issues. Once groundwater is polluted, it is not possible to clean it up easily.
2. **Surface Water:** Some basic point-source pollutants are present in surface water which are as follows:
 - High-temperature discharges
 - Nutrients (like phosphorus and nitrogen); and
 - Microorganisms (like bacteria, viruses, and Giardia)

Nutrients and a temperature rise can result in excessive plant growth and decomposing organic matter in water, reducing dissolved oxygen levels and causes a great threat to aquatic life. Microorganisms are supposed to be dangerous for both human health and aquatic life. Pesticides and other harmful substances are also considered as hazardous to

both aquatic life and human health but are less likely to be present in surface water due to high infusion rates.

❖ **Non-point sources:**

Pollution through non-point sources doesn't have a sole and identifiable source. Pollution through these pollutants occurs as water shifts across the land or through the ground and carries natural and man-made pollutants, which can then get deposited in rivers, coastal waters, wetlands, lakes and also groundwater.

The water that creates non-point source pollution can arise from natural processes like snowmelt or rainfall or human activities like lawn maintenance and crop irrigation. They are also known as 'diffuse' pollution. Its influence occurs over a large area and is not identified to a single source. The main types of non-point sources of pollution are:

STUDY NOTE

The most prominent source of water pollution in India is 'untreated sewage'. Other sources of pollution are unregulated small-scale industry waste and agricultural

1. **Sediment:** It is the runoff of fertilizer and soil during a rainstorm. Sediment includes fine particles like a slit and also suspended solids. Sediment enters surface waters from surface runoffs and eroding stream banks. Sediments generate turbidity in water resources, decreasing the amount of light going at the below depths, which can hamper the growth of aquatic plants. This, in turn, will harm aquatic animals that depend on those plants.
2. **Nutrients:** It refers to inorganic matter from runoff, livestock operations, croplands and landfills. The two major ingredients are nitrogen and phosphorus. Phosphorus is most commonly transported to water areas through soil erosion and nitrogen is transported by water as nitrate (NO₃). A surplus amount of phosphorus leads to the expansion of microscopic algae, which will lead to the huge growth of organic matter that is eutrophication, i.e., lowering of oxygen level in the water bodies.
3. **Pathogens:** These are bacteria and viruses generally found in the water and cause human diseases. They cause diseases when they get mixed in the public drinking water supplies. It contains Salmonella, Parasitic worms, Cryptosporidium parvum and Giardia lamblia. They can pollute runoff due to inappropriately managed livestock tasks, poor pet waste management and polluted storm sewers.

4. Toxic contaminants and chemicals: Compounds comprising heavy metals like lead, zinc, cadmium, mercury and organics like polycyclic aromatic hydrocarbon (PAHs) and polychlorinated biphenyls (PCBs) and other substances are insusceptible to breakdown. Toxic chemicals comprise organic and inorganic compounds like acids. Pesticides are not breakable and thus sustain in the environment. The principal sources of non-point sources of pollution are:

- **Urban land use:** In urban areas, rainfall drains as storm water, a pollutant that influences water quality. Storm water is contaminated from animal feaces, dust, car oil, sediments and industrial wastes from street surfaces.
- **Agricultural land use:** Pesticides, fertilizers, animal dung, and soil flow in the streams of rainfall-runoff ultimately results in pollution. When livestock are allowed access to watercourse areas, it pollutes the water and increases erosion risks. When huge sectors of land are ploughed to grow crops, it uncovers and loosens the soil, increasing soil erosion during rain and wind.
- **Forestry land use:** Forestry operations decrease the number of trees in the environment, thus decreasing the oxygen level also. The excessive usage of bulky machinery increases the chances of soil erosion. Thus, increasing the chances of soil erosion and sediment runoff, contributes to non-point source pollution.
- **Marinas and boating operations:** Chemicals used for the maintenance of the boat, such as paint, oils and other solvents, get to mix with water through runoff and contaminate it. Spilling fuels and leaking fuels also contaminate the water. Improper management of sanitary waste bins on the boat leads to a rise in the levels of solid waste, that hamper the quality of water.

SELF-ASSESSMENT QUESTIONS – 5

18. Water pollution is referred to the existence of _____ and _____ in water.
19. The _____ & _____ ecosystems are majorly affected due to water pollution.
20. Pollution arising from single and identifiable source is called _____.
21. Pesticides, fertilisers, and animal waste can be found in the rain-runoff waterways, which result in water pollution. . [True/False]

3.2 Effects of Water Pollution on Human Health**On the Environment:**

- A healthy ecosystem consists of a bunch of flora, bacteria, fungi, and fauna, all interacting directly or indirectly. Any problem to any of the above organisms can develop a chain effect that affects the aquatic environment. When water pollution causes algae to thrive in the lake, the increase of freshly introduced nutrients induces plants and algae to grow, decreasing the water's oxygen levels. This deficiency of oxygen is called eutrophication, which can create “dead zones” (where waters lack life) by suffocating plants and animals. These hazardous algae blooms sometimes result in neurotoxins that affect wildlife.
- Marine ecosystems are also included in the danger zone by the segregation of marine waste that can choke, starve, and restrain animals. Dumping fishing nets and other tools to clean water results in hurting more than 200 various species under marine life that also affects the environment.
- Ocean acidification turns out to be more difficult for shellfish and corals to survive under water. Ocean acidification is defined as the never-ending decline in the pH value of Earth's oceans that is caused by the absorption of carbon dioxide from the atmosphere. This process creates hurdles for shellfish and other marine species to develop and affects the nervous systems of clownfish, sharks and other marine species.

On Human Health:

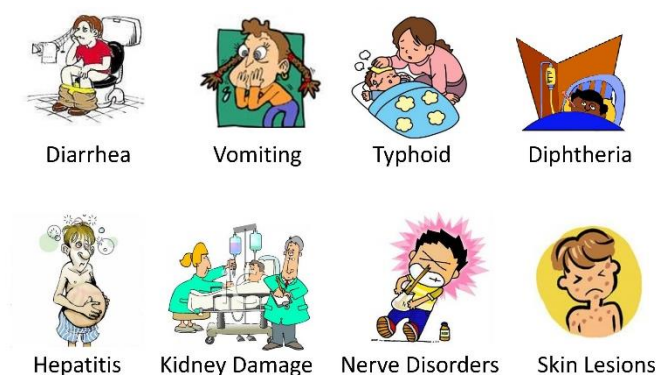
- Contaminated water makes people ill and can also cause death. Every year, contaminated water nauseates about one billion people, especially, the lower-income people, as they reside close to polluting industries.
- Waterborne pathogens are disease-causing bacteria and viruses that take birth from natural and man-made waste and are a main cause of illness from the intake of unsafe drinking water. Diseases developed by unsafe water comprise of cholera, typhoid and giardia. In developed countries, accidental or prohibited discharges from sewage treatment facilities turn out from runoff from farms and urban areas that supply dangerous pathogens to waterways. Thousands of people across the country are found ill every year by Legionnaires disease.
- A large variety of chemical pollutants like arsenic and mercury are coupled with pesticides and nitrate fertilisers and ultimately get into water supplies. Once they are mixed, these toxic elements can cause severe health issues such as cancer, hormone imbalances, weak neural system, etc. Also, from swimming, large number of people contract health issues like skin rashes, respiratory infections, pink eye and hepatitis.
- Pollutants like nitrates are especially hazardous to babies that drink formula milk. It limits the amount of oxygen in the brain, causing the 'Blue Baby' syndrome. It is also termed infant methemoglobinemia, which is a skin disease where a baby's skin turns blue. It happens when babies are fed baby food that is nitrate-rich.
- Another harmful pollutant is 'lead', which can get accumulated in the body and can cause harm to the central nervous system. Arsenic leads to liver damage, vascular disease and skin cancer. A huge number of fluorides can make teeth yellow and harm the spinal cord. An increase in the number of pollutants is the root cause of the high rate of infant mortality.

STUDY NOTE

About 80% of stomach diseases in India are caused due to unsafe drinking water.

STUDY NOTE

According to UNICEF, children die globally because of intake of polluted water that can increase the death count to 3000 per day.



Source: learnpick.in

Fig 4: Effect of Water pollution on human health

SELF-ASSESSMENT QUESTIONS – 6

22. Blue Baby syndrome is also termed as infant methemoglobinemia. [True/False]
23. What makes it difficult for shellfish and coral to survive?
- Dumping fishing nets
 - Ocean acidification
 - Both a and b
 - Neither a nor b
24. _____ is the insufficiency of oxygen due to filing of sediments in water.
25. In formula milk, _____ pollutant is very harmful as this can adversely affect the health of infants and cause major brain problems.

3.3. Control Measures for Water Pollution

Due to water pollution, water bodies are harmed and also water resources convert into waste. Prevention of water pollution and conservation of water resources is extremely important to ensure an abundance of safe water for ourselves and future generations. Reducing pollution is inevitable for the good health of humans and the environment. Following are the various ways in which water pollution can be controlled:

- The most prominent way of controlling water pollution is prevention, and minimization of gathering of pollutants into water bodies.

2. The Indian government has implemented the Water Prevention and Control of Pollution Act, in 1972. to restrict all types of water pollution in the country.
3. Each individual should show concern towards water quality and avoid polluting water. Reusing, reducing and recycling should be done by every individual wherever possible.
4. Industries should do treatment of hazardous outputs before they pour them into water bodies. They should reduce the level of water pollution through chemical processes like coagulation, reverse osmosis and ion exchange method, etc.
5. At regular intervals, the monitoring of wastewater should be conducted.
6. There should be a setup of a well-structured effluent and sewage treatment plant.
7. Effective drain line maintenance should be done at every level in the country. There should be no leakage in the drainage pipeline.
8. Both the government and the people of the nation should plant more trees around water bodies as they will naturally help them incorporate and recycle the pollutants.
9. Natural pesticides and fertilizers should be used in place of chemical ones as they are organic and safe for plants and water bodies.
10. Special plants like the Water Hyacinth must be grown in areas which are more likely to have water pollutants. It will decrease the negative effects to a greater extent. This plant absorbs dissolved toxic substances such as cadmium and mercury from water bodies.
11. A comprehensive water management plan should be developed for every affected or prone-to-be-affected region, stating goals to achieve within a stipulated time.
12. Construction of proper storm drain and settling ponds should be done to reduce the chances of water clogging and harming aquatic life.
13. Rainwater harvesting structures should be built at both household and official levels to ensure zero water wastage.
14. People should be aware of the harmful effects of water pollution and should be against dumping wastes into freshwater. The people who dump wastes should be punished through fines.
15. Throwing of paint and petroleum products into toilets and sinks should be highly penalized as they cannot be treated and adversely affect the water bodies.
16. Avoidance of oil spills in the ocean should be done.

STUDY NOTE

India is taking productive steps in rebalancing the quality of its water source. Gujarat government launched new platform as the Reuse of Treated Waste Water Policy, which aims to downsize the use of the Narmada River.

17. Landfills (the disposal of waste by burying it) should be undertaken as gases generated in the landfill can be used for the generation of electricity.
18. Bioremediation should be done. It is a waste management technique. It involves the use of organisms to neutralize or separate pollutants from contaminated sites. The organisms break down the harmful substances into less toxic or non-toxic substances.
19. Rivers should not be used for washing clothes or bathing animals in them. Strict surveillance should be made on rivers, ponds and lakes.
20. Lastly, everyone should try to minimize the consumption of water and reuse water whenever it is possible to decrease the level of pollution.

STUDY NOTE

The word pollution was derived from Latin word 'pollution' which means to make dirty. Soil pollution is a type of land pollution.

SELF-ASSESSMENT QUESTIONS – 7

26. Tertiary wastewater treatment stage is done to downsize the biological components of the waste through aerobic biological processes. [True/False]
27. _____ are used majorly in industrial wastewater treatment plants as they provide great quality.

4. SOIL POLLUTION

Pollution is very hazardous to human life and can be categorised in various types such as, air pollution, water pollution, noise pollution, litter, visual pollution, radioactive pollution, thermal contamination, and plastic pollution.

First of all, we need to clearly understand the concept of the word pollution. Individuals may purify their surroundings by implementing a variety of pollution-reduction techniques that are not harmful to the environment. However, when the number of pollutants exceed a certain threshold, it begins to destroy the environment.

Definition: Soil pollution is the pollution caused due to contamination of soil with the absorption of toxic elements. Garbage, plastics, fertilisers, pesticides, and many more are the causes of soil pollution.

Most of us know that soil supplies essential elements for human survival, such as food and water. These contaminants have an impact on the crops that grow on that soil, as well as the quality of the groundwater. As a result, we should make soil preservation a priority.

SELF-ASSESSMENT QUESTIONS – 8

28. _____ is the introduction of contaminated particles to the environment that affects its components adversely.
29. Which of the following contribute to pollution?
 - a) Deforestation
 - b) Chemical release in environment
 - c) Urbanisation
 - d) All of the above
30. Which of the following can be categorised under causes of soil pollution?
 - a) Garbage
 - b) Pesticides
 - c) Oil spills
 - d) Both A and B

4.1 Sources of soil pollution

“Essentially, all life depends upon the soil. There can be no life without soil and no soil without life; they have evolved together.” -Charles Kellogg

Let's have a look at various sources of soil pollution:

1. **Agricultural sources:** Agriculture focuses on organic farming of the products to maintain the basics of our lives. Still, it uses pesticides, herbicides, and fertilisers to grow crops and livestock much faster than organic production leads to soil pollution.
2. **Industrial sources:** Industrial waste leads to soil pollution in the form of gas, liquid, solid substances. CO₂, NO₂ are the gases that are released from industrial activities. These wastes mix with rainwater and can result in acid rain that drastically affects the soil's pH.
3. **Urban waste:** Garbage dumped in urban areas contains food waste plastic, industrial waste and general household wastes. In simple terms, urban waste refers to any waste that comes out from household or industrial places, which goes into rivers, and various other water bodies, which ultimately pollute environment.
4. **Sewer sludge:** Sewage sludge in the form of semi-solid material contributes to soil pollution. The sewage sludge is generally processed with septic tanks before being disposed of into land or water bodies.
5. **Mining and smelting sources:** Mining activities trigger soil pollution in huge numbers. Erosion of soil containing some mineral ore traces results in soil degradation. It ends up in the soil by irrigation of land.

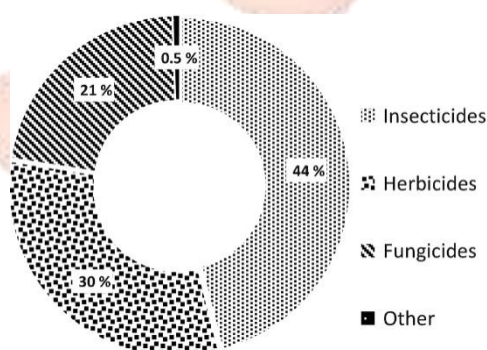
There are numerous pollutants in today's era that contaminate soil in one way or the other. Let's look at some of them for a better understanding of pollutants. Soil pollution can be caused by the following:

- ✓ Microplastics
- ✓ Oil spills
- ✓ Mining and activities by other heavy industries
- ✓ Accidental spills
- ✓ Corrosion of underground storage tanks (including piping used to transmit the contents)
- ✓ Acid rain

- ✓ Intensive farming
- ✓ Agrochemicals, such as pesticides, herbicides and fertilisers
- ✓ Petrochemicals
- ✓ Industrial accidents
- ✓ Road debris
- ✓ Drainage of contaminated surface water into the soil
- ✓ Ammunition, chemical agents, and another agent of war
- ✓ Waste disposal

STUDY NOTE

Mining activities are a major source of soil pollution as they lead to release of chemicals in the soil.



Source: Springer Link

Fig 5: Pollutants Distribution

SELF-ASSESSMENT QUESTIONS - 9

31. Industrial sources cause soil pollution by _____ of pH of soil.
32. Which of the following are agricultural sources of soil pollution?
- a) Herbicides
 - b) Pesticides
 - c) Fertilisers
 - d) All of the above

4.2. Effects of Soil Pollution

- **Soil Life:**

- Due to the presence of chemical substances, soil fertility decreases, thereby reducing soil yield.
- The contaminated soil becomes useless for producing any type of crop.
- Degraded soil leads to the production of contaminated vegetables and fruits consumed by various living organisms, and hence cause various diseases such as food poisoning, weakness, etc.

- **Agriculture:**

- It decreases the quality of the crop.
- It affects the production of crops and also the success of Civilisation.
- It also affects the quality of food produced on that soil.

- **Human Beings:**

- Soil pollution adversely affects human health.
- It also causes neuromuscular blockage.
- It leads to various diseases such as nausea, skin rashes, fatigue, eye problems, headache, etc.

- **Plants and Animals:**

- The decrease in pH of the soil leads to the sickness and death of various plants and animals.
- It also leads to the destruction of the entire aquatic food chain.
- Soil pollution adversely affects the metabolism of microorganisms and arthropods, which ends the primary level of the food chain.

- **Ecosystem:**

- Soil pollution increases soil acidity which negatively affects vegetation, making it useless and barren.
- When any kind of crop grows on these kinds of lands, it leads to the growth of poisonous crop.
- Consumption of such plants grown on degraded soil leads to various health issues and the ecosystem would be adversely affected by this.

SELF-ASSESSMENT QUESTIONS – 10

33. Soil pollution leads to the _____ of pH of soil.
34. Degraded soils are
- a) Acidic
 - b) Basic
 - c) Neutral
 - d) None of the above
35. Soil pollution leads to the destruction of food chain. [True/False]

5. CONTROL MEASURES AND IMPACT ON FUTURE OF SOIL POLLUTION

As we have studied various impacts of soil pollution on our health, and on this environment, we should start adopting various kinds of measures to make this environment safe and clean for our livelihood. Here are some minor steps which each individual can take to make improvements:

STUDY NOTE

Polluting our soil will pollute our future.

- ✓ To bring greenery, start planting more and more trees and try to avoid cutting down trees.
- ✓ Avoid various kinds of chemicals in agricultural practices.
- ✓ Try to dump the chemical wastes in a suitable place.
- ✓ Avoid the use of pesticides and chemical fertilisers as much as possible.
- ✓ Adopt the method of crop rotation in cultivation areas.
- ✓ Try to remove the weeds and various microorganisms that affect plants and soil.

Soil pollution has a very adverse impact on our future lives, which are as follows:

1. Soil pollution hurts biodiversity that leads to the extinction of flora and fauna.
2. It also reduces the soil fertility, which leads to the enhancement of barren land in our country.
3. It causes a reduction of healthy crop cultivation, an increase in food quality degradation.
4. It causes an imbalance of various nutrients in the soil responsible for healthy crop growth.

5. It also degrades groundwater quality, leading to an increase in contaminated water scale.

SELF-ASSESSMENT QUESTIONS – 11

36. Soil pollution _____ of underground water level.
37. Which of the following method can be adopted as a control measure?
- a) Crop Rotation
 - b) Crop Sedimentation
 - c) Crop Filling
 - d) All of the above

6. SUMMARY

- Air Pollution alludes to the condition where the presence of harmful substances in the air, produced by different human exercises and characteristic marvels like volcanic ejections, harms the government assistance of people and the living climate.
- Outside air contamination alone makes 4.21 million passing yearly. Generally, air contamination causes the passing of around 7 million individuals worldwide every year and is the world's biggest single ecological well-being hazard.
- The impacts of air contamination on materials, vegetation, and creatures can be estimated. The majority of the proof comes from toxins that the overall population is exposed to.
- Types of Air pollution
 - **Particular Matter:** It is the combination of two particles that are in form of solids and liquids which includes sulphate, nitrates, carbon, complex chemicals, and water suspended in the air.
 - **Nitrogen dioxide:** This is gas and it comes from man-made sources such as vehicles, heating plants, and power stations. Mostly Diesel vehicles are responsible for these gases.
 - **Sulphur dioxide:** This is a colourless gas that cannot be seen and has a very suffocating smell. This type of gases originates from burning oil and coal. The major source of gases is industries as they burn fossil fuels mostly.

- Water pollution is referred to as the existence of harmful chemicals and biological components that is mixed with freshwater.
- Primarily, two sources of Water Pollution exist i.e., Point and Non-point sources. Point sources include groundwater and surface water while non-point includes urban, agricultural, and forestry land use and marinas and boating operations.
- The effects of water pollution are Eutrophication, Contamination of waterways, Threat to aquatic and human life, Ocean Acidification.
 - The types of water contaminants are Physical, Chemical, Radiological, Biological, Organic, Inorganic, Macroscopic, Thermal and Microbial Contaminants.
- Soil pollution is the pollution caused by the contamination of soil with incongruous absorption of toxic elements.
- The contaminated soil becomes useless for producing any type of crop.
- Soil pollution increases soil acidity, which negatively affects vegetation, making it useless and barren.

7. CASE STUDY

DELHI POLLUTION (Air Pollution)

This has made an endeavour to dissect the effect of air contamination on wellbeing, its causes and level of mindfulness by directing a field overview on the young people of Delhi, concentrating in various colleges and schools in Delhi. The gathered information has been evaluated with numerous strategies like engaging measurements, recurrence tables, cross classification and chi-square trial of autonomy on sources, impacts, and answers for air contamination. At long last, the examination reasons that an enormous number of mindfulness missions ought to be coordinated including a more noteworthy cooperation from all areas of the general public alongside better usage of present laws. More seasoned individuals, children and patients with previous respiratory illnesses like lung irritation, flu and asthma are defenseless to the specific matter suspended noticeable all around. Furthermore, it's not just outside air contamination, the suspended particulates clear a path to indoor air in homes and stay suspended indoor. There were additionally expanded rates of prohibitive and obstructive lung capacities. The high commonness of metaplasia and dysplasia of aerial route epithelial cells likewise puts the population of Delhi in danger. The sputum showed moderate to serious cytological changes, though the country's control populace in West Bengal was ecstatically unmindful of this symptom of living in a metropolitan city.

Conclusion:

While there has been enormous advancement on foundation and examination front, nonetheless, the predicament of Delhi occupant isn't coming to an end. While a few investigations referenced above suggest an ecological relationship and can be utilized to draw causal derivations, best case scenario, these are the best system and outlines accessible to date to cause to notice this consuming issue. Lower emanation vehicles and usage of a few rigid measures have brought about some improvement and alleviation from Delhi air contamination. All things considered, the public authority and other power figures need to send war-like measures to guarantee the prosperity and strength of the residents. Wellbeing is an all extensive and all-inescapable viewpoint. While clean air is a resident's correct, it can't be inside an administration's domain to control all the components. The residents

additionally should be dependable. The vision of Clean Delhi is absurd without local area cooperation.

Source- Times of India newspaper, 10 February 2021

Discussion Questions:

1. Is air pollution principally a neighbourhood issue or would it being able to travel significant distances?
2. Does air pollution make Delhi an uninhabitable place?

E-Waste Polluting Delhi's Groundwater and soil

Due to the advancement of technologies, the number of electronic wastes has increased tremendously. Electronic wastes are produced by electrical devices such as PCs, mobile phones, and many other electronic devices.

Improper dumping of these wastes leads to the contamination of groundwater quality and soils. Researchers of Jamia Milia Islamia studied this.

The researchers examined the soil nearby Mandoli's Krishna Vihar – where you can easily find considerable amounts of unprocessed electronic wastes. Among various pollutants – Copper was present in large quantities, i.e., approx. 283 mg/kg at the top soil. But just 4-5 km from the reference site (where an examination was going on), the amount of Copper was just about 8.39 mg/kg.

In the same way, Lead (Pb) in the subsoil (the layer where the plant takes its roots) was found to be 183 mg/kg and a few km away from that, the amount was only 0.43 mg/kg. This contamination of subsoil results in the absorption of these contaminated particles into plants and can reach human tissues.

Heavy metals such as lead, Cadmium, Copper in these areas were about 20 times more than the normal amount as declared by Central Pollution Control Board (CPCB).

The risks of consumption of heavy metals were also analysed. Organs that are vulnerable to heavy metals are the liver and kidney. The consumption of these metals in long-run leads to various diseases such as Alzheimer's disease, Parkinson's disease, muscular dystrophy, and multiple sclerosis.

However, in March 2016, India made it compulsory for electronic goods manufacturers to ensure the safe and proper disposal of these electronic wastes. This quantity was attached to their annual sale and promised a per-year increment.

Still there is a situation such that nearly 80% of electronic goods are dumped unsafely from which costly metals such as gold and silver were obtained and rest got disposed in the open area leading to leaching of metal constituents into the soil.

Source: Article from The Hindu newspaper

Discussion Questions

1. How can you manage the waste disposal made on lands?
2. How do you think electronic waste affects human health?

8. TERMINAL QUESTIONS

SHORT ANSWER QUESTIONS

1. Define Air pollution. Who is at risk from Air pollution?
2. What are the two sources of water pollution?
3. What do you mean by the greenhouse effect and eutrophication?
4. What do you mean by food chain?
5. What do you mean by pollutants and what are their two types?

LONG ANSWER QUESTIONS

1. What is the impact of air pollution on human health?
2. What are the drastic effects of Water Pollution on the environment as well as on human health?
3. Define Soil pollution. Write about its sources and what are the pollutants responsible for its contamination?
4. How can we control soil pollution? What is its impact on our future lives?
5. What are the impacts of soil pollution on soil life, agriculture, human beings, ecosystem, plants and animals?

9. ANSWERS

Self - Assessment Questions

1. Air pollution
2. True
3. False
4. D. Both A and B
5. C. Colourless
6. Portable or mobile sources
7. False
8. False
9. D. Wood-consuming chimneys, farming territories
10. C. stationary sources
11. True
12. Poisonous
13. D. All of the above
14. A. Aggravation
15. Afforestation
16. True
17. A. Use eco-friendly products and packaging
18. Chemical & Biological
19. Human & Aquatic
20. Point source pollution
21. True
22. True
23. B. Ocean acidification
24. Eutrophication
25. Nitrates
26. False
27. Construction wetlands
28. Pollution
29. D. All of the Above

- 30. D. Both A and B
- 31. Altering
- 32. D. All of the above
- 33. Decrease
- 34. Acidic
- 35. True
- 36. Degrade
- 37. A. Crop Rotation

Terminal Questions

SHORT ANSWER QUESTIONS

Answer 1: Air pollution is brought about by gases and particles radiated to the environment by an assortment of human exercises, like the wasteful ignition of energise agribusiness, and cultivating. Additionally, regular sources add to air contamination, remembering particles of soil residue and salt for ocean splash.

Individuals' well-being hazard from air contamination changes generally relying upon age, where they live, their hidden well-being, and different variables. Numerous investigations show that individuals with lower financial status and minority

- Individuals with lung sicknesses, like asthma, constant bronchitis, emphysema, and ongoing obstructive aspiratory infection
- Babies and little youngsters

Answer 2. The two sources of water pollution are the point and non-point sources of pollution. Pollution that arises from a common, recognizable source, like a drainpipe from a sewage plant, vessel or factory, is commonly termed as point-source pollution. Pollution through non-point sources doesn't have a sole and identifiable source.

Answer 3: The greenhouse effect is the process of warming the earth's atmosphere by trapping sunlight. Eutrophication is the increase in the concentration of nitrogen, phosphorus and other nutrients in the aquatic ecosystem such as a pond, lake, and so on.

Answer 4: A food chain is a series of organisms, each dependent on the next as a source of food. It is generally a link of who eats whom. Soil pollution adversely affects the metabolism of microorganisms and arthropods, which ends the primary level of the food chain.

Answer 5: The unwanted substance which causes various kinds of pollution is called pollutants. Two types of pollutants are: naturally occurring and man-made pollutants. Due to industrialisation, the introduction of various chemicals, garbage, and many more day-to-day activities play a major role in creating various kinds of pollutions in which we are used to some kinds of pollution and it's difficult to deal with some others.

Long Answer Questions:

Answer 1: The impact of air pollution on human health is as follows:

- It causes both intense infection and persistent illness. There is solid proof connecting long haul exposure to air contamination with an expanded danger for ischemic coronary illness, stroke, persistent obstructive aspiratory infection (COPD), lung and upper aero digestive and others.
- The WHO's International Agency for Research on Cancer (IRCA), has assigned air contamination as a cancer-causing agent. Some quick well-being impacts of its exposure include disturbance of the eyes, nose and throat, windedness, hacking, and compounding of previous conditions, similar to asthma assaults and chest torment.
- Carbon monoxide (CO) limits the exchange of oxygen to tissues and can be deadly in exceptionally high focuses. Sulfur dioxide (SO₂), is a powerful lung aggravation influencing the soundness of those with a previous respiratory infection (asthma and COPD), particularly those living and working near SO₂ sources.

Answer 2. The effects of water pollution on the environment and human health are:

- A healthy ecosystem consists of a bunch of flora, bacteria, fungi, and fauna, all interacting directly or indirectly. Any problem to any of the above organisms can develop a chain effect that affects the aquatic environment.
- Marine ecosystems are also included in the danger zone by the segregation of marine waste that can choke, starve, and restrain animals.
- Ocean acidification turns out to be more difficult for shellfish and corals to survive under water.

Answer 3: Soil pollution is the pollution caused with the contamination of soil with incongruous absorption of toxic elements. Garbage, plastics, fertilisers, pesticides, and many more are the cause of soil pollution. Human-made chemicals are pesticides, fertilisers and improper disposal of waste.

Sources:

1. **Agricultural sources:** Agriculture focuses on organic farming of the products to maintain the basics of our lives, but the methods like pesticides, herbicides, and fertilisers used to grow crops and livestock much faster than organic production leads to soil pollution.
2. **Industrial sources:** Industrial waste arises from soil pollution and can be caused in the form of gas, liquid, solid substances. CO₂, NO₂ are the gases that are released from industrial activities. These wastes mix with rainwater and can cause the generation of acid rain that drastically affects the soil's pH.
3. **Urban waste:** Urban waste is simply garbage that has already been deposited and it includes plastic, food waste, industrial waste, and regular household waste. It would seem that organic garbage and non-biodegradable rubbish may be recycled and discarded in places meant for natural breakdown.

Pollutants: Soil pollution can be caused by the following:

- ✓ Microplastics
- ✓ Oil spills
- ✓ Ammunition, chemical agents, and other agents of war

Answer 4: Control Measures:

- ✓ Reduce deforestation and begin afforestation.
- ✓ Avoid intensive farming and reduce your footprint.
- ✓ Maintain a vegetation cover, especially in vulnerable areas.

Impact on the Future:

1. Soil pollution hurts biodiversity that leads to the extinction of flora and fauna.
2. It also reduces the fertility of the soil, which leads to the enhancement of barren land in our country.

3. It causes a reduction of healthy crop cultivation, an increase in Food quality degradation.
4. It causes an imbalance of various nutrients in the soil which are responsible for healthy crop growth.
5. It also degrades groundwater quality, leading to an increase in contaminated water scale.

Answer 5: Impact of soil pollution on:

1. Soil life:

- Due to the presence of chemical substances, the fertility of soil decreases which decreases the soil yield.
- The contaminated soil becomes useless for producing any type of crop.
- Degraded soil leads to the production of contaminated vegetables and fruits consumed by various living organisms and hence, causes various diseases such as food poisoning, weakness, etc.

2. Agriculture:

- It decreases the quality of the crop.
- It affects the production of crops and also the success of Civilisation.
- It also affects the quality of Food produced on that soil.

3. Human Beings:

- Soil pollution adversely affects human health.
- It also causes neuromuscular blockage.
- It leads to various types of diseases such as nausea, skin rashes, fatigue, eye problems, headache and so on.

4. Plants and Animals:

- The decrease in the pH of the soil leads to the death and sickness of various plants and animals.
- It also leads to the destruction of the entire aquatic food chain.
- Soil pollution adversely affects the metabolism of microorganisms and arthropods, which ends the primary level of the food chain.

5. Ecosystem:

- Soil pollution increases soil acidity, which negatively affects vegetation, making it useless and barren.
- When any kind of crop grows on these kinds of lands, it leads to the growth of a poisonous crop.
- Consumption of such plants that are grown on degraded soil leads to various health issues and the ecosystem would be adversely affected by this.

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BACHELOR OF COMPUTER APPLICATIONS

SEMESTER 1

ENVIRONMENTAL SCIENCE

Unit 6

Environmental Pollution & Policies -II

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1. INTRODUCTION

In this unit, we will introduce the notion of Noise Pollution—another type of pollution that exists in the environment. This topic will help you to get in-depth insights into noise pollution-its meaning, examples, basic concepts, sources, effects, examples of controlling different noise pollution, and measures to be adopted to control noise pollution.

The word noise refers to the illness in which one feels vomiting and originates from the Latin word 'Nausea'. Hence, noise is an unwanted and nasty sound that causes discomfort to living organisms.

The slightest sound that a human ear can hear is 1 dB. Noise pollution is increasing daily; thus, one has to take measures to reduce noise pollution. The major sources of pollution are firecrackers, vehicles, loudspeakers, aircraft, machines, and many more. Other devices, such as radio, transistor, television, etc., pollute noise when used at high volume.

In the present times, our environment faces many environmental issues such as loss of natural resources, global warming, pollution due to development, etc. Environmental studies help us to understand the various environmental phenomena occurring naturally within our environment.

We know that our lives are directly linked with the surroundings; thus, the production of wastes is inevitable. However, it is our responsibility to manage them efficiently and take care of the environment. An environmental study is a basic and an applied science, which mostly talks about the social sciences.

STUDY NOTE

The source of outdoor noise worldwide is mainly caused by machines, transport, and propagation systems.

In environmental studies, we study each component of the environment interact with the surrounding physical environment and with each other. The ecosystem is expansive and spread out over a wide region. Wastes are products that are unwanted or unacceptable. Every material that is wasted after use, or is no longer useful, is referred to as a waste. Municipal industrial waste, drainage, hazardous waste, nuclear waste, and other forms of waste are some examples.

Environmental policies and their elements play a crucial role to give importance to identifying objectives for each challenging area towards relevant implementation. This includes designing long-term vision documents for forecasting a sustainable and healthy utilization of natural resources. The current environmental standards entirely revolve around the concentration of pollutants in emission and effluents. The guidelines and norms of environmental policies are to be further revised for laying down mass-based regulatory standards.

This will then set the necessary limits for promoting the limitation of waste, encourage reusing and recycling materials, along with conserving natural resources. Standards aren't solely a regulatory tool but are the very key to encouraging technological up-gradation for conserving resources, reducing or mitigating pollution, and further managing waste. For this very purpose, environmental practices and policies are modified for individual processes.

The environmental impacts from the disposal of varied products that are toxic and dangerous are to be analyzed carefully and monitored in the regulations. In this chapter on environmental policies and practices, you will learn in-depth about varied sub-topics like global warming, climate change, environmental laws and acts, ozone layer depletion, and so on.

The main purpose of this chapter is to educate you on the harmful impacts of pollution on the environment. Changes in the climate are a fundamental challenge for the global nation today. Each one of us plays a role in contributing to the adverse consequences on the environment. The chapter offers deep insights into topics like international agreements, tribal population and rights, and even human-wildlife conflicts in India.

We will comprehensively study what has shaped the environment today. All types of pollution, be it air, water, soil, etc., play a quintessential role in impacting not only our environment but also animals and other living creatures. It further creates a harmful atmosphere for us to live in. We will study the different environmental policies and practices that are being implemented for conserving the natural resources and restricting pollution.

1.1 Learning Objectives

After studying this chapter, you will be able to:

- ❖ *Describe the Effects of Noise Pollution*
- ❖ *Explain the Control Measures for Noise Pollution*
- ❖ *List the types of solid waste.*
- ❖ *Recognize the impacts of environmental damage and pollution*
- ❖ *Recall the regulatory and legal frameworks that promote sustainability*



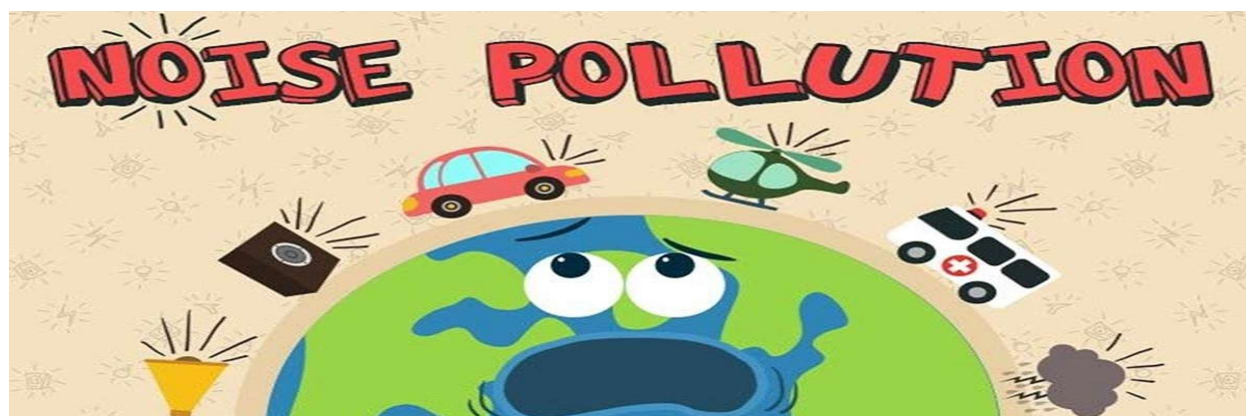
2. OVERVIEW AND SOURCES OF NOISE POLLUTION

Noise pollution is an invisible danger. Noise pollution is any unpleasant and disturbing sound that affects human creatures' physical condition and well-being.

The word noise refers to the illness in which one feels vomiting and originates from the Latin word 'Nausea'. Hence, noise is an unwanted and nasty sound that causes discomfort to living organisms. The intensity of sound is measured in decibels (dB). The slightest sound that a human ear can hear is 1 dB. Noise pollution is increasing daily; thus, one has to take measures to reduce noise pollution. The major sources of pollution are firecrackers, vehicles, loudspeakers, aircraft, machines, and many more. Other devices, such as radio, transistor, television, etc., pollute noise when used at high volume.

Noise pollution refers to high-level sounds that irritate living organisms' hearing capacity. It can result from machines, high music, and elevated vehicle sounds. Noise pollution is a disturbance that leads to severe health hazards and environmental problems, which should be controlled. Noise pollution refers to the undesired and unpleasant sounds that are hindering modern day development and disrupting the activity or stability of animal and human life.

Noise pollution is the pollution caused due to higher volumes or intensities of noise. Each organism or thing possesses a different type of noise, and these noises severely impact people. Due to the functioning and operations of urban regions, the environment gets polluted. This leads to higher volumes of factory noise, vehicle noise, etc. Noise pollution typically creates several risks like diseases, death, etc. In addition, excessive noise pollution may cause the person to lose hearing ability.



Source: thelivenagpur.com

Fig 1. Noise Pollution

Listed below are examples of noise pollution:

- Unnecessary usage of fireworks
- Construction noise
- Unnecessary usage of horns
- Noise from aircraft and railway
- Using loudspeakers for political or religious purposes
- Industrial noise

2.1 Basic Conceptualities Related to Noise Pollution

Noise is referred to as an unwanted sound. In addition, it is deemed as a disruptive and loud sound. Noise Pollution is an unfortunate, uncomfortable, and loud sound that prompts uneasiness and discomfort in individuals. Noise pollution is usually produced in some workplaces and industrial facilities and comes from aeroplane traffic, construction activities, trains, highway traffic, and heavy vehicles. It affects the well-being and comfort of all living creatures, both ashore and in the ocean, from road traffic noise to live concerts; many sounds cause stress, hypertension, hearing activities, and so on. Here are four concepts that are related to noise pollution.

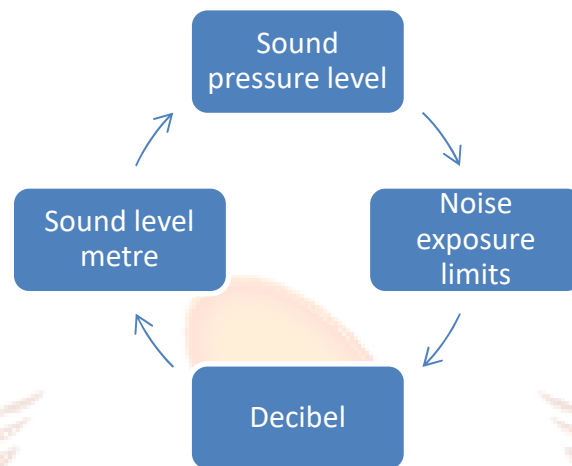


Fig 2. Basic Concepts Related to Noise Pollution

- Sound pressure level – This level indicates the human perception of sound. This level varies among people.
- Noise exposure limits – It highlights the desirable and acceptable limits of noise that a human being can undertake; the World Health Organization provides these limits.
- Decibel – It is depicted as the standard unit for measuring sound and sound vibrations. It is abbreviated as dB.
- Sound level meter – This meter is depicted as an instrument that facilitates control over the sound.

2.2 Sources of Noise Pollution

Sources mean how noise pollution can be created. There is much noise in a region but identifying the source of noise pollution needs to identify the decibel level of these noises. Mechanical hardware and cycles are made from different sources such as interior ignition motors, stators, rotors, electrical machines, gears, tempestuous liquid streams, vibrating boards, fans, sway measures, and so on.



Source: eschooltoday.com

Fig 3. Sources of Noise Pollution

Industrialization: Industrialization has brought a rise in noise pollution. Industries use laborers and capital goods in large quantities in their production process. Various noise-producing machines are used in the production processes. Various machines or ornaments, such as compressors, generators, etc., are used for production purposes. These machines produce noise to excessive limits. The designing plants, factories, metal enterprises, and print machines have been a leading cause of noise pollution. Most of these sources use heavy machinery, which results in heavy noise. In addition, they have hardware such as crushing factories, blowers, generators, and exhaust fans which raise the noise levels in the surrounding.

Poor urban planning: The people living in urban areas possess luxurious vehicles, which may cause severe problems. Most people live with bigger families but in smaller places. The parking lots constructed in urban areas generate a hub of noise pollution. People unnecessarily honk in public places, which results in noise pollution. Thus, these things create immense noise pollution, which disrupts the effective functioning of the environment. Developing countries habitually lack proper urban planning, which leads to small industry proliferation, small spaces, congested housing and a lack of enough parking spaces. Poor urban planning thus adds to environmental noise through fights or social and basic amenities, wrangles over parking spaces, noise from small manufacturing industries, noise from playing children, and family quarrels from the neighbouring houses.

Social events: Social events such as parties, marriages and so on include music played at high volumes. The administration and concerned authorities have issued certain rules restricting music in high volumes. However, people do not follow such rules, creating a huge nuisance. Due to the higher intensities of noise pollution, people's health conditions have worsened in urban areas. Weddings, festivals, public gatherings, and occasions involve loudspeakers playing music, thus producing unwanted noise in the surroundings.

Transportation: The automobile industry has become huge, especially in urban areas. In the modern era, there is a rise in traffic due to the increase in the number of vehicles such as

cars, buses, ambulances, jeeps, trucks, trains and so on. As a result, many vehicles are plying on the roads. People stuck in traffic jams are also habitually impatient and continuously blow their horns to alert the drivers ahead of them to move. Their frequent honking creates immense noise pollution. As a result, people's hearing abilities get hampered. Moreover, continuous honking produces unbearable noise in the neighbouring areas and the environment. Aeroplanes also create a problem of noise in main urban towns. Most airports are placed near residential areas producing a high sound level while taking off and landing.



Source: www.bing.com

Fig 4. Noise Pollution by Vehicles

Construction activities: Construction cannot be avoided anywhere as it has become pivotal in each part of the world. Activities such as the construction of houses, mining and so on take place so that the overall demands of the increasing population can be fulfilled. The noise from the construction of buildings, highways, dams, streets, and houses is a foremost contributor to noise pollution. For instance, the construction of buildings, highways, and houses require equipment such as air hammers, bulldozers, loaders, air compressors, and dump trucks. This scenario tends to create a higher level of noise pollution. The different machines and equipment used in constructing buildings and roads are also crucial causes of environmental noise pollution. The machinery used for construction, such as loaders, bulldozers, air compressors, dump trucks, pneumatic hammers, pavement breakers, etc., is also a great contributor to noise pollution.

Household chores: These days, households are often equipped with gadgets such as vacuum cleaners, mixer grinders, mobile phones, etc. People majorly use these gadgets and create higher intensities of noise pollution. This noise pollution often negatively impacts the neighbours and society as a whole. Household activities such as furniture movement, loud

doors banging, crying infants, noises from playing children, loud arguments and fights, noises from entertainment equipment like loudspeakers, television sets, video games, and so on also contribute to the overall noise pollution.

In addition, household equipment such as vacuum cleaners, pressure cookers, exhaust fans, washing machines, mixer grinders, sewing machines, air-conditioners, and desert coolers produce much noise.

Fireworks: Fireworks are a common scenario in various events such as occasions, marriages, festivals, etc. They hamper the quality of life by creating painful noises and producing too much pollution, both sound and air. Indian festivals such as Diwali, Dusshera, marriages, and so on use fireworks for celebrations. Bursting firecrackers results in health hazard as well as noise pollution. The noise produced is hazardous and can sometimes lead to non-recoverable hearing loss.



Source: th.bing.com

Fig 5. Noise Pollution by Firecrackers

Agriculture machines: Agricultural practices have become mechanized due to the technological revolution. As a result, several machines and tools such as drillers, tractors, harvesters, powered tillers, thrashers, and tube wells are used for farming, causing heavy noise pollution.

Defence equipment: Defence equipment is used by almost all countries for progress and national protection. Explosives and military equipment are used to protect the nation. Besides that, satellite launches facilitate countries toward progress. However, the use of such

equipment creates immense noise pollution. Military aeroplane drills, artillery tanks, explosions, rocket launching, and shooting practices are serious noise polluters. Likewise, the sounds of jet engines produce deafening impacts.

Miscellaneous causes: Boilers, air conditioners, plumbing equipment, fans, and generators greatly pollute noise. Also, the noise created by people in public places affects the health of living beings. For instance, loudspeakers create high degree/amplitude noise.

- **Automobile repair shops:** Automobile repair shops and workshops also contribute to noise pollution by using heavy equipment such as air compressors, oil drains, jacks, engine hoists, spanners, tools, strut compressors, air conditioning machines, and many more.
- **Market places:** Besides the huge traffic jams and parking troubles, continuous honking has made the lives of people in some localities very tough. Vendors selling fruits and vegetables, chit-chat of ladies, vehicles in market areas, machines, generator noise, and shouting voices of people contribute to the noise pollution.
- **Schools:** Students talking with each other, laughing and shouting, the noise of the teacher, the sound of footsteps and bell ringing, sound of various objects such as pencil, fan, pen, spoon, water bottle, air conditioning, etc., creates noises.
- **Bus stands:** People waiting at the bus stops are exposed to high noise levels. In addition, bus drivers are more vulnerable to noise pollution's auditory and non-auditory ill effects. Also, buses use air horns in the cities that do not follow standards, thus polluting noise.
- **Office hardware:** In workplaces, there are printers, scanners, and typewriters, among other hardware, which add to noise pollution in the working spots and their environment.

SELF-ASSESSMENT QUESTIONS – 1

1. The intensity of sound is measured in _____.
2. Which among the following is correct in regard to noise pollution?
 - a) Parties and marriages include music played in high volumes.
 - b) Military aero plane drills, artillery tanks, explosions, rocket launching, and shooting practices are serious noise polluters.
 - c) Furniture movement and crying infants also contribute to noise pollution.
 - d) All of the above

3. EFFECTS OF NOISE POLLUTION

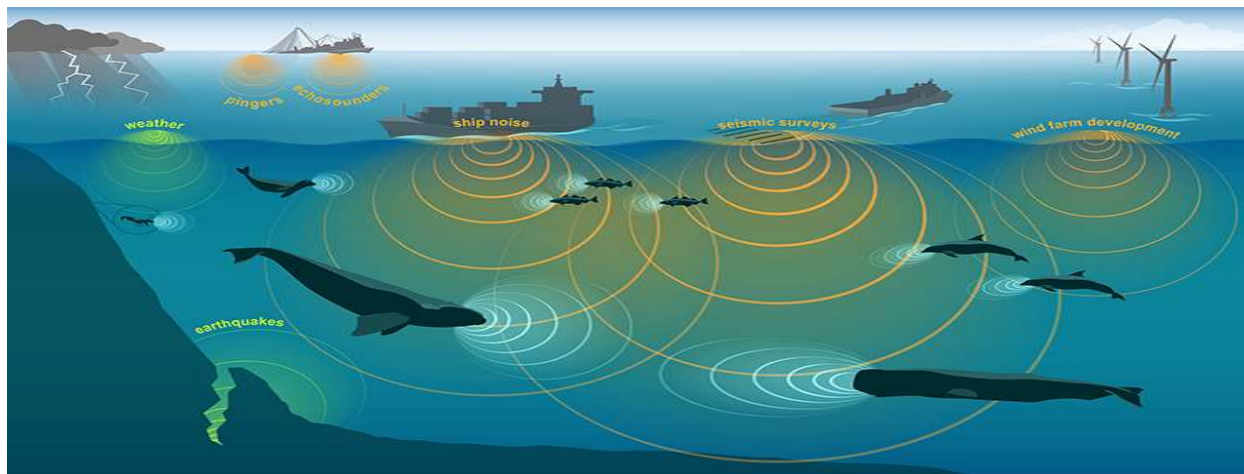
Effects of noise pollution refer to the constant interference of noise or unwanted sound in the environment, which can influence human health.

Noise Pollution impacts a large number of individuals constantly. The most widely acknowledged medical problem it causes is NIHL which stands for Noise-Induced Hearing Loss. These diseases can impact all age gatherings, especially youngsters. Several youngsters who live close to roads or loud airport terminals have experienced the ill effects of stress, pressure, and several issues like consideration level, understanding of expertise, and memory loss.

3.1 Effect of Noise Pollution on the Environment

- Noise pollution is also dangerous for both plants and animals. The sources of noise in the environment are trolleybuses, cars, compressors, aeroplanes, trains, machinery, and many more. The acceptable noise level, which is easily bearing, ranges from 30 dB to 60 dB. But the statistics generally go beyond these figures and reach 100 dB. Moreover, numerous tests have established that some plants died because the leaves released too much moisture, owing to high sound. Again, this happens because noise pollution crosses the acceptable range. Therefore, the life spans of trees in big cities are much less, and with the sound of a jet engine, bees lose their ability to navigate space.
- Human activities on the seas like infrastructure, shipping, wind farm development, fishing, and mining, are increasing, thus increasing the level of sound and noise in the oceans. The most extensive pollution source is ship noise, which occurs everywhere

and every time. The most hazardous forms generate the most extreme noises, i.e., underwater bomb explosions of dynamite for fishing. All these activities affect marine animals to a great extent. As a result, there is a bigger urgency to take suitable actions appropriately and quickly.



Source: wildwhales.org

Fig 6. Effects of Noise Pollution on Marine Animals

- The noise pollution limits the freedom of birds, insects, and animals. It desists them from roaming, sleeping, and eating. The environment refers to the surroundings which are naturally created or made for animals, plants, or humans. Unfortunately, noise pollution disrupts the whole environment. It constrains the movements and interactions of all living organisms. Thus, it affects animals, and through them, the stability of ecosystems, both marine and terrestrial, is negatively distorted.

EFFECT OF NOISE POLLUTION ON CROPS:

Noise pollution reduces the quality of crops because plants require a calm and quiet environment to grow. Additionally, noise pollution affects bugs and insects, which are essential for natural resources. Scientists have found that long-lasting noise pollution can hinder normal plant growth, with the effects showing in habits years later.

EFFECT OF NOISE POLLUTION ON BUILDINGS:

Noise pollution affects the whole structure of the building. The noise pollution cracks the framework of the building. The high density of noise, such as vibrations emitting from heavy equipment and machinery, causes the shattering of window panels and glasses, breaking

down the hangings in the house, cracks in household crockery walls, and losing the plaster of house walls. It also damages the paintings and so on.

EFFECT OF NOISE POLLUTION ON HUMAN HEALTH:

The living organism's physical, mental, and social activity represents human health. Human health is affected by several types of environmental pollution. Noise Pollution can be dangerous to human health in the following manner:

STUDY NOTE

According to the reports of BioMed Central in 2017, about 36% of people exposed to noise pollution areas suffer from a change in pulse rate, heart rate and

- **High Blood Pressure:** The loud noise causes the heart to beat fast, which results in high blood pressure. The music played on high-volume sound instruments increases blood pressure. Work environments include factories and construction sites that increase human blood pressure. High noise levels raise blood pressure and pulse rates, making humans susceptible to heart attacks. Hypertension is an immediate consequence of noise pollutants caused by raising blood pressure for longer. In addition, noise pollution causes heart attacks due to sleep interference.
- **Loss of Hearing Ability:** Noise pollution damages cells and membranes. Long-term use of music systems harms the ear cells. Loud noise harms a human's inner ear and can lead to death. Continuous exposure to loud noise beyond the acceptable range can damage the eardrums, thus resulting in hearing loss. Hearing impairment induced by noise pollution causes temporary or permanent deafness.



Source: www.fashionlifestylefood.com

Fig 7. Effects of Noise Pollution on Human Health-Hearing Loss

- **Disturbance:** Effects of noise pollution include not only disturbed sleep but also affects mental health. Unnecessary sounds can have a major effect on mental health as well. Noise pollution causes sleep disturbance, leading to depression and anxiety. Noise pollution leads to disturbance in the concentration level of humans in the workplace. Some psychologic more problems than humans due to noise pollution because al effects of noise pollution are losing peace of mind, depression, stress, getting irritated more frequently, anxiety, waking too early, and so on. Lack of concentration can also be examined due to unwanted sounds and noise.

SELF-ASSESSMENT QUESTIONS – 2

3. Wildlife animals faces more complications as compared to humans due to noise pollution because animals depend on _____.
4. The acceptable noise ranges from 35 dB to 65 dB. [True/False]
5. The impact of noise pollution on human health is governed by _____.
 - a) Intensity of noise
 - b) Sensitivity of human ear
 - c) Socio-economic status of an individual
 - d) Both A. & B.
6. Which among the following are adverse effects of noise pollution?
 - a) There is a partial loss of hearing.
 - b) Fetus may develop defects.
 - c) It affects human performance in task in communication skill and thinking.
 - d) All of the above

3.2 Control Measures for Noise Pollution

It includes the usage of any measure that will decrease clamor being created and diminish the noise transmission through the air or the design of the work environment. Such measures incorporate hardware changes, the working environment activities, and the design of the workroom.



Source: www.wikihow.com

Fig 8. Control Measures for Noise Pollution

Examples of Controlling Different Noise Pollution

Examples of Controlling Construction Noise:

- Using bulldozers and air compressors and with less noise. These are to be avoided in highly populated regions.
- Using low-noise loaders and trucks.



Source: contractoradvantage.ca

Fig 9. Controlling Construction Noise

Examples of Controlling Transportation Noise:

- Stop road traffic noise by issuing challans.
- Aircraft noise to be decreased by developing the newest engines.
- Noise from railroads to be reduced using new ways to run rails. For instance – Metro trains.

Examples of Controlling Social Events Noise:

- Avoid using loudspeakers and DJs.
- Use musical instruments wisely.

3.3. Measures of Effective Control of Noise Pollution

Noise pollutants can be adequately constrained by taking the following measures:

Control at Receiver's End: For individuals working in loud establishments, ear-security helps such as noise caps, earplugs, earphones, earmuffs and so on should be given to lessen occupational openness. People working in noisy regions should be provided with ear protection aids.

Controlling at Source: The best control method for noise pollution is to reduce the noise from the source. Noisy machines with sound-absorbing materials should be installed. This is appropriate if methods and strategies are improved by:

- Installing noisy machines in sound verification chambers.
- Using glass covered with a sheet of punctured metal with the aim of mechanical assurance.
- Designing, manufacturing, and using calmer machines to replace noisy ones.
- Reducing the commotion delivered from a vibrating machine by vibration damping; for instance, making a layer of damping material such as plastic, neoprene, elastic, stopper, underneath the machine.

STUDY NOTE

Psychology Today reports state that hearing loss is the topmost cause of noise pollution. It is not with age but the noise. Older people are at a higher risk of hearing damage.

The measures for effective control of noise pollution are as follows:

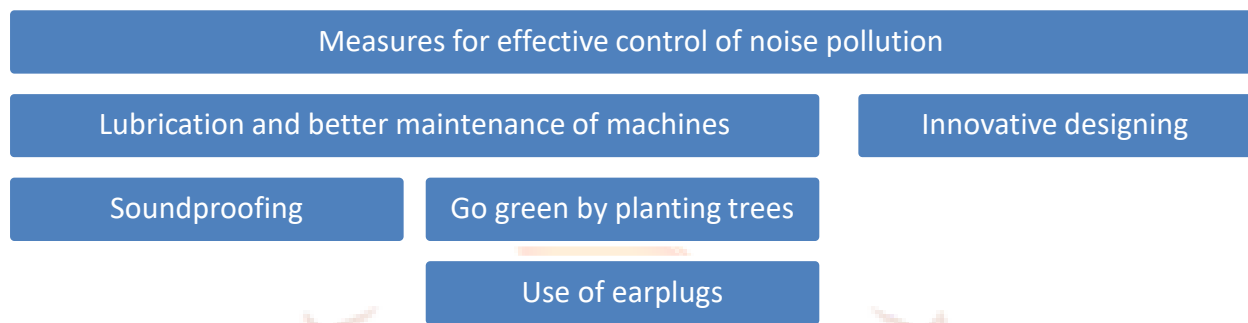
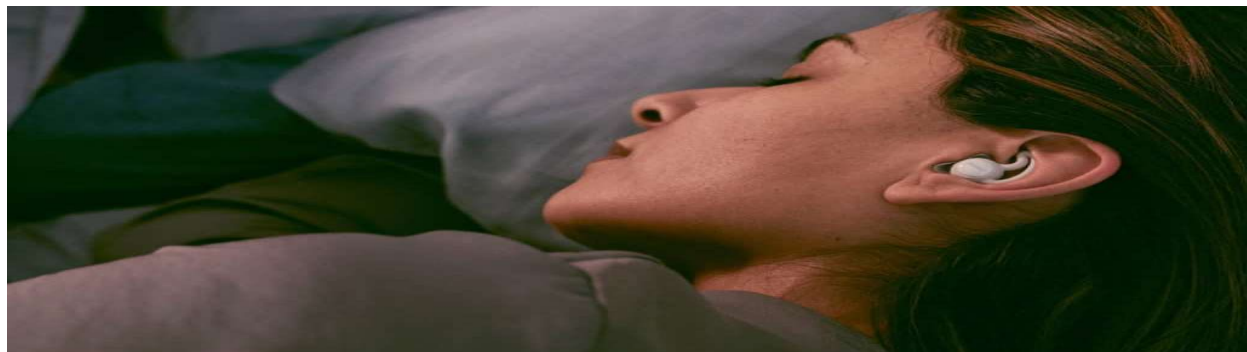


Fig 10. Measures For Effective Control of Noise Pollution

- **Innovative designing:** Properly designing windows and doors and sealing them well will help in reducing noise. Innovative building patterns and designs can reduce the intrusion of noise.
- **Soundproofing:** Soundproof windows and doors need to be installed to prevent unwanted noise from outside. Industrial plants that generate much noise from using the machinery must use soundproof materials to lessen high sound frequencies. In addition, one can install soundproofing items at home that stop the noise and unwanted vibrations. Double-pane windows, for instance, can be a unique method for blocking noise pollution.
- **Use barriers or go green by planting trees:** Planting bushes and trees effectively reduces noise pollution. It is technique popularly called as green muffler. However, a simple way to decrease strong sound waves and vibrations is by using barriers like planting trees around buildings, houses, and companies. These barriers will absorb the waves and lessen environmental noise drastically.
- **Lubrication and better maintenance of machines:** Regularly servicing automobiles can successfully reduce noise pollution. When the components of engines and machines are well lubricated and well maintained, it enhances efficiency and helps decrease noise. The cause is that lubrication and proper maintenance decrease friction between moving parts. Noise mufflers are to be used to avoid noises.
- **Use of earplugs:** One of the most effective solutions to reduce noise pollution is to wear earplugs. They could be worn while sleeping or in working places to decrease the

amount of noise from the environment. In addition, they have health benefits as well, like healthy sleeping habits and preventing damage to the eardrum.



Source: th.bing.com

Fig 11. Using Earplugs to Reduce Noise Pollution

With the projections of rapid urban growth and the consequent rise in transport demand, it is possible to estimate a simultaneous rise in noise exposure and, thus, the adverse effects.

Subsequently, it is important to implement policies that protect our health and wildlife instantly. These policies and actions are particularly required during festive seasons when noise pollution is more obvious.

SELF-ASSESSMENT QUESTIONS – 3

7. Which among the following sound is pleasant to our ears?
 - a) Transportation equipment
 - b) Musical instruments
 - c) Heavy machinery
 - d) None of the above
8. Which among the following measures will help in reducing noise pollution?
 - a) Automobiles to be fitted with silencers.
 - b) Factories need to be located near residential areas.
 - c) Use loudspeakers only when required.
 - d) Both A. & C.

4. SOLID WASTE AND ITS MANAGEMENT

Solid waste refers to the rubbish that consists of wasteful items discarded by the public on a daily basis. It comes from various human activities in the form of wasteful material during household activities, industrial activities, and other commercial activities by organisations. The increase in the utilisation of different resources has led to the production of more waste. In this section, we will study what comprises solid wastes generated in our environment, we will also study the origins and the types of different resources and their management. It is essential to understand the importance of the disposal of solid wastes, and identify the problems that may arise related to each type of waste.

We should understand the mechanism of waste disposal so that it has a minimal effect on our environment. It is also critical to understand that as humans, it is our utmost responsibility to take care of the wastes we generate. The processing of urban solid wastes, along with the high organic content of wastes, and its often inaccurate disposals, result in substantial biological emissions, as well as the emission of methane (CH₄) and carbon dioxide (CO₂), all of which add to the greenhouse effect.

STUDY NOTE

Waste is regarded as something undesirable or not in use anymore. When we buy or produce something, the wastes left after their use becomes solid wastes.

As a result of this environmental challenge, local governments are attempting to introduce more efficient techno-economic, social, and political strategies to handle the growing amounts of urban solid wastes generated in the environment.

4.1 Meaning and Definition of Solid Waste Management

Every day, thousands of tonnes of solid wastes are disposed at landfills. These solid wastes are created at homes, offices, and by a number of other companies. If wastes are not adequately handled, these landfill sites emit a foul odour. When hazardous wastes such as lead-acid batteries, poisons, washing solvents, cadmium, arsenic, or zinc, as well as radioactive products, are disposed and combined with paper and other scraps, and burned, gases such as dioxins are released.

These gases are carcinogenic and poisonous, and they pollute the air and waters, posing a significant threat to human, animal, and environmental health. Certain kinds of wastes are

considered dangerous because they have an immediate impact on humans or the environment. Municipal solid waste (MSW) or refuse, on the other hand, refers to any non-hazardous solid waste from a city that requires collection and transport to an appropriate recycling or disposal site.

Garbage and rubbish are used with refuse. Garbage is normally decomposable food waste; however, it may also be dry materials such as glass, paper, or wood. Trash will also contain heavy items such as couches or old refrigerators. Trash necessitates special treatment and collection. Solid waste is a term used to describe the unnecessary or harmful solid materials produced by humans in industrial and residential areas.

Strong waste disposal is important because it limits the detrimental effects on public health and the environment. A municipality's waste materials are efficiently managed using a variety of approaches. Monitoring, collection, transportation, sorting, recycling, and eventually, disposal, are all included. An urban population's socio-economic status directs its lifestyle, which leads to the different quanta of waste generation.

Solid Waste

Garbage disposal is referred to as solid waste management. Garbage or solid waste has been a huge problem for people since they began to live in cities. Global cultures accumulate more solid waste than any prior human society.

Industrialization has impacted people's daily lives significantly, as they accumulate several pounds of solid waste per user. The production of wastes take place both directly inside homes and indirectly in factories that manufacture consumer products. Solid wastes, also known as trash, is broadly classified as follows:

1. **Organic waste:** Kitchen wastes, vegetables, flowers, leaves, and fruits are all included in organic wastes. Because of an inefficient waste collection scheme, urban solid wastes accumulate on the highways. People clean their own homes with care, but they litter their surroundings, which has an adverse impact on the communities and, indirectly, on them.
2. **Toxic waste:** Paints, chemicals, old medicines, bulbs, spray cans, fertilizers, pesticide containers, batteries, shoe polish, etc. are some examples of toxic waste.
3. **Recyclable:** Paper, glass, metals, plastics are some examples of recyclable waste.

4. **Hospital waste:** Hospital waste includes products used in hospitals such as clothes, gloves, hand rubs, syringes, etc.

Dumping of solid wastes contributes to the decomposition of biodegradable products in unsanitary and unregulated environments, resulting in foul odours, which provides a breeding ground for a number of infectious organisms and insects. Hazardous wastes and radioactive metals that are released from industrial solid wastes also cause changes in the physiological and biological characteristics, and affects the fertility of soil as well. Biological characteristics of solid waste include human and animal excreta, which can be decomposed, as compared to other solid waste items.

STUDY NOTE

Hospital supplies are severely considered as harmful solid wastes, which cannot be easily recycled or disposed.

Wastes like paper scraps and non-toxic materials, which are recyclable but are not recycled properly, hampers the productivity of soil. Furans, dioxins, and polychlorinated biphenyls are generated when products like cans, cleaning solvents, pesticides, radioactive materials, plastics, etc. are burned, causing a variety of cancers. Such kinds of adverse effects have made it more necessary to understand waste management.

The following are some of the methods to classify and manage solid wastes:

1. **Centralized method:** This approach entails the disposal of urban wastes using the land filling method. The solid wastes are collected by a Municipal Corporation's waste keeper and later handed over to a recycling team. The waste is then disposed in a dump by a recycling crew.
2. **De-centralized method:** Waste is obtained by wards and divided at a source into non-biodegradable and biodegradable materials. The process of composting is used to transform biodegradable wastes into compost. Non-biodegradable wastes are divided further into paper, plastic, metal, and other wastes before being processed by recyclers for up-cycling. This model is observed in south India-Andhra Pradesh, and Karnataka.



Source: Wikipedia.org

Fig 12: Solid Waste

SELF-ASSESSMENT QUESTIONS – 4

9. Which of the following items do not fit in a trash?
- a) Micro-organisms
 - b) Plastics
 - c) Papers
 - d) None of the above
10. In the decentralised method, a non-biodegradable waste is _____.
11. Toxic wastes include _____, _____ and _____.

4.2. Classification of Solid Waste

We have discussed so many types of wastes that are produced at different sites. The enormous amount of wastes produced can be classified into a number of categories. The classification can be done into two ways:

- Classification based on the source of an origin
- Type-based classification

The classification based on the source of an origin directs wastes based on their origins. They are categorised as residential, commercial, institutional, municipal, industrial, agricultural, and open areas.

The other type of classification is based on types, i.e., physical, chemical, and biological characteristics of a waste. They can be categorised as animal waste, sewage waste, hazardous waste, construction and demolition wastes, abandoned vehicles, dead animals, street wastes, and bulky wastes.

4.2.1 Classification Based on the Source of Origin

1. **Residential:** This contains waste from residential areas, such as homes and apartments, which include leftover fruits, vegetable peels, plastics, garments, and ashes, among other items.
2. **Commercial:** This includes waste caused by supermarkets, restaurants, markets, hotels, motels, auto-repair shops, medical services, and other businesses, such as leftover food, glasses, metals, and ashes.
3. **Institutional:** This mostly consists of paper, rubber, glasses, and other materials originating from instructional, administrative, and public buildings such as classrooms, universities, offices, and prisons.
4. **Municipal:** This includes dust, leafy matter, building wastes, leftover sludge from treatment plants, and other civic operations such as renovation and demolition, street sweeping, landscaping, and so on.
5. **Industrial:** Industrial wastes are the waste from an industry's operations, which include process wastes, ashes, renovation and building wastes, toxic wastes, and so on.
6. **Agricultural:** This includes ruined food grains and vegetables, agricultural waste, trash, and other materials produced by vineyards, farms, fields, orchards, among other locations.
7. **Open areas:** This covers pollution from parks, sidewalks, alleys, vacant lots, playgrounds, beaches, roads, and recreational areas, among other locations.

STUDY NOTE

There have been some recent initiatives taken by the people to classify dry and moist residential wastes separately to recycle them more easily. Indore is one such city where the municipal corporation has taken this initiative.

Waste category	Source
Residential	Food wastes, plastics, paper, glass, leather, cardboard, metals, yard wastes, ashes, tires, batteries, old mattresses
Industrial	Packaging wastes, ashes, chemicals, cans, plastics, metal parts
Commercial	Thin and thick plastics, food wastes, metals, paper, glass, wood, cardboard materials
Institutional	Wood, paper, metals, cardboard materials, electronics
Construction and Demolition	Steel materials, concrete, wood, plastics, rubber, copper wires, dirt and glass.
Agriculture	Agricultural wastes, spoiled food, pesticide containers
Biomedical	Syringes, bandages, used gloves, catheter, urine bags, drugs, paper, plastics, food wastes, sanitary napkins and diapers, chemicals.
E-Waste	Electronic items like used TVs, transistors, tape recorders, computer cabinets, mother boards, CDs, cassettes, mouse, wires, cords, switches., chargers.

Source: brainkart.com

Fig 13: Classification based on source

SELF-ASSESSMENT QUESTIONS – 5

12. Which of the following is not an origin-based classification?

- a) Residential
- b) Industrial
- c) Municipal
- d) Hazardous waste

13. Pesticide containers are a part of:

- a) Institutional wastes
- b) E-commerce wastes
- c) Agriculture wastes
- d) None of the above

4.3. Solid Waste Management Methods

A solid waste management system (SWM system) is a set of different functional elements that work together to handle solid wastes. The system assists in the storage and recycling of solid wastes at low prices while protecting public health. The functional aspects of SWM are:

- 1. Waste generation:** Wastes are inevitable as they will be generated at every stage—from production to consumption. Households, factories, industrial areas, institutions, commercial areas, street sweeping, and other public utilities, for example, produce

wastes. The detection of wastes is the most important feature of this section of the SWM scheme.

2. Waste collection: This involves storing wastes and transporting them to a recycling station, dump site, or recycling plant after a collection vehicle has been cleared. The amount of waste disposal, the number of containers used, and the types of collection systems affect waste collection. Waste collection is offered in a variety of management arrangements, ranging from local to franchised services, as listed below:

- Transfer and transport
- The type of transport operation equipment, accessories, capacity, and environmental specifications have an impact on the design of a transfer station
- Recovery and recycling
- Sorting

3. Waste storage: Since waste is never collected at its source, storage is a necessary functional feature. The heterogeneous wastes generated in households must be taken off within eight days because of a lack of storage space. Onsite storage is very critical because of aesthetic consideration and public health. Some of the storage options used are oil drums, plastic containers, used oil drums, big storage bins, etc.

4. Waste disposal: All solid wastes, including household wastes, agricultural wastes, urban semi-solid wastes, incinerator residues, composts, or other items of no other utility to a community, will eventually be disposed of. As a result, land serves as a preparation area for landfill operations, being a prime determinant in their collection, nature, and service. As a result, the final use of reclaimed land is one of the most important practical aspects of SWM.

4.1 Landfills

Garbage is distributed in thin layers and enclosed with plastic form or clay in landfills. The bottom of a modern landfill is covered with a water-proof liner, different layers of sand, clay, and thick plastics. Because of the percolation of leachate, which is pumped and sent for treatment, this impermeable liner serves as a barrier of protection for the groundwater.

When a landfill is completed, it is surrounded by grit, mud, asphalt, and topsoil to prevent water from seeping in. Many wells are dug near landfills to see if there is any groundwater

pollution due to leaks. Methane is created by anaerobic decomposition, and is extracted and burned to generate heat or electricity. The principles for selecting a Sanitary Landfill Site are:

- ✓ It cannot be used in a rock quarry because water will leak through a water fracture system through the cracks in rocks.
- ✓ It should not be put in gravels or sandpits because of their high leaching ability.
- ✓ It must not be situated in a flood plain. The bulk of a garbage is less compact than water, but if it rains in a landfill, the garbage will rise to the surface and flow away downstream.
- ✓ It should not be close to human and animal habitats as it may adversely affect the health of living beings due to the emission of various types of gases while decomposing.

Landfill activities can have a number of detrimental effects. These consequences may include fatal injuries, contamination, and infrastructure damage; the rotting of organic wastes create methane, which is a hazard to residents; and a poorly built landfill harbours disease vectors such as flies or rats.



Source: study.com

Fig 14: Sanitary landfills

4.2 Incineration

Incineration refers to an act of burning something until only ashes remain. An incinerator, which is a device or facility that burns garbage and other forms of waste until it is reduced to ash, is a tool used in this process. To avoid the release of excessive quantities of exterior heat, an incinerator should be made of thick, well-

STUDY NOTE

A landfill is an area for the disposal of solid wastes as a method of filling and excavation after a certain period.

insulated materials. The high temperatures of heat are retained in the furnace, which is used to rapidly and effectively burn the waste.

If we don't regulate the heat, the waste in an incinerator won't burn as thoroughly. It is a form of waste management, in which a large amount of solid organic wastes are subjected to burning and converted into gaseous materials or residues. This method aids in 20 to 30 percent of the reduction of solid waste loads of their initial volume.

Incineration, also known as 'thermal treatment', is a heat-based waste treatment technique. Wastes are converted into coal, fire, smoke, and steam in incinerators. Industries perform incineration on a large scale, while individuals perform it on a smaller scale. Strong liquid and gaseous wastes are all disposed in an incineration process. It is often used to dispose off hazardous waste products in a practical way. Incineration is a contentious waste disposal process because it results in the release of gaseous contaminants. These gases are hazardous for life, due to which they cannot be reused as they are the smallest form of waste.

4.3 Composting

Since garbage capacity is scarce in major cities, biodegradable wastes (which is kept separate from urban wastes) are permitted to decompose in a medium. To boost soil fertility and conditions, environmentally sustainable and high-quality manure are mixed. Organic matters account for 35 to 40 percent of India's urban solid wastes, which can be recycled using this composting process.

In the Indian history, composting is one of the oldest ways for the disposal of waste. The natural biological mechanism of organic waste decompositions creates manure or compost, which is extremely nutrient-dense. Composting is a biological mechanism in which microorganisms such as fungi and bacteria degrade food waste into a humus-like material.

This finished product, which resembles compost, is rich in carbon and nitrogen, making it a perfect growing medium for plants. A composting process means that kitchen wastes should not be dumped carelessly and left to rot, but it should be recycled as the nutrients from a compost contributes nutrients to the soil. Composting will dramatically minimise discarded waste, in addition to being clean, affordable, and safe.

Instead of artificial fertilisers, we should continue to use organic fertilisers, which are safer for crops. It enhances the soil's ability to absorb water while also making it easy to grow crops. Composting helps in keeping a plant's nutrients in the soil.

Vermicomposting has gained a lot of attention in the recent years. Worms are added to a compost in this process, which aids in the breakdown of waste and also adds worm excreta, which provides additional nutrients.

Every responsible citizen should aim to make a compost pit; the steps for doing so are as follows.

- You must first pick a shady and cool spot in your garden or in your institution's grounds, and then dig a pit that is approximately 3 feet deep. For aerobic composting, this depth is ideal.
- To prevent nitrite pollution of the highly radioactive subsoil water, the pit should be lined with brick or granite. When you add an organic matter to the pit, make sure to cover it with a thin layer of dirt or dry leaves. This encourages oxygen to penetrate the pit, which helps to control the foul odour. After nearly 40 days, the rich pure organic matter is available for use. Composting has many benefits, like:
 - ✓ It increases the retaining power of plant nutrients.
 - ✓ It provides the plants with vital nutrients.
 - ✓ It aids in the reduction of the negative effects of high acidity, alkalinity, or chemical fertilisers' use.
 - ✓ It makes it easy to farm soil.
 - ✓ It aids in maintaining the soil's temperature.
 - ✓ It aids in the prevention of soil erosion and the growth of weeds.

4.4 Recycling

Recycling is a method of converting garbage into something valuable. It reduces the volume of garbage that needs to be handled by lowering the cost of handling and landfill disposals. It also helps to save natural resources by lowering the amount of energy used to create new materials. Upcycling and downcycling are two words for recycling.

Upcycling a product entail updating it by various recycling systems. Making roads out of cheap plastic is an example of upcycling. Downcycling is a practise of lowering the value of a product by multiple recycling methods. Downcycling is a method of breaking down high-quality plastics at high temperatures into lower-quality plastics, which can be used to produce a range of products.



Fig 15: Recycling

SELF-ASSESSMENT QUESTIONS – 6

14. The most famous method of composting is _____.
15. We should use _____ when vermi-composting.
16. Incinerators should be made from the following things. Choose the correct answer.
 - a) Conductors
 - b) Insulators
 - c) Both of these
 - d) None of these

5. CLIMATE CHANGE

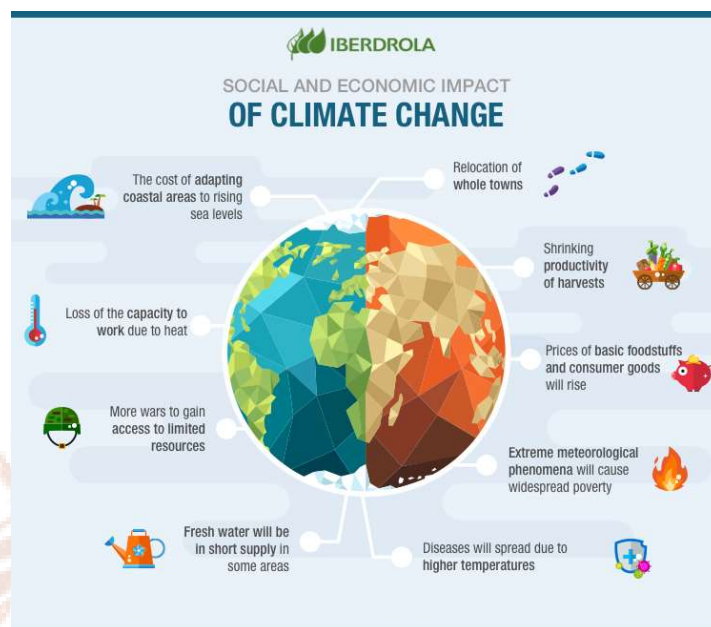
Climate change has become one of the most critical and significant issues that have put the global environment at a risk. Climate change, with time, will have several hazardous effects on the environment worldwide, socio-economic and other relevant sectors, including food security, water resources, human health, agriculture, coastal zones, biodiversity, and ecosystems. The alterations in rainfall patterns will further result in extreme water shortages and even flooding. Climate change as a whole is a severe threat to the citizens across the globe.

STUDY NOTE

The UN Framework Convention solely aims at stabilizing greenhouse gas concentrations present in the atmosphere at a level that can limit dangerous interaction with the climate system.

The melting of glaciers tends to cause soil erosion and flooding. Increasing temperatures will further lead to changes in the crop growing seasons, which will impact food securities, and even increase disease causing vectors like mosquitoes. This can put people's health at an extensive risk. The climate change crisis will make individuals more vulnerable to diseases like dengue and malaria. Rising temperatures may simultaneously enhance extinction rates for a wide range of species and habitats.

Climate change is a rather paradoxical matter. While several scientific details suggest that the climate change crisis is an absolute threat to humanity and its future, the public responses, and political discussions haven't played their role in truly recognizing climate change and its criticality.



Source: iberdrola.com

Fig 15: Social and Economic Impact of Climate Change

Potential Causes of Recent Climate Change:

The recent climate change crisis is not a result of recent activities altogether. It is a result of human activities from the last few decades. However, the potential causes of climate change can be categorized into three elements – Natural internal climate mutability, natural effects that force shifts, and man-made effects that force shifts.

Natural factors may be anything from the sun's energy output, the presence of volcanic elements in the stratosphere, and the earth's orbit around the sun. Man-made factors, however, are factors such as greenhouse gases, small particles like soiling effects of sulfates, and so on.

SELF-ASSESSMENT QUESTIONS – 7

17. The amount of greenhouse gas emissions that often develop depends on _____.
18. Compared to today's condition, more greenhouse gases may warm us in the near future.
[True/ False]

5.1. Global Warming

In simple terms, global warming refers to the process of a gradual temperature increase near the earth's surface. This phenomenon has been in existence for the last two centuries, and has severely affected the climate of the planet. There are over a multitude of causes of global warming, and all these causes negatively impact the human species, animals, and even plants. The factors responsible for global warming can either be the result of human activities or due to natural circumstances. To reduce the phenomenon of global warming, it is crucial that every individual understands the adverse impacts of global warming.

3.1 Effects of Global Warming

The prime effects of global warming are briefly explained below.

- **Climate change:** The most severe effect of global warming is climate change. Global warming has resulted in a critical impact on climatic conditions. Several places are experiencing droughts, which is further causing a climatic imbalance.
- **Rise in Temperature:** Global warming has also resulted in an extensive rise in the temperature of the surface of the earth. Ever since 1880, the temperature of the earth has increased by at least 1.1 degree Celsius. This has additionally caused a deliberate increase in the melting of the glaciers. Due to this, there has been a rapid rise in the sea level, and this can severely affect the coastal regions.
- **The threat to the ecosystem:** Global warming has adversely impacted the coral reefs, which has resulted in the loss of animal and plant lives. A rise in the global temperature has put coral reefers in a vulnerable position.
- **Spread of diseases:** Global warming has unfortunate and terrifying impacts on human lives. With the alteration of heat and humidity patterns, global warming has put the lives of humans at a greater risk. For instance, the rise in the temperatures because of global warming also increases the population of mosquitoes, which puts humans in a vulnerable environment.
- **Loss of natural habitat:** Due to the global shifting of the climate, there has been a massive loss of animal and plant habitats – animals have been forced to migrate from

STUDY NOTE

Asthma, allergies, infections disease outbreaks, etc., are prone to become more widespread and common due to the rise of pollen-producing parthenium, increased levels of air pollution, and the transfer of conditions ideal to mosquitos and pathogens.

their homes/natural habitats. On the other hand, some plants and animals have also become extinct.

- **Increased mortality rates:** Due to the rise of tsunamis, floods, droughts, and other natural calamities, the average death rate has rapidly increased, thereby hampering human life as a whole.

3.2 Causes of Global Warming

1. **Man-Made Causes:** The prime man-made causes that have led to global warming are briefly explained below.

- **Chlorofluorocarbon:** With extended use of refrigerators, air conditioners, etc., humans have increased chlorofluorocarbons (CFCs) into the environment. This has further affected the atmospheric ozone layer. The ozone layer plays a fundamental role in securing the earth's surface from the sun's ultraviolet rays.
- **Deforestation:** Plants are the most significant contributors of oxygen in the atmosphere. They release oxygen by taking in carbon dioxide and thus maintain the harmony of the environment. With the excessive rise of deforestation, the environment is faced with an imbalance. Forests are being cut down or depleted for a wide range of commercial and domestic purposes, thereby leading to global warming.
- **Agriculture:** Several farming activities release methane gas and carbon dioxide, which further contribute to the greenhouse gasses present in the atmosphere, thereby raising the earth's temperature.
- **Industrial growth:** The growth of industries has also given rise to global warming. Industrialization has resulted in the rise of the temperature of the earth. Factories emit harmful gases, thereby impacting the earth's temperature severely.
- **Overpopulation:** Overpopulation is one of the most prime causes of global warming. With more people breathing and living, there is a rise in the level of carbon dioxide. An increase in population also means an increase in the resources used for sustenance of a human life, which will further impact the global warming in one way or the other.
- **Usage of Vehicles:** Excessive use of vehicles results in the emission of a wide range of gasses, thereby increasing the carbon dioxide gas content in the atmosphere.

2. **Natural Causes:** The crucial natural causes of global warming are briefly explained below.

- **Water Vapor:** Water vapor is recognized as a type of a greenhouse gas. With the rise of the earth's temperature, there is a rise in the evaporation from water bodies. This process contributes to global warming.
- **Volcanic eruptions:** Volcanic eruptions emit excessive ash and smoke. This leads to an imbalance of the climate.
- **Forest fires:** Forest fires or forest blazes secrete a broadened amount of carbon-containing smoke, which further affect the temperature of the earth.
- **Melting permafrost:** Another natural cause of global warming is the melting of permafrost. Permafrost is present where glaciers are situated. As the permafrost melts, it excessively releases gases into the atmosphere, leading to the rise of the earth's temperature.

STUDY NOTE

The ozone layer was discovered in 1913 by Charles Fabry and Henri Buisson, who were French physicists.

SELF-ASSESSMENT QUESTIONS – 8

19. Early snowmelt is not a cause of global warming. [True/ False]
20. Which of the following leads to the extinction of flora and fauna?
- a) Improved weather condition
 - b) Advancement in agricultural activities
 - c) Improvement in the current habitat
 - d) Disruption of habitats like alpine meadows and coral reefs

5.2. Ozone Layer Depletion

The ozone layer is recognized as one of the prime strata of the atmosphere. The ozone layer consists of comparatively higher concentrations of O₂ (oxygen). This layer absorbs over 93% – 99% of the ultraviolet rays of the sun. About 91% of this layer is present in the earth's atmosphere. It is chiefly situated in the stratosphere's lower portion, about 10 to 50 Kms above the Earth. The thickness of this layer entirely depends on geographic and seasonal factors.

The ozone layer present in the stratosphere takes a part of the sun's radiation, thereby limiting it from reaching the earth's surface. The most fundamental characteristic of this layer in the stratosphere is that it absorbs the ultraviolet rays of the Sun. Ultraviolet B (UVB) is correlated to harmful effects like cataracts, skin cancers, and other critical effects on marine life and crops.

Ozone concentrations present in the atmosphere primarily vary with seasons, sunspots, and latitudes. These procedures are predictable and well analyzed. Every natural reduction that occurs in the ozone layer has been monitored and assessed by a recovery. Back in 1970, scientists found out that the ozone shield was exhausted above all natural processes. Atmospheric concentrations of the ozone layer depend on the weather, altitude, latitude, and even temperature. Such factors are also dependent on natural calamities like droughts and volcanic eruptions, which impact the ozone levels.

'No natural phenomenon resulted in depletion of ozone levels'. The scientific evidence suggested that a majority of the cause of the depletion of the ozone layer was due to man-made impacts. These ozone-depleting elements were first established in the 1970s. The wide use of consumer and industrial appliances like extinguishers, refrigerators, air conditioners, and so on were found to be major factors that severely caused the depletion of the ozone layer. When bromine and chlorine atoms blend with each other and are in contact with the stratosphere's ozone layer, they exhaust the ozone molecules.

SELF-ASSESSMENT QUESTIONS – 9

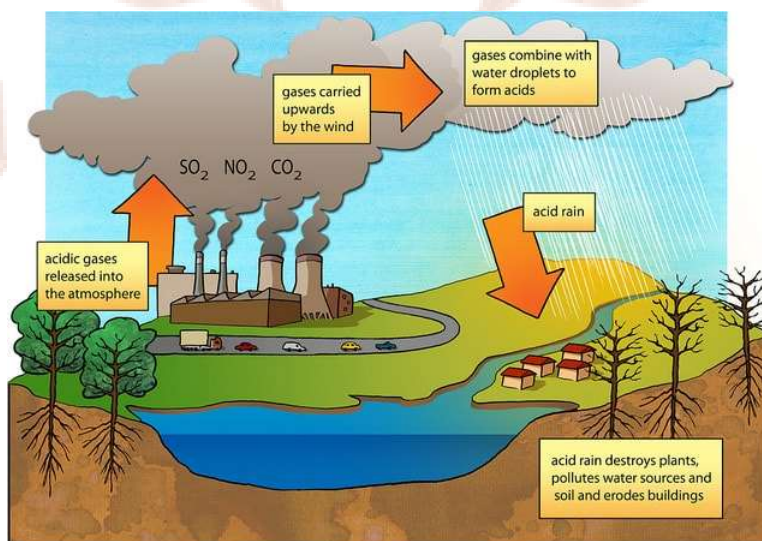
21. The depletion of the ozone layer is found in which layer of the atmosphere?
- a) Lithosphere
 - b) Ionosphere
 - c) Stratosphere
 - d) None of the above
22. Which of the following causes the depletion of the ozone layer?
- a) Chlorofluorocarbon
 - b) Methane
 - c) Carbon dioxide
 - d) Nitrous oxide

5.3 Acid Rain

When fossil fuels like oil, natural gas, coal, etc., are burned, several chemicals like sulfur trioxide, nitrogen oxide, sulfur dioxide, etc. are released into the air. One of the prime results of air pollution is acid rain. It often occurs when factories, heating boilers, cars, and other such elements emit gases into the atmosphere. When in contact with water, these gases become sulphuric acid, sulfurous acid, and nitric acid. Such phenomena also occur when natural calamities like volcanic eruptions take place.

5.1 Effects of Acid Rain

- Acid rains are one of the most critical factors that can affect agriculture, animals, and even plants. It eradicates all the essential nutrients that are necessary for the survival and development of plants. Acid rain impacts agriculture by changing the compositions of a soil. It even causes respiratory issues among humans and animals.
- When acid rain flows down ponds and rivers, it impacts the aquatic ecosystem. Acid rain changes a water body's chemical composition and causes harm to an aquatic ecosystem.
- Acid rains cause severe damage to monuments and buildings made up of metals and stones.
- Acid rains erode water pipes. It then leads to the leaching of extensively heavy metals like lead, iron, copper, etc., into drinking water.



Source: epa.gov

Fig 16: Acid Rain Pathway**SELF-ASSESSMENT QUESTIONS – 10**

23. Which of the following occurs when emissions from cars and factories come in contact with atmospheric water?
- a) Burning of Fossil fuels
 - b) Ozone layer depletion
 - c) Acid Rain
 - d) None of the above
24. Acid rain erodes _____.

6. ENVIRONMENT LAWS

The environmental challenges faced by humanity (be it an individual, private companies, communities, governments, etc.) all throughout the world are tremendous and complex. Many civil societies, governments, and companies have come to realize that environmental issues are strongly linked with social, economic, and cultural issues. To promote an economy, most governments have recognized that the only path forward is by safeguarding human health, their culture, and the environment. But due to decades of mismanagement of environmental resources, there are legacy issues in most countries, which need to be rectified.

Here is a glimpse of the global environmental problems in a nutshell:

- Water and air pollution
- Deforestation
- Climate changes
- Land and soil degradation
- Depletion of non-renewable energy sources
- Non-sustainable depletion of natural resources
- Environmental impacts of reservoirs and water abstraction
- Ocean and freshwater deoxygenation
- Persistent toxins and waste management
- Impacts of mining

STUDY NOTE

The history of environmental laws goes back as early as 80 AD, when the senate of Rome aimed to secure clean supply of drinking water in the city.

- Loss of biodiversity and habitat degradation and fragmentation
- Ozone depletion
- Pressure on the environment due to population explosion
- Nuclear risks

Many of the issues mentioned above transcend national and other administrative boundaries. This nature of trans-boundary challenges makes it imperative that there are management agreements and processes devised for decision-making. Environment conservation needs laws around dispute resolutions that go beyond individual and national borders. A socially acceptable and ecologically sustainable development requires the creation of legal and administrative structures.

In the following sections we will look into some important environmental laws imposed in India such as:

- The Environment (Protection) Act, 1986
- The Wildlife Protection Act, 1972
- Water (Prevention and Control of Pollution) Act, 1974
- Air (Prevention and Control of Pollution) Act, 1981
- The National Green Tribunal Act, 2010

6.1 Important Environmental Laws in India

The constitutional framework of India facilitates the conservation and protection of our environment. The same is reflected in international commitments. In the constitution under Part IV A (which is a part of Article 54 – Fundamental Duties), every person has the duty of securing and enhancing the natural environment including forests, rivers, wildlife, and lakes. The Article 51-A (g) Act says that the citizens of India must have compassion for all living creatures around them. This is further strengthened by Article 48A (Directive Principles of State Policies), Part IV by stipulating that the state will strive to improve, preserve, and protect the environment by taking care of the wildlife and forests of the country.

The Ministry of Environment, Forest and Climate Change (MoEF), along with CPCB (Central Pollution Control Board) and SPCBs (State Pollution Control Board) has laid down the following important environmental legislations:

- **The Wildlife Protection Act, 1972:** This Act protects wildlife that includes plants, wild animals, birds, and other species. This act's objectives are as listed below:
 - Killing, capturing, poisoning, or trapping wild animals is prohibited
 - Hunting is prohibited
 - Wildlife habitats are to be protected and managed
 - Protected areas for wildlife should be established
 - Management of zoos
 - The trade of products that are derived from wildlife should be regulated and controlled
 - No part of a wild animal's body can be destroyed, injured, or removed
 - For reptiles and wild birds, the act forbids disturbing or damaging their eggs.
 - This act opposes taxidermy

This Act has provisions for the formation of wildlife wardens and advisory boards and specifies their powers and duties. The National Board for Wildlife is an Advisory Board constituted as a statutory organization under the provisions of the Wildlife protection Act. It is the apex body for reviewing and approving all matters pertaining to development and conservation of forests and wildlife. Agencies like the police, CBI, Customs, and the forest departments can punish offenders of this protection Act.

- **The Water (Prevention and Control of Pollution) Act, 1974:** It is also known as the 'Water Act' and was established to ensure the prevention of water pollution and restore the wholesomeness of water all through the country. The objectives of this act are:
 - Penalties are laid down if this law is violated.
 - The CPSB is at the center of the Water Act, which forms the foundation for controlling and preventing water pollution.
 - The SPCBs function under the supervision of the state government and the CPSB.

In 1977, the Water (Prevention and Control of Pollution) Cess Act decided to levy a certain tax on people who consume water for carrying out specific types of industrial activities. This tax aims at augmenting CPSB and SPSBs' resources to implement the Water Act. The Water Act was last amended in the year 2003.
- **The Air (Prevention and Control of Pollution) Act, 1981:** It provides control, inhibition, and mitigation of air pollution. It also provides for the establishment of

Boards in order to carry out the objectives of this Act. Air pollutants can be in any form – solid, liquid, and gas, which are present in such a concentration in the air that it can prove to be harmful to the well-being of humans and other living creatures.

- **The Environment Protection Act, 1986:** It aims at enhancing and protecting the environment. Under this Act, a framework is provided on how to study, plan, and implement long-term requirements of environment safety. Here are the salient features of this Act:
 - The environment consists of air, water, plants, and the land and their inter-relationships amongst themselves as well as human beings, animals, microorganisms, and property.
 - It sets standards for the discharge and emission of pollutants in the atmosphere by people who carry out industrial activities.
 - It regulates where an industry can be located.
 - Lays down rules of managing hazardous wastes.
 - Any non-compliance with this Act is punishable and a person can be in jail for up to 5 years or pay a fine of INR 1, 00,000 (or both).
 - If the punishable act is continued even after the conviction of the act is over, then an additional charge of INR 5000 per day is levied on an offender for as long as the contravention of the Environment Act continues.
- **The National Green Tribunal Act, 2010:** It provides for the creation of the National Green Tribunal (a specialized judicial body that has the expertise solely for adjudicating citation related with the environment of the country). The Tribunal was set up based on the recommendations from the Law Commission of India, the Supreme Court of India, and India's International law obligations. The need for the tribunal arose from the fact that environmental cases are best addressed in a specialized forum since these cases involve multi-disciplinary issues. The Tribunal addresses issues related to the conservation of forests (and other natural resources) and environmental protection. It also enforces any legal rights related to the environmental protection. The key features of the tribunal are:

- The orders passed by the Tribunal are binding and it has the right to grant relief to the affected parties in the form of compensations.
- The tribunal is present in 5 zones in India – the North, Central, East, South, and the West
- The principal tribunal bench is located in the North Zone with its headquarter in New Delhi.
- The central zone bench of the Tribunal is in Bhopal, South Zone in Chennai, East zone in Kolkata, and West Zone in Pune.
- The head of the Tribunal is its chairperson who sits at the principal bench and it has at least 10 members (but not more than 20).
- The subjects that the Tribunal deals with are:
 - The Water (Prevention and Control of Pollution) Act, 1974.
 - The Water (Prevention and Control of Pollution) Cess Act, 1977.
 - The Air (Prevention and Control of Pollution) Act, 1981.
 - The Forest (Conservation) Act, 1980.
 - The Environment (Protection) Act, 1986.
 - The Biological Diversity Act, 2002.
- The Tribunal's decisions can be cited in the Supreme Court within 90 days of a decision.

STUDY NOTE

Convention on International Trade in Endangered Species (CITES) is a multilateral treaty, which aims to protect endangered plants and animals. The Wildlife Protection Act helped India become a part of CITES.

STUDY NOTE

The Vienna Convention on the law of treaties, held in 1969 contains the definitions for the international treaties.

SELF-ASSESSMENT QUESTIONS – 11

25. The _____ ensures that any compliance with the act is a punishable offence and a person can be in jail for up to 5 years or pay a fine of INR 1, 00,000.
26. Which of the following acts restricts hunting?
- Environment Protection Act
 - National Green Tribunal Act
 - Air (Prevention and Control of Pollution) Act
 - Wildlife Protection Act
27. Who is responsible for testing the air quality in pollution control areas?
- State pollution control board
 - The central pollution control board
 - Police in the area
 - The Forest department

7. INTERNATIONAL AGREEMENTS

International agreements are signed between two or more participating countries, who formalize certain understandings and commitments on a specific issue. A 'bilateral' agreement is an agreement where only two countries are involved, whereas a 'multilateral' agreement is established between many countries. A treaty is a legally binding agreement between countries, based on international law.

Many Multilateral Environmental Agreements (MEAs) at the international level have been introduced in the past. These agreements aim to create cooperation between its members by creating procedures and rules that are based on international laws. Some of the important international agreements that India has signed are outlined below.

- **The Antarctic Treaty (Washington, 1959):** This treaty had the aim to keep the Antarctic a peaceful zone and to prevent it from being an object of international discord. The treaty covers the range area of the south of 60 degrees S Latitude, which is called the ATA (Antarctic Treaty Area). According to this treaty, there can be no territorial claims or military activities in this region. It also imbibes the freedom of scientific inquiry with international cooperation in scientific undertakings. India signed the Antarctic treaty in 1983 as a CEL (Consultative party member).

- **Montreal Protocol on Substances that deplete the Ozone Layer, 1987:** Famously known as the Montreal Protocol, it aims at reducing the production and consumption of ODS (Ozone-depleting substances). The protocol also identifies countries that produce more ODS than other countries, and recognizes that nations should reduce ODS (in terms of technological and financial abilities). This protocol came into force in 1989 and was adopted by India in 1992. The MoEF has formed an Ozone cell to supplement the Montreal Protocol. The cell comprises a steering committee to implement the India country program.
- **UN Framework Convention on Climate Change (UNFCCC), 1992:** In an endeavor to regulate and contain greenhouse gas emissions, the UNFCCC sought international cooperation through this agreement. This framework aimed to bring down the emission of greenhouse gases to a level that the consequences of the change in climate and global warming are offset. India became part of the convention in 1992, and ratified it in 1993. At the time of the convention, India was a developing nation (according to the UNFCCC) and not bound to commitments to mitigate greenhouse emissions. But this changed after the Paris agreement in 2015 and India commits, under the UNFCCC, to take part in multilateral negotiations. India has shown leadership in this commitment with policy frameworks like NEP (National Environment Policy) and NAPCC (National Action Plan on Climate Change).
- **Convention on Biological Diversity, 1992:** It is an enforceable framework geared towards conserving biodiversity. The convention was put in force in 1993 and has a set of complex requirements to promote a sustainable way of using biological resources and preserve natural habitats. In article 6 of the Convention on Biological Diversity (CBD), nations that are part of the convention are instructed to form national strategies and programs for the conservation of biodiversity. In 2002, India passed the Biological Diversity Act, which provides a legal framework for addressing concerns regarding biodiversity in the country.
- **UN Convention on Desertification, 1994:** This convention seeks the participation of local users of land and non-governmental organizations to alleviate desertification and its effects. India, along with 7 South Asian countries, has signed this convention with a regional action program to combat desertification issues and address them in drought-prone areas.

SELF-ASSESSMENT QUESTIONS - 12

18. A _____ international agreement is an agreement between more than two countries.
19. Which amongst the following protocols deals with the origination and ingestion of ODS (Ozone-depleting substances)?
- a) The Antarctic Treaty
 - b) The Montreal Protocol
 - c) UN Framework Convention on Climate Change
 - d) Convention on Biological Diversity
20. Why do we need international agreements for the environment?
- a) An international agreement has more value than a national agreement.
 - b) The harmful effects on the environment are not restricted to a specific country and it is not in isolation.
 - c) An international environmental agreement can be enforced while national ones are not binding.
 - d) International agreements put more pressure on participating countries to enforce an agreement.



8. CASE STUDY

DELHI POLLUTION, INDIA

Back in 2011, an examination by the Center of Science and Environment (CSE) has affirmed that New Delhi is the most intense city in India. The level of noise in the roads can go over 110 decibels, which is a few times louder than Singapore. The noise level has arrived at hazardous levels, past the suggested rules of 50-55 decibels for private zones.

Delayed openness to this degree of commotion has brought about the expansion of danger in hearing misfortune for the residents. As per the studies, the normal age of residents in New Delhi are 12 years older in terms of hearing, which implies they are at more serious danger of losing their hearing in their 50s or mid-60s.

The uproarious noise is frequently created by the honking of cars, which means changes in behavior and attitude can decrease the key source of the noise. Though, this is an obstacle as the habit of honking is ingrained into their everyday routine.

The roads of New Delhi are shared by people, vehicles, cyclists, individuals, and many more. Traffic on the roads is very heavy, thus the use of honk of an approaching vehicle is necessary to warn people. As this concerns individual security, the honking behavior will be a strong internal hurdle as the drivers can't just quit honking.

Source: Article from The Times of India newspaper

Discussion Questions:

- Q1. Being a General Manager to control noise pollution, what activities according to you can the state of Delhi restrict to reduce noise pollution?
- Q2. Highlight the sources and the impact of noise pollution discussed in the above case.

9. SUMMARY

- Noise pollution refers to high-level sounds that irritate living organisms' hearing capacity. It can result from machines, high music, and elevated vehicle sounds. Noise pollution is a disturbance that causes severe health hazards and environmental problems and should be controlled.
- Noise pollution refers to the undesired and unpleasant sounds that are hindering modern day development and disrupting the activity and stability of animal and human life.
- Noise pollution is usually produced in some workplaces and industrial facilities and comes from aeroplane traffic, construction activities, trains, highway traffic, and heavy vehicles. It affects the well-being and comfort of all living creatures, both ashore and in the ocean, from road traffic noise to live concerts; many sounds cause stress, hypertension, and hearing activities.
- Various measures can be taken to reduce the level of noise like covering noise delivering machines, leaving behind sound-retaining materials to check noise creation; installing noisy machines in sound verification chambers; using silencers to control noise from autos; designing, manufacturing, and utilizing calmer machines to supplant the uproarious ones; and proper oiling and better support of machines.
- To lessen noise pollution, one must use earplugs at construction sites. In addition, soundproof windows and doors need to be installed to prevent unwanted noise from outside. Finally, an effortless way to decrease strong sound waves and vibrations is by using barriers like planting trees around buildings, houses, and companies.
- Any undesirable or unusable thing is called a waste.
- Developing industrialisation has led to the generation of more waste.
- Wastes are inevitable; however, we should know how to manage the waste in our environment.
- There are various types of wastes, and they are classified on the basis of their origin.
- Municipal solid wastes are a type of waste that are generated from household activities and industrial activities.
- There is a need to create public awareness to think about the proper disposal of wastes.
- Proper waste disposal methods should be designed.

- Landfills, incineration, recycling, composting are different methods used for disposing waste.
- The earth's climate has drastically changed in the past few decades. Climate change is real and visible in temperature, wind patterns, precipitations, and others.
- Global warming is the rise of the average world temperature as a result of the greenhouse effect. Global warming adversely impacts ecosystems through water shortages, temperature increases, raised fire threats, droughts, insects and pests upsurge, severe storm damage, and salt invasion, among many others. Global warming is caused by natural causes like forest fires, glacier melting, volcanic eruption, etc., and human activities like deforestation, industrial growth, farming, burning of fossil fuels, overpopulation, etc.
- The depletion of the ozone layer refers to the increase in the amount of ultraviolet (UV) rays that enter the surface of the earth.
- Acid rains greatly affect agriculture, animals, and even plants. It drains aluminum from the soil that may be dangerous for plants and animals. It also eliminates nutrients and minerals from the soil that plants and trees need for their growth.
- Environmental laws are based on the following principles:
 - The precautionary principle: This principle states that if an activity has a high probability of causing environmental damages, it must be controlled immediately rather than looking for empirical proofs.
 - The prevention principle: Most of the environmental legislations are passed after a catastrophe has occurred. But preventing a hazardous situation is a much cheaper and wiser decision than reacting or managing a disaster or harm, which has already been done.
 - The 'Polluter pays' Principle: This environmental regulation forces a producer to bear the cost of pollution.
 - The integration principle: This principle says that all the concerns of environmental safety should be fully considered while decisions on any activity are taken.
 - The public participation principle: When a decision is taken regarding environmental protection, it is often done by integrating public opinion on a particular matter.
- Some of the important environmental laws in India are:

- The Wildlife Protection Act, 1972: This act protects wildlife, which includes plants, wild animals, birds, and other species.
- The Water (Prevention And Control Of Pollution) Act, 1974: It is also called the “Water Act” and was established to ensure water pollution is prevented and controlled with the wholesomeness of water restored all throughout the country.
- The Air (Prevention and Control of Pollution) Act, 1981: It provides control, security, and preservation from air pollution.
- The Environment Protection Act, 1986: The Environment Act aims at enhancing and safeguarding the environment.
- The National Green Tribunal Act, 2010: It provides the development of the National Green Tribunal (a specialized judicial body that has the expertise solely for adjudicating cases related to the environment in the country). The need for the tribunal arose from the fact that environmental cases are best addressed in a specialized forum, since these cases involve multi-disciplinary issues.
- International agreements are signed between two or more participating countries and understandings and commitments on a specific issue are formalized.

10. TERMINAL QUESTIONS

SHORT ANSWER QUESTIONS

- Q1. How do agricultural machines contribute to noise pollution?
- Q2. Mr Frank's family wants to purchase a house. They have been offered two houses, one facing a busy street and another away from the busy road. If you were Mr Frank, which house would you recommend to your family to purchase? Justify the answer with a suitable reason.
- Q3. Name the top 6 effects of climate change.
- Q4. List the top 5 causes of global warming.
- Q5. What is the prime gas that destroys the ozone layer?

LONG ANSWER QUESTIONS

- Q1. 'Constant interference of noise or unwanted sound in the environment influences human health.' Explain.
- Q2. Explain the main sources of noise pollution.
- Q3. Why it is important to dispose waste properly?
- Q4. How is acid rain harmful?
- Q5. Explain the need for international environmental agreements.

11. ANSWERS

Self-Assessment Questions

1. Decibels
2. D. All of the above
3. Sound
4. False
5. D. Both A. & B.
6. D. All of the above
7. B. Musical instruments
8. D. Both A. & C.
9. A. Micro-organism
10. Recycled
11. Materials with lead, mercury
12. Hazardous
13. Agriculture
14. Vermicomposting
15. Organic fertilizers
16. Insulator
17. Energy use, new technologies, population growth, and so on.
18. True
19. False
20. D. Disruption of habitats like alpine meadows and coral reefs
21. C. Stratosphere
22. A. Chlorofluorocarbon
23. C. Acid rain
24. Water pipes
25. Environment Protection Act
26. D. Wildlife Protection Act
27. A. State pollution control board
28. Multilateral
29. B. The Montreal Protocol

30. B. The harmful effects on the environment are not restricted to a specific country and it is not in isolation.

Terminal Questions

Short Answer Questions

Answer 1: Agricultural practices have become mechanized due to the technological revolution. As a result, several machines and tools such as drillers, tractors, harvesters, powered tillers, thrashers, and tube wells are used for farming, causing heavy noise pollution.

Answer 2: I would recommend purchasing the house far away from the busy road. As houses facing busy streets will have a lot of noise and air pollution, houses away from busy roads will have less traffic and less noise. The concentration of noise reduces with the decrease in the distance between the receiver and the source.

Answer 3: Effects of Climate change:

- Increase in maximum temperatures
- Increase in minimum temperatures
- A rise in ocean temperatures
- Increase in sea levels
- Shrinking glaciers
- Thawing permafrost

Answer 4: The top 5 causes of global warming include:

- Agricultural activities
- Industrial activities
- Deforestation
- Natural calamities
- Variations in the intensity of the sun.

Answer 5: Ozone depletion destroys the ozone layer. When bromine and chlorine atoms are exposed to the ozone layer in the stratosphere, they harm and further destroy the ozone molecules.

Long Answer Questions

Answer 1: Noise Pollution can be hazardous to human health in the subsequent manners:

- **High Blood Pressure:** High noise levels raise blood pressure and pulse rates, making humans susceptible to heart attacks. In addition, noise pollution causes heart attacks due to sleep interference.
- **Loss of Hearing Ability:** Continuous exposure to loud noise beyond the acceptable range of sound can harm the eardrums, thus resulting in hearing loss. Hearing impairment induced by noise pollution causes temporary or permanent deafness.
- **Disturbance:** Effects of noise pollution include not only disturbed sleep but also affects mental health. Some psychological effects of noise pollution are losing peace of mind, depression, stress, getting irritated more frequently, anxiety, waking too early, and so on. Lack of concentration can also be examined due to unwanted sounds and noise.

Answer 2: The sources of noise pollution are as follows:

- **Social events:** Weddings, festivals, public gatherings, and occasions involve loudspeakers playing music, thus producing unwanted noise in the surroundings. The administration and concerned authorities have issued certain rules restricting music in high volumes. However, people do not follow such rules, creating a huge nuisance.
- **Fireworks:** Indian festivals such as Diwali, Dussehra, marriages, and so on use fireworks for celebrations. Bursting firecrackers is a health hazard and creates noise pollution. The noise produced is hazardous and can sometimes lead to non-recoverable hearing loss.
- **Defence equipment:** Military aeroplane drills, artillery tanks, explosions, rocket launching, and shooting practices are serious noise polluters. Likewise, the sounds of jet engines produce deafening impacts.

Answer 3: It is important to dispose waste properly because the ill disposal of waste can lead to a number of effects like:

- **Soil Contamination:** Contamination happens when toxic components are buried or spilled in dirt. As a result, we must exercise caution when treating liquid hydrocarbons, solvents, chemicals, heavy metals, and lead.

- **Air contamination:** Hazardous wastes, such as bleach and acids, must be disposed off safely and only in designated bins with clearly labelled labels. The ozone layer is affected by the burning of papers and plastics in landfills, which releases chemicals and methane.
- **Water contamination:** Hazardous wastes in the atmosphere seep deep into the earth, eventually contaminating groundwater, and is used for a variety of purposes including drainage and drinking.
- **Bad impact on human health:** Inadequate waste management may have a major effect on the health of the people who live near contaminated areas or landfills. Landfills facility stakeholders are also at a higher risk.

Answer 4: Acid rain is extremely harmful and hazardous to forests. It seeps into the ground and can further dissolve nutrients like calcium and magnesium. These nutrients are required by plants to stay healthy and grow. Acid rain can moreover release aluminum into the soil. This can hinder the growth of trees and get in the way of trees absorbing water.

Answer 5: Since the human-induced effects on the environment are profound, they need to be measured and an assessment of the performance of processes involved is needed. The harmful effects on the environment are not restricted to a specific country and it is not in isolation. Hence, these far-reaching effects affect the earth in complex ways, and there is a need for some level of organization that is not just local or national but is formed at an international level. This organization must bring about a sense of how human beings can approach ecological processes. Laws of international agreements must be initiated by an international organization, which aims at seeking the cooperation of diverse participants irrespective of their concerns or agendas.

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BACHELOR OF COMPUTER APPLICATIONS

SEMESTER 1

ENVIRONMENTAL SCIENCE

Unit 7

Human Communities and the Environment

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1. INTRODUCTION

The word 'population' has its origin in the Latin word 'populus', which means people. A population is used to describe a group of individual species who inhabit a definite and same geographic area at a given time.

A population is a dynamic entity, and recent years have seen massive changes in birth rates and population growth across countries and continents. The global population is now growing by more than 80 million in a year, and is estimated to grow till the rest of this century.

Human societies have few fundamental aspects like the size of a population, the composition of the people in it, and the rate of the growth of a population. All these aspects and trends of a population affect economic prosperity, education, health, crime patterns, family structures, cultures, languages, etc in India.

Let us look at some terms and definitions related to population before we delve into it in detail.

- Population growth rate: At any given time, the change in population per unit area is called population growth. The formula for finding the growth rate is given by:

Population growth rate = (Population at the end of the time period - Population at the beginning of the time period)/Population at the beginning of the time period

- Population density: The number of people residing in per unit area or unit volume of an environment gives the population density of that environment.
- Carrying capacity: The maximum number of populations that can be organized by the available resources of an environment is termed as a carrying capacity.
- Demography: This is a study of human populations. Demography draws insight from multiple disciplines like history, economics, statistics, sociology, biology, medicine, and anthropology.
- Closed population: A population where there are no emigrations and no immigrations. Additions and reductions in such a population can happen only by births and deaths, respectively.

- Overpopulation: This happens when the population of a place exceeds its carrying capacity that its environment has.
- Doubling time: The time required to double the size of a population is referred to as doubling time period.

The primary parameters that affect trends in a population are:

- Natality (Birth rate): The number of new born per thousand persons in a given year constitutes the birth rate of a population.
- Death rate: The number of deaths per thousand persons in a year with a possible count is termed as the death rate of a population.
- Immigration: When people arrive in a population from neighbouring regions, it is called immigration.
- Emigration: When people migrate or disperse from their original population area to a new area, it is called emigration.

We will now look into where the human population stands today and the reasons for the tremendous growth in population. We will also consider how this population trend affects human welfare and the various measures taken by the governments to control and manage growing populations.

1.1 Learning Objectives

After studying this chapter, you will be able to:

- ❖ *Explain the meaning of population and other terms surrounding this concept.*
- ❖ *Define the various reasons for population growth and how it affects our environment.*
- ❖ *Explain the different measures taken by the governments to slow down the population explosion.*
- ❖ *Describe resettlement and rehabilitation of populations and issues around them.*
- ❖ *Explain why we need environmental communication and maintain environmental ethics.*
- ❖ *Describe what disasters are and how they can be caused.*

STUDY NOTE

The UN has projected that the global population would be 9.7 billion in 2050, and by 2100, we will be 10.9 billion in size.

2. HUMAN POPULATION GROWTH

The world population stands at 7.3 billion people as per the data collected in the year 2015. This means that we have more than 7 billion bodies that need food, clothes, education, jobs, entertainment, etc. Apart from the enormous resources to maintain so many lives, we also have vast amounts of waste being produced and disposed every day. As a result, the increase in human population affects the environment mainly by two ways:

- Resources such as land, fossil fuels, water, air, food, minerals, etc. are being consumed in huge quantities.
- The massive production of waste products, air and water pollutants, greenhouse gases, and many other toxic materials are pervading our ecosystem.

Apart from the size of a population, another issue is the issue about what is happening within a given population. The population distribution (migration, urbanisation, density, etc.), population composition (age, gender, income levels, etc.), and their consumption patterns also play an equal role in affecting our environment adversely. We shall look at these issues closely now.

2.1 Aspects of Population that Affect the Environment

It is not enough to focus solely on a population's number to get a clear picture of what is happening around us. We must understand the multifaceted relationship between human communities and the ecosystem.

- Population size: We are all aware of the fact that with growing populations, there will not be enough fertile land to produce food, forests, fisheries, or drinking water. The human carrying capacity of the earth has been a matter of debate for hundreds of years. There is still non consensus over and how many is too many and it has a wide range from 500 million humans to more than a trillion.

STUDY NOTE

The amount of environmental resources consumed in the last 50 years is more than what the whole of humanity consumed before that period.

The huge variability in the carrying capacity of the earth is because scientists do not agree on the accuracy of the methods that are used to determine these numbers. The consumption of resources has huge variations across the globe. If everyone on the Earth consumed

resources at the same rate, then the carrying capacity of the Earth could be only 2 billion. But if people consumed only as much as needed to survive, the Earth could support a much larger number of human lives.

- **Population Distribution:** The manner in which a population is distributed also affects the ecosystem. For instance, the birth rates in developing countries are more than that in the developed countries due to less access to education and family planning as well as high poverty levels. This rapid growth in the population of developing countries adds more pressure on the environmental resources.
- **Population Composition:** If there are more young people in an environment, there are higher chances of their migration to urban areas, which will result in an intensification of urban environmental concerns. Currently, India has the highest percentage of youth in urban cities. Another part of this component is an uneven distribution of incomes, which puts pressure on the environment.

To survive, the poorest people resort to using resources in unsustainable ways like burning tires, rubbish, plastics, etc. To feed their families, the poorer population may also be forced to deplete forests or animal populations. On the other side of the spectrum are the rich, who consume large proportions of resources by driving cars, living in big houses, and other environment unfriendly lifestyle choices.

- **Population Consumption:** Governments and advertisers in many developed countries, in their pursuit to increase their economy, have promoted a culture where the pattern of consumption and production has become unsustainable. It has gotten imbibed deeply in their culture to own bigger homes, buy commodities for luxury, own a faster and costlier car, etc. These are seen as signs of success in these nations, but this pattern has a negative impact both on our environment and on our emotional wellbeing. This kind of production leads to excessive usage of energy and the release of excess toxins and pollution. The impacts of such high consumptions of energy are not only confined to the local areas where the energy is consumed, but its impacts are felt globally in the form of increase in the CO₂ levels or the depletion of the ozone layer. The ecological footprints of developed countries are much bigger in size than that of developing countries.

2.2 Factors Affecting Population Growth

Overpopulation has many reasons attached to it; some of the most prominent ones are briefly explained below.

- **High birth rates and reduced death rates:** With the development in the world of medicines and technology, the standard of living has significantly improved. There are many immunisations, antibiotics, clean water, air, and better food production available to people, which has increased their longevity. The rate of birth is higher than the death rate, ultimately leading to more people alive in a given region.
- **Social customs, beliefs, literacy level, and needs:** In some parts of the world where the literacy rate is low, people are not well informed about the benefits of having fewer children. Some need more children so that their offspring can help them in agriculture, growing crops, or other fieldwork. Some are bound by the desire to have a male child, so they keep bearing children till they have a boy in their family.
- **Public health programs:** Many government health programs have improved the living conditions and life expectancy of people.
- **Fertility rates:** Total Fertility Rate (TFR) is defined as the number of children a woman can give birth to, in a lifetime. Some countries experience rapid population growth temporarily with an increase in the average fertility rates of women of that country. After that rapid growth, the TFR comes down. Currently, the average global fertility rate is 2.5 children per woman. In the pre-modern era, the average fertility rate used to be 6.5 to 7 children, and the decline in this rate can be attributed to education and increasing labour market participation, which has empowered women so that they can exercise their rights in deciding how many kids they want to have. Another reason for the lowering in the fertility rates is the rising cost of bringing up children.
- **Decline in Maternal mortality rate (MMR) and Infant mortality rate (IMR):** IMR and MMR have declined all around the world due to the availability of more health centres and health programs. The infant mortality rate is the number of children under 1 year of age dying per 1,000 live births in a particular year.

STUDY NOTE

The highest Total Fertility Rate in the world is 4.6, which is in sub-Saharan Africa.

- Rise in food production: After the World War II, there has been a steep rise in food production with the advent of the green revolution.

SELF-ASSESSMENT QUESTIONS - 1

1. Based on the discussions above, which of the following is not an aspect of population that affects our environment?
 - a) Population consumption pattern
 - b) Population Composition
 - c) Population Distribution.
 - d) The male and female ratio in a population.
2. How does a younger composition of a population affect environment?
 - a) A younger population consumes more food, which puts pressure on the environment.
 - b) A younger population takes up more jobs, which contributes to environmental pollution.
 - c) A younger population is more likely to migrate to cities for jobs, which will intensify urban environmental concerns.
 - d) Young people are more negligent towards environmental degradation.
3. The term that defines the number of kids that a woman can possibly bear in a lifetime is_____.
4. The decline in death rates is one of the reasons for the rise in population.
[True/False]

3. HUMAN HEALTH AND WELFARE

Human welfare in a nutshell means health, happiness, good fortune, and prosperity of individuals, groups, communities, etc. Human welfare is closely related to their health, which is inseparably intertwined with the environment. Population growth, aging population, urbanisation, migration patterns, etc. pose severe challenges to the overall wellbeing of the world's population. Amongst all these reasons, population growth coupled with a disparity in population distribution is one of the biggest reasons for the degradation in health and well-being.

Human health impacts the economic development of a country in the form of market failures and distortions. The advancement in technology, use of fertilizers and pesticides have degraded the environment and caused several health issues. Environmental pollution causes millions of deaths every year. The major factors influencing human health are briefly discussed below.

- **Malnutrition:** WHO (World health organization) published a report in 2005 that defined overpopulation as the breaking down of the ecological balance due to a population's size exceeding the carrying capacity of its environment. Such a situation results in a weakened food production system. With less food, there is inadequate food consumption and people suffer from malnutrition. The WHO had also cautioned in 2011 that dangerous strains of man-made viruses are likely to cause a huge count of deaths in the future since a lot of people have weakened immune systems, owing to malnutrition.
- **Migration:** Population growth can make job opportunities a struggle for people, which results in their migration to other countries that have better job prospects. Migration causes 'brain drain', which means that there may be a lack of access to skilled healthcare workers in certain areas from where they have migrated.
- **Easy spread of diseases:** When the population in a certain area increases, it becomes easier for infectious airborne illnesses to propagate fast and wide. In 2012, there were 8.6 million cases of tuberculosis.

STUDYNOTE

In 2019, a bill called the 'Population Control Bill' was proposed in the Indian parliament and signed by 125 members of the parliament. This bill is yet to become an act of law. The bill states that people who opt for a smaller family will be given incentives in the form of tax relaxations, free education, employment, etc.

- Water pollution: Overpopulation pollutes the water supply at a faster rate.

3.1 Government Family Welfare Programme

India was the first country in the world that launched a family planning program in 1951 to control the population growth of the country. Many pilot activities were done during this family welfare program, which were geared towards helping women who were at risk of unwanted pregnancies. The slogan of this program was 'We two, ours two' (hum do, hamare do). The government utilised all forms of modern as well as traditional communication systems to propagate and develop social support for this program.

The main objectives of this program were to:

- Reduce fertility to slow down population explosion. The program aims to bring down the TFR in India to 2.1.
- Provide clinical and educational services to remote villages to help couples decide and choose how many children they want to have and when.
- The program also provides information on how to manage birth spacing, control birth, and proper healthcare for a mother and her new born.
- The program has also reduced the number of illegal abortions in a year and reduced the rate of deaths due to pregnancy. Maternal Mortality Rate (MMR) is the number of women who die per 100,000 live births in a given time period. The major reasons for MMR have been identified as toxemia (Hypertension during pregnancy), haemorrhage (both ante and postpartum), puerperal sepsis (infections after delivery), anaemia, obstructed labour, and unsafe abortion haemorrhage. These deaths can be prevented if:
 - Good antenatal care is provided.
 - There is timely identification of pregnant women who have complications related to their pregnancy, and are referred to healthcare professionals.
 - Timely provision of emergency obstetric care

STUDY NOTE

Malnutrition has increased by 37% from the year 1950 to 2007 and is one of the main reasons for the deaths of 6 million children in a year.

Activity 1

Collect the maternal mortality rate and malnutrition rate of your state for the last five years. Check for the programmes initiated by your government to control the prevailing rates and assess the success of these programmes using the data collected.

SELF-ASSESSMENT QUESTIONS - 2

5. One of the aims of the family welfare program launched in India is to reduce TFR. [True/False]
6. The name of the proposed bill in the Indian Parliament in 2019 that aims at controlling population is _____.
7. According to the 'Human Health and Welfare' section that was discussed, an aging population affects the environment because:
 - a) Older people are not aware of sanitation rules.
 - b) Older people tend to contaminate the environment more.
 - c) The elderly consume more resources and put pressure on the environment.
 - d) The elderly are more likely to suffer from disease like diabetes, arthritis, heart diseases, dementia, etc., which puts pressure on the healthcare system.
8. What is maternal mortality rate?
 - a) Number of women who can bear children at any given time.
 - b) The number of women who die every year.
 - c) The number of women who die per 100,000 live births, in a given time period.
 - d) The number of women who give stillbirths every year.

4. RESETTLEMENT AND REHABILITATION

STUDY NOTE

One of the recent nuclear disasters has been the 'Fukushima nuclear disaster' in Japan, which forced many people to leave their homes for their safety.

Due to natural and human-caused disasters, people are forced to move out of their original land. For instance, when a Tsunami happened in South Asia in 2004, or the massive earthquake that happened in Latur and Gujrat, and many other incidents of flood or drought in different parts of the world that rendered thousands of people jobless and homeless. One

prime example of a human-made disaster is the Bhopal gas tragedy, which happened in the union carbide factory. Apart from disasters, there are other reasons that displace people from their homes such as many development projects (construction of dams, roads, flyovers, canals. etc.).

The process of settling in a new location is termed as resettlement. The rehabilitation of an environment occurs when a lifestyle restores the environment to its former state.

STUDY NOTE

One of the biggest natural disasters in India was the drought in 1987, which affected 15 states and displaced 300 million people.

4.1 Causes for Displacement of People

People are displaced from their original place of living due to various reasons. In the past, some of the most prominent reasons for displacements, resettlements, and rehabilitation have been:

- Natural disasters: Earthquakes, cyclones, tsunamis, volcanic eruptions, floods, prolonged drought conditions, hurricanes, etc.
- Industrial accidents (Man-made disasters): Nuclear accidents like the Bhopal gas, tragedy, Oil spills (Exxon Valdez oil spill), etc.
- Relocation while searching for better job prospects.
- Development projects like construction of irrigation canals, dams, reservoirs, etc.
- Infrastructural projects such as the construction of flyovers, roads, bridges, etc.
- Transportation activities like highways, roads, etc.
- Energy-related projects such as oil explorations, power plants, mining activities, pipelines (for example HBJ pipeline), etc.

- Projects that are linked with conserving wildlife like national parks, biosphere reserves, sanctuaries, etc.
- Agricultural projects.

4.2 Issues with Resettlement

The World Bank has estimated that close to 10 lakh people are displaced around the world for varied reasons. There are development projects happening that only have economic efficiency as their objective, and many a time, such projects fail to see that there are many people who stand to lose all they have because of such developmental projects.

Rehabilitation is a continuous and slow process and needs a lot of organised effort. One of the strategies of rehabilitation is to take preventive actions so that resettlement is not needed. For example, building earthquake-proof houses, be informed about cyclones in advance so that evacuation can be planned ahead of time, construct appropriate bunds in areas that are prone to floods, etc. All these measures can avoid a plethora of issues surrounding resettlement as outlined below:

- Lack of support: Displacements mainly affect rural and tribal people who do not get much support from the government.
- Meagre compensation: Many a time, no compensations are given on the land lost by people in the name of mega developmental projects. Even if compensations are paid, they are much delayed and too low both in monetary terms as well as the social changes that are forced upon the displaced people.
- Loss of livelihood: Displacement is a huge phenomenon in the lives of people who get displaced. The ancestral lands and forests on which they have depended all through their lives are lost. Most people do not have enough skills to pick up any other activity or occupation to earn. Usually, the land offered to them is of low quality and cannot be used to earn a living.
- Lack of facilities: There is no basic infrastructure and amenities provided in the new area offered to the displaced people. Most of the time they live in temporary tents, which later become their permanent abodes.
- A rise in stress levels: The shock of purposelessness created in human beings who are resettled is humungous. They are filled with emptiness and there are quarrels within a

family on the share that each must receive from the resettlement amount given to them. Their family lives wither with their prestige of land ownership after they are displaced. Often, the daughters of refugees find it difficult to get good matches for marriages from the neighbouring villages.

- Increased health issues: With their forest and agriculture-based livelihood gone, there is a general decline in the health conditions of refugees. Most tribal and rural people depend heavily on their herbal homemade remedies, which are submerged due to the development projects.
- Identity loss: Tribal people have unique lifestyles, which are community-based. Displacement hampers their culture and spiritual existence in multiple ways. Their loss of connection with their environment and their indigenous knowledge about wildlife are the biggest losses in a resettlement process.
- Exploitation due to lack of knowledge: Tribal people are not well versed with market trends, policies, price of commodities, and other details of a city life. This lends them to exploitation from shopkeepers and others in new and foreign areas.

STUDY NOTE

During the process of exploiting oil and constructing a connected pipeline in the Eastern Upper Nile region, 48 villages were burnt, and 55,000 people were displaced.

4.3 Objectives of Resettlement

Before giving an alternative site to a displaced group of people, there are few objectives of resettlement that must be met.

- Tribal people should have the freedom to live their lives on their own accord and nothing should be imposed or forced upon them.
- Villagers should be given an option to move out and stay together as a community.
- They must be given means and facilities so that they can pursue and develop their traditional art and culture in ways that suit them.
- Rehabilitation must focus on removing poverty.
- The fruits of a development must be shared appropriately with the people who were displaced due to a development project.
- Job opportunities must be provided to the displaced people.
- Resettlement must be provided in an area nearby their original land.

STUDY NOTE

Due to the fire episode in Jharia coal fields, 3 lakh people were affected and about 115 crore INR was spent in putting out the fire.

- In case resettlement is not possible in neighbouring areas, the new area must be well equipped with all amenities and infrastructure like schools for children, irrigation facilities, wells, supply for basic agricultural activities, primary healthcare units, drinking water, grazing grounds, and other necessary facilities.
- Every stage of rehabilitation must be done by taking the respective villagers into confidence. Villagers should be well informed and educated through public meetings about the different legalities around the land acquisition act and other provisions of rehabilitation.
- Any sort of decision-making must be taken with the view and advice of the elderly people of the village.

SELF-ASSESSMENT QUESTIONS – 3

9. Rehabilitation is a slow process. [True/False]
10. The difference between resettlement and rehabilitation is:
 - a) Resettlement means settling in a new area and rehabilitation is the compensation received.
 - b) Resettlement and Rehabilitation are the same.
 - c) Resettlement is a slow process whereas rehabilitation is a faster process.
 - d) Resettlement is a process of settling in a new location and rehabilitation is when the lifestyle of displaced people restores to the former state.
11. Which of the following is not true about resettlement and rehabilitation?
 - a) There is a loss of livelihood of displaced people.
 - b) Marriages are difficult for resettled people as neighbouring villages do not want to marry refugees.
 - c) There is a rise in health-related issues with a resettled population.
 - d) Resettled people easily get jobs to restore their livelihood.
12. Which of the following is not a natural disaster that causes resettlement?
 - a) Cyclones
 - b) Earthquakes
 - c) Development projects
 - d) Tsunamis

5. DISASTER MANAGEMENT

Disaster management is a methodology employed to deal with the impacts of a disaster.

Disaster management is a process that defines:

- how we prepare for a disaster
- how we respond to a disaster
- how we learn from these disastrous happenings.

The main causes of disasters are mostly natural though there have been many cases of disasters that have human origins. A disaster is a combination of a hazard, the vulnerability of the people involved, and their incompetence in the reduction of the negative impacts of the hazard. Mathematically we can depict it as:

Disaster = (Vulnerability + Hazard)/Capacity, where:

- **Hazard:** A dangerous event or condition that has the potential to damage lives, properties, or the environment.
- **Vulnerability:** The extent to which a community or geographical area is affected by a hazard. Vulnerability depends on the construction of an area, proximity to hazardous terrains, etc.
- **Capacity:** The resources or means that are present within households and communities to prevent, cope with, and mitigate the negative effects of a hazard.

The first people who respond to a disaster in an area are the ones who live in that local community. They are the first to start rescue and relief operations. Therefore, it is important for societies to focus on preparing communities for a potential disaster. This will reduce their vulnerability and strengthen their capacities to overcome a disastrous event.

5.1 Types of Disasters

Disasters can be caused by nature, could be human-induced, pandemic emergencies, or could be a result of multiple hazards. We will now look into the details of each of these types of disasters.

STUDY NOTE

The International federation advocates with international organisations, governments, and humanitarian donors to have better practices and accountability in managing a disaster, which shows greater respect for the affected and vulnerable people.

- **Natural disasters:** A natural disaster could be any rapid or slow onset of a natural or physical phenomenon, which has an immediate impact on human health and can cause deaths and sufferings of various kinds. The UN Office for disaster risk reduction categorises natural disasters based on their intensity or magnitude, speed of their onset, the area or extent to which they cover, and duration of their occurrence. For example, earthquakes have a short duration and cover small areas, whereas droughts develop slowly but take time to vanish and affect huge areas. The different types of natural disasters are:
 - Geophysical (earthquakes, tsunamis, landslides, and volcanic eruptions)
 - Climatological (droughts and wildfires)
 - Biological (disease epidemics, insect or animal plagues)
 - Hydrological (avalanches and floods)
 - Meteorological (storms or wave surges and cyclones)
- **Man-made disasters:** Disasters that occur close to a human settlement, which is often caused by technological or environmental emergencies are man-made disasters. They include:
 - Accidents (industrial, transport that involves production, technological, using or transporting hazardous materials, etc.)
 - Environmental pollution and degradation
- **Complex emergencies:** There are disasters caused by a hoard of hazards that are a combination of man-made and natural events. This results in authority breakdown, looting, attacks on strategic locations, wars, etc. Reasons for such disasters could be food insecurities, armed conflicts, epidemics, displaced populations, etc. As per the International Committee for the Red Cross (ICRC), such emergencies are often accompanied by:
 - extensive violence
 - loss of lives
 - need for large-scale humanitarian support, which can require multiple agencies
 - A threat to the security of humanitarian relief workers
 - Constraints (political or military) that restrict or prevent humanitarian assistance.
- **Pandemic emergencies:** Pandemic is derived from two Greek words, 'Pan', which means 'all' and 'demos', which means 'people'. A pandemic occurs when an unknown disease

spreads and covers a large population of an area (covering human and animal populations), which affects their health and disrupts economic and social activities. The disease might have existed for a while in a specific region, but a pandemic happens when there is an unforeseen and abnormal increase in cases of such a disease. A certain region could be free from a disease but, in a pandemic, such a region will also see a sudden appearance and rise of that particular disease, which is plaguing other areas. The many pandemics that have occurred in the past include diseases like Ebola, malaria, cholera, avian flu, zika virus, yellow fever, and the recent COVID-19.

5.2 Four Aspects of Disaster Management

There are four key aspects of a disaster that allows for a more organised management of resources and responsibilities.

They are:

- **Disaster prevention:** As per the United Nations Office for Disaster Risk Reduction (UNISDR), disaster prevention is a phenomenon of engaging in those activities that help in preventing the potential adverse effects of a disaster. These actions are taken in advance, which provides protection from the occurrence of a disastrous event. The World Confederation for Physical Therapy (WCPT) says that it is not possible to prevent all disasters but one can reduce the risks to life and mitigate major risks by having a good evacuation plan, good risk management, and better environmental planning and design. In India, the apex body for disaster recovery management is The National Disaster Management Authority (NDMA), which is headed by the Prime Minister of India.
- **Disaster preparedness:** The International Committee of the Red Cross (ICRC) defines disaster preparedness as the measures taken that prepares a community to reduce the effects of a disaster. If the disaster preparedness of a society is improved, it can result in maximum lives being saved and enable an affected population to get back to normalcy soon after a disaster is over. Physical therapy practice management must emphasise enhancing community-based preparedness and management of disasters.

STUDY NOTE

13th October is celebrated as the world disaster recovery day all over the world. It denotes how communities are

- **Disaster response:** Disaster response or relief is focused primarily on the short-term and immediate needs of an affected population. Some of the response actions are the provision of temporary housing and water, which could also extend till the recovery stage. A disaster response's main goal is to rescue survivors from immediate perils and stabilise their physical and emotional conditions. It also includes recovering the dead and restoring essential services like power and water supply. This stage of disaster management requires a coordinated multi-agency response with many relief activities like relocation, rescue, prevention of diseases, provision of food and water, provision of temporary shelter, restoration of essential services like transport and telecommunications.
- **Disaster recovery:** After the initial crisis is over, an affected crowd continues to be vulnerable for a longer period of time. A disaster recovery is that part of a disaster management that goes beyond the immediate relief activities. It includes the following activities:
 - Rebuilding important infrastructures like schools, hospitals, roads, etc.
 - Developing policies and procedures aimed at mitigating similar situations in the future
 - Health care
 - Other development activities such as creating human resources for health

STUDY NOTE

In 5 seconds, avalanches can acquire a speed of 80 miles an hour.

SELF-ASSESSMENT QUESTIONS – 4

13. Which of the following is an example of a geophysical disaster?
- a) Floods
 - b) Droughts
 - c) Avalanches
 - d) Earthquakes
14. What is the difference between a disaster response and a disaster recovery?
- a) A disaster response means providing accommodation while a disaster recovery means providing provisions of basic amenities like water.
 - b) A disaster response is how a society prepares for a disaster and a disaster recovery is a process of using those resources after the disaster has occurred.
 - c) A disaster response is an immediate and short-term response like providing temporary shelters while a disaster recovery is a process after the initial response is over like building schools and hospitals.
 - d) Both disaster recoveries and disaster responses are the same processes.
15. The term used to describe a usually unknown disease spreading over the huge population is _____
16. Complex emergencies result from a mix of natural and man-made disasters and are often accompanied by violence. [True/False]



6. ENVIRONMENTAL ETHICS

Environmental ethics is an arm of the environmental philosophy that considers how human ethics are deeply related to the environment. Environmental ethics believe that human beings are as much part of nature as other plants and animals. These ethics talk about the moral obligations that we as human beings have towards the environment we live in.

STUDY NOTE

There are few other philosophies around environmental ethics and one of them is Psychocentrism, which believes that human beings are the most valuable species in the environment since their mental capacities are better developed and more complex.

When we think of the environment, we think of the current degradation like global warming and global rising sea levels. We must understand that these issues are strongly linked with our activities, which cause air pollution and depletion of natural resources like forests, water, etc.

Humans have now (in the past few decades) started understanding how important it is for our environment to be sustainable and started studying the ethical and moral relationships that integrate an individual with the environment. There are sustainability specialists, environmentalists, conservationists, and geoscientists who spend a major part of their careers preserving the non-human world.

Experts started finding a connection between technology, commercial industries, global population growth, and their impact on the quality of the environment. If people do not value the environment, they will never get involved with various movements like preservation and conservation. Some of the basics of environmental ethics are:

- Humans and other organisms on earth are interdependent.
- Human beings are not superior to other species.
- Humans should not harm any natural entity.
- We should not try to change, control, or manipulate our ecosystem.

6.1 Environmental and Ethical Philosophies

The purpose of sustainability might be still unclear to those who are not familiar with environmental ethics. Many approaches and principles have been invented in order to determine how we should value our environment.

In order to understand what environmental ethics is all about, we must look into the different philosophies which are related to it.

- **Conservation ethics and conservatism:** Conservation is an activity of protecting and safeguarding what is available in the environment. Conservation depends on how much respect humans have for their environment. It works towards protecting available resources for the future and improving the ecosystem and human communities. This ethics recognises the ongoing relationship of nature with humans.
- **Anthropocentrism:** This is in contrast to the conservation ethics and believes that humans are the most important species in the world. Since this philosophy does not put other plants and animals on the same level as humans in the hierarchy, it allows humans to exploit nature for their own benefits. There are two distinct types of anthropocentrism given by Peter Vardy:
 - **Strong Anthropocentrism:** This holds that human beings are the centre point of the world and most important since they rightfully deserve to be there.
 - **Weak Anthropocentrism:** This holds that human beings are most important since environmental situations can be interpreted only through human perspectives.
- **Libertarian extension philosophies:** Libertarian extension philosophies are a complete opposite to Anthropocentrism and extend the same importance to animals and the environment as given to humans.
- **Holism:** Jan Smuts first developed the term holism in his book 'Holism and Evolution'. According to the holism theory, the environment is a whole system rather than having many individual parts like humans, plants, animals, etc.

STUDY NOTE

The word 'Ecofeminis' was coined by the French writer Françoise d'Eaubonne in her book 'Le Féminisme ou la Mort'.

6.2 Types of Environmental Ethics

The disclosure of various philosophies around environments has given rise to different environmental ethics. There are primarily three types of environmental ethics:

- **Social Ecology:** This studies people and their environment and how they are interdependent.
- **Deep Ecology:** This promotes an intrinsic value and worth of every living being, regardless of whether they have any instrumental utility to human beings.

- **Ecofeminism:** This type of ethics considers the Earth as a strong woman with the same qualities of a woman, so that we can relate and respect it.

SELF-ASSESSMENT QUESTIONS – 5

17. Environmental ethics is an arm of an environmental philosophy that considers how human ethics are deeply related to the _____.
18. Anthropocentrism believes that all living beings hold the same importance in the world. [True/False]
19. The _____ philosophy believes that the environment should be considered as a whole and not in parts.
20. Which of the following choices best describe the Libertarian extension philosophies?
 - a) It studies how people and the environment are interdependent.
 - b) It considers human beings as the most important part of the entire ecosystem.
 - c) It considers the earth as a woman so that it can be respected better.
 - d) It considers all parts of the environment (humans, plants, animals, etc.) to have equal importance.

7. ENVIRONMENTAL COMMUNICATION AND PUBLIC AWARENESS

Environmental communication encompasses all the diverse forms of mediated communication between organisations, interpersonal, public, and different groups. Such communications give rise to social debates about issues pertaining to the environment and human connection to it.

Anyone who participates in discussions about the environment is part of the environmental communication system. It could be passionate environmental advocates or those who fiercely propagate ecological preservation. It could also be a casual activity that anyone can partake in, though it is also a field of practice created by professional environmental communicators.

Environmental communication can also be deemed as an interdisciplinary stream of study that looks at the techniques, influence, and the role of communication in environmental matters. Environmental communication draws its theory from environmental studies,

communication, sociology, psychology, and political science. It is a pragmatic approach to persuade, educate, alert, mobilise, and help solve environmental problems.

Developing an effective environmental message requires communicators to outline the goals that an intended communication wants to achieve, identify stakeholders, and identify key messages and communication methods to disseminate information.

7.1 Why Do We Need Environmental Communication?

If more than 7 billion people are constantly using and abusing our environment, then an apocalypse might come sooner than we think. Some studies suggest that if the current pattern of waste disposal continues, by 2050, there will be 4 billion people living in areas facing severe water-related issues.

One can grow environment sensitivity in a country only through intensive public awareness campaign utilising several tools like the press, electronic media, schools, and colleges.

Public awareness would bring about the following improvements:

- Protection of the atmosphere.
- Management of biotechnology and hazardous waste would become more environment friendly.
- Land resources would be protected.
- Biological diversity will be preserved.
- The living and working conditions of the poor would improve by stopping the degradation of the environment. This would further help in eradicating poverty.

7.2 Important Advocacy Campaigns and Landmarks

Some of the landmarks in the environment advocacy campaigns that have shaped and made an indelible mark on the current environmental movements are briefly discussed below.

STUDY NOTE

Guerrilla gardening is a protest directed at land rights and land reforms. It aims to reclaim land that is either being misused or neglected and promotes reconsideration of land rights and land ownerships.

- **Guerrilla gardening:** In this type of gardening, gardeners create gardens in areas in which they do not have a legal right to do so. They utilise an abandoned area or uncared area to create an 'urban foodscape', which allows foraging. It increases local food security, self-sufficiency, and brings back the food growing skills in urban civilisations. One thing to note is that Guerrilla gardening is illegal by definition and it is a civil trespass against a person who owns the land.
- **Greenpeace campaign:** Such campaigns design campaigns to protect forests.
- **Green consumers:** Green consumers gain information on an enterprise's environmental behaviour and boycott buying products from companies that adversely impact the environment and society. They reward companies that show green policy and approach through their daily shopping preferences.

7.3 Organisations for Climate Movements

Many climate organisations have come up and are striving to keep the environment clean and create a sustainable ecosystem. Some of the famous ones are:

- **350.org:** It is an international organisation that aims to end the era of fossil fuels and build a world that works solely on renewable energy sources. They want to stop any new fossil fuel projects through community resistance and local resolutions. They work by providing support to community-led energy solutions and cutting licenses and stopping finances to fossil fuel industries.
- **Grist.org:** This is a news outlet consisting of a network of innovators. The goal of this organisation is to get people to start talking, thinking, and taking actions to better the ecosystem. They reach out to more than 2 million people in a month out of which 65% start working on guidelines based on the grist.org content. Their publication takes out a daily digest of environmental news and provides a forum for green advocates, and diverse voices and influencers to promote green living.

SELF-ASSESSMENT QUESTIONS – 6

21. Environmental communication is not a necessity since most people are globally aware of issues around the environment. [True/False]
22. _____ boycott buying products from companies that adversely impact the environment and society.
23. 350.org aims to reduce the usage of _____ and promote renewable energy sources.
24. Which of the following is not true about guerrilla gardening?
 - a) Gardeners utilise an abandoned areas or uncared areas to create an 'urban foodscape'.
 - b) It increases local food security.
 - c) It brings back the food-growing skills in urban civilisation.
 - d) It is a legal activity.



8. ENVIRONMENTAL IMPACT ASSESSMENT (EIA)

The Environmental Impact Assessment (EIA) is a tool used by governments across the globe to assess the impacts of a present or a future project or development proposals on the environment as well as the well-being of humans. A project could be a legislative proposal, policy, operational procedure, development project, etc. While undertaking the EIA, many interrelated aspects like socio-economic, human-health, cultural, etc. are taken into account. The EIA looks at both the positive and negative consequences of a project. At the end of an assessment, the EIA also interprets and communicates the information to the decision-makers.

The objectives of the EIA are:

- Recognise, foresee, and analyse the environmental, economic, and social effects of a development activity.
- The EIA technique does a technical assessment that systematically presents the impacts on the environment that a project is likely to cause and explains the significance of impacts predicted.
- As a result of this assessment, there is also an indication of the scope of mitigation or modification to a project.
- The EIA provides information on consequences to the environment for the purpose of decision-making to the concerned ministries and agencies.
- The EIA enhances public participation.
- It fosters inter-agency coordination.

Some features that characterise the EIA and their significance are stated below:

- The EIA is not a procedure for preventing actions, but its main objective is to authorise project actions with full knowledge of how they would impact the environment.
- The EIA is not a science, but it uses an amalgam of sciences in an interdisciplinary and integrated manner.
- The EIA should be treated as an integrated part of a project and not just an appendage or an add-on. The cost of the EIA must be calculated as part of a planning and not as an additional cost.

- The EIA does not give out decisions but is considered while making policies and decisions, which reflect in the final choices.
- The EIA is a government policy, and all industrial projects need an EIA clearance from the Ministry of Environment before they get the approval of the Planning Commission.

8.1 History of Environmental Impact Assessment

The EIA is one of the best policy innovations that happened in the 1900s. The EIA's main goal is to conserve our environment and also bring about the best combination of economic and environmental benefits. EIA started in the US in 1970 and its brief history is presented below:

- Section 102 is the first comprehensive environmental legislation in the US that came into force on the 1st of January 1970. It came in the form of the National environmental policy act (NEPA).
- The EIA was initially a practice only in developed countries but later came to developing countries like India.
- The earliest examples of developing countries that made the EIA a part of their legislation are Columbia (in 1974) and the Philippines (in 1978).
- The World Bank adopted the EIA as a major development project in 1989.
- The Indian EIA notification was amended on the 4th of May 1994 to include a self-explanatory note on how to obtain an environmental authorisation from the Ministry of Environment, Forest, and Climate Change, a list of documents that needed to be submitted to get a clearance, and technical information.

8.2 Contents of Environmental Impact Statement (EIS)

The EIS is a government document that describes the impacts of a future project on the environment around it. An EIS must contain the following data and information:

- The nature and intensity of environmental effects that can be caused by the proposed actions.
- Possibility of cyclones or earthquakes.
- The effects of a project on endangered species, wildlife, and vegetation.
- A project's possible effects on the quality of surface or groundwater, air, and soil qualities.

- It must identify any relevant human concerns related to the proposed actions.
- Demographic and economic factors.
- It must specify if a project could cause noise pollution.
- There must be a section on recycling and how waste products can be reduced in the process.
- Techniques for disaster management and risk analysis.

8.3 EIA Methodology

Whenever a new project is developed, it needs to pass through the EIA, which comprises the following steps:

- **Screening:** In the very first step of the EIA, it is determined if a project under consideration follows the relevant acts and regulations. If it does, then it is further examined whether the project can cause significant environmental disruptions.
- **EIS preparation:** If a project is found to impact the environment, the EIA is undertaken with the preparation of an EIS.
- Many countries have made the EIA open to public scrutiny and they are reviewed at public hearings.
- **Scoping:** At this stage, it is decided which impacts are necessary to be evaluated considering international conventions, legal requirements, expert knowledge, etc. The alternative solutions that reduce the impacts of a project are also considered at this stage. Any alternative design or sites that can mitigate the effects of the project is also scrutinised in this stage.
- **EMP preparation:** The Environmental Management Plan (EMP) is a non-technical briefing of the impacts of a project, which is prepared for the common public.
- **Decision making:** In the last stage, a political decision is taken, which could be any of the following:
 - A proposed development project is accepted.
 - A proposed development project is accepted with a few amendments.
 - The alternative proposed for a project is accepted.

STUDY NOTE

An EIS is an impact statement that needs to accompany an application to get planning permissions, whereas the EIA is a process that is to be carried out by a planning authority before a planning permission is granted.

- A proposed project is rejected.
- Monitoring: An ongoing project is monitored to see whether the mitigation efforts and predicted impact are as per the EMP.

SELF-ASSESSMENT QUESTIONS - 7

25. Which of the following is not true about the EIA?

- a) The EIA technique does a technical assessment that systematically presents a project's impacts on the environment.
- b) An EIA process can indicate the scope of mitigation or modification to a project.
- c) An EIA is a must for every industrial project.
- d) An EIA is not mandatory for every industrial project.

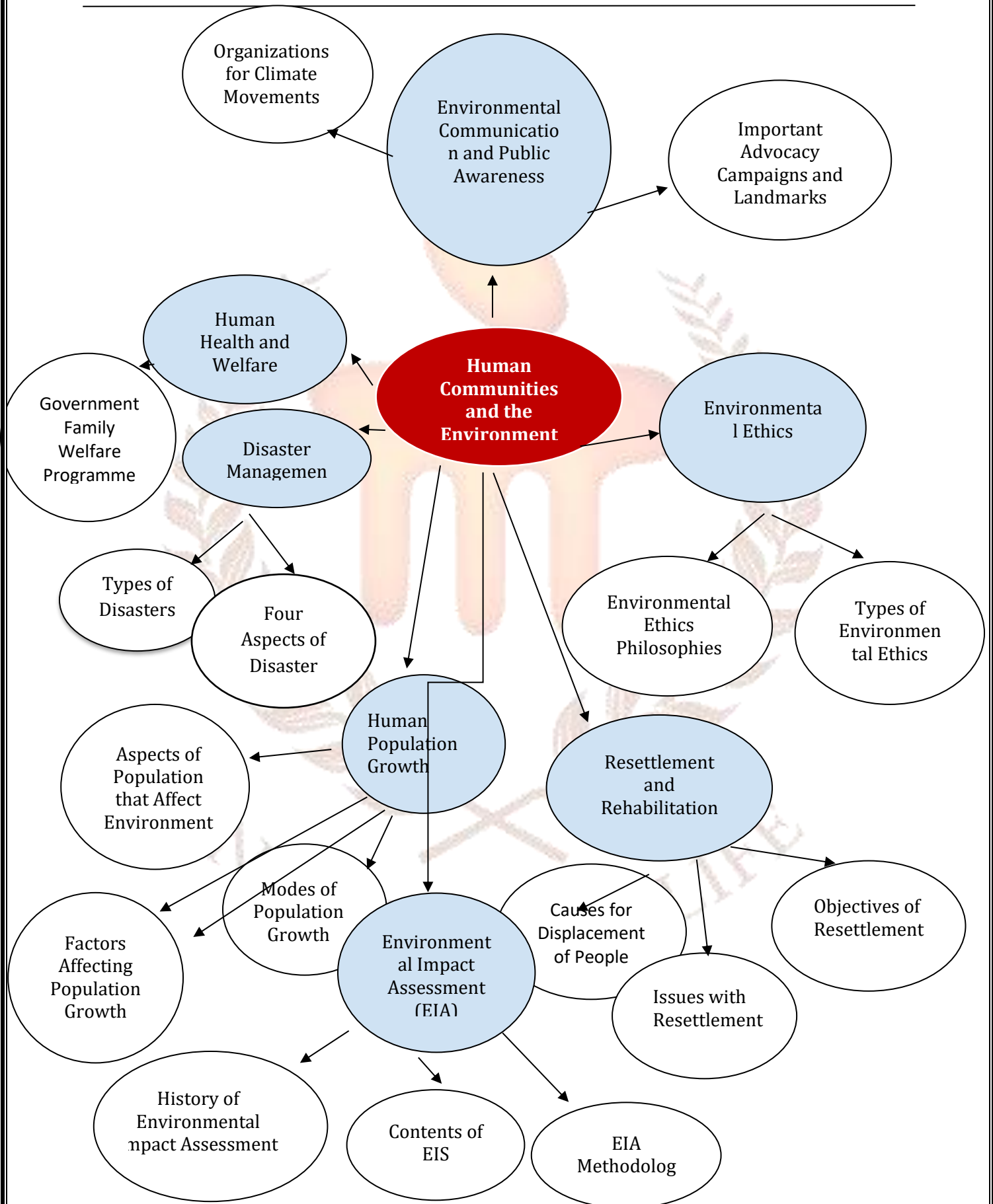
26. India adopted the EIA on _____

27. The EIS is a government document that describes the impact of a proposed project on the environment around it. [True/False]

28. Which of the following is not an EIA guideline?

- a) Timely access is given to all interested parties and stakeholders.
- b) The basis of all assessment decisions should be open and accessible.
- c) The assessment is done with objectivity and complete professionalism.
- d) The EIA process is not flexible.





9. SUMMARY

- A population is used to describe a group of individual species who inhabit a definite and same geographic area at a given time.
- The global population is now growing by more than 80 million in a year and is estimated to grow till the rest of this century.
- The environment gets affected by population growth in mainly two ways:
 1. The resources such as land, fossil fuel, water, air, food, minerals, etc. are consumed in huge quantities.
 2. The massive production of waste products, air and water pollutants, greenhouse gases, and many other toxic materials are released into the environment at a much bigger scale.
- At any given time, a population growth can happen in two ways depending on the resources around.
 - Exponential growth: This type of growth happens when there are unlimited resources. Exponential growth means a great increase in a population's growth rate and the rate of change of population is positively related to the population at that time.
 - Logistic growth: In logistic growth, the rate of change of a population varies with time, and is not a constant. It depicts that as resources start depleting, there is a decline in the rate of growth of the population.
- Overpopulation has many reasons attached to it, some of the most prominent ones are:
 - High birth rate and reduced death rates: With the developments in the world of medicines and technology, the standard of living has significantly improved.
 - Social customs, beliefs, literacy level, and needs: In some parts of the world where the literacy rate is low, people are not well informed about the benefits of having fewer children.
 - Public health programs: Many government health programs have improved the living conditions and life expectancy of people.
 - The TFR or total fertility rate is defined as the number of child a woman can bear in her lifetime.

- The infant mortality rate is the number of children under 1 year dying per 1,000 live births in that particular year.
- Rise in food production: After World War II, there has been a steep rise in food production with the advent of the green revolution.
- The major factors influencing human health are:
 - Malnutrition
 - Ageing Population
 - Migration
 - Easy spread of diseases
 - Water pollution
 - Unemployment
- India was the first country in the world that launched a family planning program in 1951 to restrict the population growth of the country.
- The slogan of this program is 'We two, ours two' (hum do, hamare do). The government utilised all forms of modern as well as traditional communication systems to propagate and develop social support for this program.
- Due to natural and human-caused disasters, people are forced to move out of their original inhabited land. For instance, when a tsunami happened in South Asia in 2004, or the massive earthquake that happened in Latur and Gujrat, and many other incidents of flood or drought in different parts of the world have rendered thousands of people jobless and homeless.
- The process of settling in a new location is termed resettlement and rehabilitation is when their lifestyle restores to the former state.
- Some of the major reasons for resettlements are:
 - Natural disasters: Earthquakes, cyclones, tsunamis, volcanic eruptions, floods, prolonged drought conditions, hurricanes, etc.
 - Industrial accidents (Man-made disasters): Nuclear accidents like the Bhopal gas tragedy, Oil spills (Exxon Valdez oil spill), etc.
 - Development projects like construction of irrigation canals, dams, reservoirs, etc.
 - Infrastructural projects such as the construction of flyovers, roads, bridges, etc.
- There are several issues around resettlements as listed below:

- Lack of support: Displacements mainly affect rural and tribal people who do not get much support from their respective governments.
- Meagre compensation: Many a time there is no compensation given to the land lost by people in the name of mega developmental projects.
- Loss of livelihood: Displacement is a huge phenomenon in the lives of people who get displaced.
- Lack of facilities: There is no basic infrastructure and amenities provided in the new areas offered to the displaced people.
- Identity loss: Tribal people have unique lifestyles, which are community-based. Displacement hampers their culture and spiritual existence in multiple ways.
- Resettlements must consider the comfort and freedom of tribal people while providing them resettlement options.
- Disaster management is a methodology employed to deal with the effects of the aforementioned disaster. Disaster management is a process that defines:
 - how we prepare for a disaster
 - how we respond to a disaster
 - how we learn from these disastrous happenings.
- The main causes of disasters are mostly natural though there have been many cases of disasters that have human origins.
- Disasters can be caused by nature, could be human-induced, pandemic emergencies, or could be a result of multiple hazards.
- Disaster recovery: After the initial crisis is over, an affected crowd continues to be vulnerable for a longer period of time. Disaster recovery is that part of a disaster management that goes beyond the immediate relief activities. It includes activities like rebuilding important infrastructures like schools, hospitals, roads, etc.
- Environmental ethics is an arm of the environmental philosophy that considers how human ethics are deeply related to the environment. Environmental ethics believe that human beings are as much part of nature as other plants and animals.
- Environmental ethics became a reality in the 1970s as around this time experts started finding a connection between technology, commercial industries, global population growth, and their impact on the quality of the environment.

- There are many other philosophies surrounding environment ethics, the main ones being:
 - Anthropocentrism: This is in contrast to environmental and conservation ethics and believes that humans are the most important species in the world.
 - Libertarian extension philosophies: Libertarian extension philosophies are a complete opposite to Anthropocentrism and extend the same importance to animals and the environment as given to humans.
 - Holism: According to the holism theory, the environment is a whole system rather than having many individual parts like humans, plants, animals, etc.
- There are primarily three types of environmental ethics:
 - Social Ecology: This studies people and their environment and how they are interdependent.
 - Deep Ecology: This promotes the intrinsic value and worth of every living being, regardless of whether they have any instrumental utility to human beings.
 - Ecofeminism: This type of ethics considers the Earth as a woman so that we can relate and respect it in a much better way.
- Environmental communication encompasses all the diverse forms of mediated communication between organizations, interpersonal, public, and different groups.
- One can grow environmental sensitivity in a country only through intensive public awareness campaigns utilizing several tools like the press, electronic media, schools, and colleges, etc.
- Public awareness would bring about many improvements like protection of the atmosphere, better management of biotechnology and hazardous waste, etc.
- Environmental Impact Assessment is a process that identifies the impact of a current or proposed project on the environment and the well-being of humans.
- The objectives of the EIA are to recognise, assume, and analyse the environmental, economic, and social effects of a development activity.
- An EIS is an impact statement that needs to be accompanied by an application to get planning permissions, whereas the EIA is a process that is to be carried out by a planning authority before planning permission are granted.

10. GLOSSARY

- **Anthropology:** It is the study of humanity, human behaviour, cultures, and societies.
- **Anthropocene Epoch:** A unit of geological time when in Earth's history (most recent period) human activity had a major impact on earth's ecosystem and climate.
- **Green revolution:** It is the 3rd agricultural revolution that started in the 1960s that increased the agricultural production worldwide. It introduced high-yielding varieties of grains like rice and wheat and the use of pesticides.
- **Greenhouse gases:** Any gas, such as carbon dioxide, which contributes to the greenhouse effect when released into the atmosphere.
- **Ozone layer depletion:** It is the lowering of the ozone in the Earth's atmosphere. The depletion of the ozone layer increases the amount of harmful UVB rays that reach the Earth's surface.
- **Subsistence level:** A way of living where a wage can provide only the bare minimum necessities of life.

11. CASE STUDY

Communication of Environmental Problems in Delhi - CNG Vehicles in Delhi

One can learn about exposure to ecological problems only when people of that area effectively communicate. Effective communication means selling environmental conservation concepts via social communication with different set of people from diverse backgrounds. With this idea in mind, communication efforts have been examined regarding environmental problems in Kerala.

The magnitude of these numbers has resulted in encroachment in forests, wetlands being reclaimed for food production, etc. Due to the huge number of raw materials being consumed, there is dwindling on non-renewable resources and accumulation of waste.

In recent years, all these problems have become the entertaining media for grabbing the public attention. One of the most commonly cited environmental problems of this area is pollution in Delhi and to reduce air pollution a report on the use of CNG vehicles in Delhi is presented below which has come through reports of Centre for Atmospheric Sciences, Indian Institute of Technology.

For this case study, we consider the implementation of CNG vehicles in Delhi. Delhi is considered as one of the highly polluted cities based on many known and unknown parameters. Delhi has a large quantity of suspended particles in the air, according to the White Paper of the Ministry of Environment and forests. To deal with the issue of air pollution, the Supreme Court directed the DPCC to present a list of industries that are located in non-conforming areas.

The DPCC presented a report with approximately 97,600 industries in 1977. The relocation or closure of these industries was difficult, time-consuming, and costly. Still, many of them are relocated, and many of them are closed, but the problem of air pollution still exists due to the inefficient use of vehicles, which generates air pollution in a large amount.

The Supreme Court had decided to improve the transportation system due to which it supplemented various vehicles like autos, taxis, busses, and so on, with new CNG-run vehicles that generate less pollution,

After the implementation of CNG vehicles in Delhi, the report of the Centre for Science and Environment (CSE, 2001) concluded that the use of CNG gas for various vehicles like low-duty engines and high-duty engines will have numerous beneficial impacts on the environment.

CNG is the composition of methane, ethane carbon dioxide, propane, i-butane, i-pentane, n-pentane, nitrogen, and n-butane. CNG vehicles are considered as more environmental-friendly than fuels like diesel. When comparing the reduction in the pollution emission between CNG and diesel, it can be seen that CNG reduces pollution by a greater percentage, such as it reduces more carbon dioxide than diesel by 56%, reduces hydrocarbons by 55%, reduces particulates by 86%, and reduces NO_x by 56%.

As a result, CNG vehicles in Delhi reduce vehicle pollution to half, due to which the use of CNG should be encouraged in public as well as private vehicles.

Discussion Questions:

1. Discuss the environmental problems in Delhi before the implementation of CNG vehicles.
2. Discuss the advantages and disadvantages of CNG vehicles in Delhi.

12. TERMINAL QUESTIONS

SHORT ANSWER QUESTIONS

- Q1. Mention some of the primary factors that affect population growth rate.
- Q2. Explain how the growth of a population affects our environment.
- Q3. What are the characteristics of exponential population growth?
- Q4. How does overpopulation cause malnutrition?
- Q5. What are some of the main objectives of the government family welfare program in India?

LONG ANSWER QUESTIONS

- Q1. What is population distribution and how does it affect the environment?
- Q2. What are the main causes of resettlement?
- Q3. Explain the disaster response stages of a disaster management process.

12.1 Answers

SELF-ASSESSMENT QUESTIONS

- 1. d. The male and female ratio in a population.
- 2. c. A younger population is more likely to migrate to cities for jobs, which will intensify urban environmental concerns.
- 3. Total fertility rate
- 4. True
- 5. True
- 6. Population Control Bill.
- 7. d. The elderly are more likely to suffer from diseases like diabetes, arthritis, heart diseases, dementia, etc., which puts pressure on the healthcare system.
- 8. c. The number of women who die per 100,000 live births, in a given time period.
- 9. True
- 10. d. Resettlement is a process of settling in a new location and rehabilitation is when the lifestyle of displaced people restores to the former state.
- 11. d. Resettled people easily get jobs to restore their livelihood.
- 12. c. Development projects
- 13. d. Earthquakes

14. c. Disaster response is an immediate and short-term response like providing temporary shelters while a disaster recovery is a process after the initial response is over like building schools, hospitals.
15. Pandemic
16. True
17. Environment
18. False
19. Holism
20. d. It considers all parts of the environment (humans, plants, animals, etc.) to have equal importance.
21. False
22. Green consumers
23. Fossil fuels
24. d. It is a legal activity.
25. d. An EIA is not mandatory for every industrial project.
26. 27th January 1994
27. True
28. d. The EIA process is not flexible.

TERMINAL QUESTIONS

SHORT ANSWER QUESTIONS

Answer 1: The primary parameters which affect trends in a population are:

- Immigration: When people arrive in a population from neighbouring regions, it is called immigration.
- Emigration: When people migrate or disperse from their original population area to a new area, it is called emigration.

Answer 2: A population growth affects the environment primarily in two ways:

- The resources such as land, fossil fuel, water, air, food, minerals, etc. are consumed in huge quantities.
- The massive production of waste products, air and water pollutants, greenhouse gases, and many other toxic materials are released into the environment at a much bigger scale.

Answer 3: An exponential growth happens when there are unlimited resources. An exponential growth means a great increase in a population and the rate of change of population is positively related with the population at that time. In nature, a population might grow at an exponential rate but after a while, it gets limited by the availability of resources. In exponential growth, the per capita (i.e., per individual) growth does not change irrespective of the size of a population.

Answer 4: Overpopulation results in the breaking down of the ecological balance due to a population's size exceeding the carrying capacity of its environment. Such a situation results in a weakened food production system. With less food, there is inadequate food consumption and people suffer from malnutrition.

Answer 5: The main objectives of the government family welfare program are to:

- Reduce fertility to slow down population explosion. The program aims to bring down the TFR to 2.1.
- Provide clinical and educational services to remote villages to help couples decide and choose how many children they wanted to have, and when.
- The program also provides information on how to manage birth spacing, control birth, and proper healthcare for a mother and her newborn.

LONG ANSWER QUESTIONS

Answer 1: The manner in which population is distributed also affects the ecosystem. For instance, the birth rates in developing countries are more than that in the developed countries due to less access to education and family planning as well as high poverty levels. This fast growth of developing countries adds more pressure on the environmental resources. As per 2015 data, more than 80% of the world's population lives in developing countries.

Another reason is that every country is getting more urbanised. As opposed to less than 30% of the population living in the cities in the 1960s, we now have almost 54% of the world population living in urban areas. This leads to more slums and centralised pollution, which are not effectively regulated. Modern luxuries like additional cars, heaters, etc. add to the localised environmental issues.

Answer 2: In the past, some of the most prominent reasons for resettlements and rehabilitation have been:

- Natural disaster: Earthquakes, cyclones, tsunamis, volcanic eruptions, floods, prolonged drought conditions, hurricanes, etc.
- Industrial accidents (Man-made disasters): Nuclear accidents like the Bhopal gas tragedy, Oil spills (Exxon Valdez oil spill), etc.
- People relocate while searching for better job prospects.
- Development projects like construction of irrigation canals, dams, reservoirs, etc.
- Infrastructural projects such as the construction of flyovers, roads, bridges, etc.
- Transportation activities like highways, roads, etc.
- Energy-related projects such as oil explorations, power plants, mining activities, pipelines (for example HBJ pipeline), etc.
- Projects that are linked with conserving wildlife like national parks, biosphere reserves, sanctuaries, etc.
- Agricultural projects.

Answer 3: Disaster response or relief is focussed primarily on the short-term and immediate needs of an affected population. A disaster response's main goal is to rescue survivors from immediate perils and stabilise their physical and emotional conditions. It also includes recovering the dead and restoring essential services like power and water supply. This stage of disaster management requires a coordinated multi-agency response with many relief activities like relocation, rescue, prevention of diseases, provision of food and water, provision of temporary shelter, repairment of essential services like transport and telecommunications.

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BACHELOR OF COMPUTER APPLICATIONS

SEMESTER 1

ENVIRONMENTAL SCIENCE

Unit 1

Introduction to Environmental Science

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1. INTRODUCTION

The environment consists of the surroundings where we live, and it is the basic necessity of all living beings. The environment helps an individual to survive and provides natural resources for livelihood. In this chapter, we will learn about the multidisciplinary nature of environmental studies, the importance of understanding the concept, and the ways in which it develops a sustainable environment. In the present times, our environment is facing many issues such as loss of natural resources, depletion of the ozone layer, global warming, etc.

In the multidisciplinary approach, we study the different disciplines that help understand the concept in-depth, i.e., we study this topic from various other points of viewer disciplines. The branches of environmental studies include medicine, agriculture, chemistry, geography, and so on. Environmental studies help understand various environmental changes and how to make a clean, safe, and healthy ecosystem for survival.

As we know, our daily lives are directly linked with the surroundings and the environment. It is our responsibility to take care of the environment and understand the importance of available resources. Environmental study is basic and applied science. It is based not only on scientific principles but also interlinked with various social sciences concepts.

In this chapter, we will study various concepts related to our daily lives, such as making decisions, with the environment as the pivot of all these activities. Used resources in the industries and households are disposed into the environment. The disposal of such items that cannot be reused or recycled in any form will harm the environment, which further causes many health, physical, and environmental problems. Therefore, it is important to know that there is a proper procedure for recycling and decomposing the products that we consume.

Ecology is a widely used word in environmental concepts—the word is derived from the Greek word "oekologue", which means the study of the surroundings. Ecology is a concept that studies organisms, animals, plants, and their relationships.

In environmental studies, we study the ecosystem as a whole, including nature's functional units, such as living organisms that interact with the surrounding physical environment and each other. The ecosystem is very wide and is spread across a large space. This is because it

includes small ponds as well as large forests. The multidisciplinary approach gives a broad view of the ecosystem, that is, the role that every plant, animal, or living organism plays in the environment.

In environmental studies, the ecosystem's structure works to create a final output using inputs such as productivity. This also includes the process of energy transformation, such as in the food chain, nutrient cycling, etc. This ultimately results in the output, i.e., energy loss and degradation. The whole cycle works in a circular flow.

1.1. Learning Objectives

After studying this chapter, you will be able to:

- ❖ *Explain the meaning of the Environment.*
- ❖ *Define Environment Studies*
- ❖ *Explain the scope and importance of Environment Studies*
- ❖ *Describe the concept of Sustainability*
- ❖ *Explain the importance of Sustainable Development*

2. MULTIDISCIPLINARY NATURE OF ENVIRONMENTAL STUDIES

The multidisciplinary approach means understanding one concept from different angles or different points of view to identify factors, causes, solutions, etc., with ease. It is a combination or mixture of different disciplines to provide relevant information on the concept. We can study the environment from different perspectives or various disciplines.

We will study the environment and environmental studies through a multidisciplinary approach. The environment comprises of natural resources that are present in our surroundings. It helps earn a livelihood for some individuals. We will study natural resources and their types in more detail in further sections of this unit. Survival of the human race on this planet is impossible without natural resources.

It is essential to understand the importance of natural resources in the environment and identify the problems related to the exploitation of natural resources that may arise in the near future. A multidisciplinary approach is a wide concept of environmental studies consisting of living and non-living species. We must study the ecosystem from various points of environmental studies. In a multidisciplinary nature, we make decisions to protect our ecosystem from external environmental changes and disturbances.

The ecosystem has a different lifecycle as it works with the help of biogeochemical cycles and mechanisms of energy transfer. It consists of abiotic features in an ecosystem, which are known as non-living organisms such as air, climate, water, etc. It also includes biotic features or living organisms such as plants and animals.

Environmental studies are a multidisciplinary approach which aims at defining the correlation between the human beings and the nature. The study of physics, chemistry, geography and geology talks about the functional and structural organization of the environment. It also reveals the physical characteristics of the environment. The concepts of statistics and computer application are applied for data simulation and performing interpretation of data.

Economics help to develop the knowledge about the social background for growth and development. The policies formulation for the implementation of all the subject knowledge

is done with the help of management studies. Thus, environmental studies can be thought as a multidisciplinary approach.

2.1. Meaning & Definition of Environment

The environment is the outer or external surroundings that help maintain the lifecycles of human beings, animals, and plants. The environment provides us with natural resources that are further used for production activities. Various factors influence and affect the growth of people, animals, and plants in the environment.

The environment also explains the interrelationships among different organisms, factors, etc. For example, in the environment, man is the geological force that helps maintain balance in the environment, reshapes the landscape, and favours some organisms. Humans help protect the natural resources in the environment necessary for survival and growth on the earth.

2.2. Meaning & Definition of Environmental Studies

Environmental studies deal with the key aspects and issues affecting the natural environment, different organisms, and human life. Environmental studies incorporate scientific studies and applied science to seek practical solutions to make everyone's life more sustainable on earth. The basic necessities for human beings' survival are provided by nature, i.e., food, shelter, air, and water.

Environmental studies are systematic studies that observe a human's effect on the environment. In simple words, it studies the effect of the decisions or actions that an individual or organisation takes every day with respect to the environment, and how it affects the surroundings.

Environmental study is a single concept itself, but when we talk about the systematic study, it means involvement of science, which discusses the relationship between human beings and the environment. Science also focuses on the multidisciplinary view. It is made from the different disciplines, and it includes the following fields, as shown below:

- Physics
- Biological science
- Geography

- Philosophy
- Agriculture
- Medicine
- History
- Geology
- Ecology
- Chemistry

Environmental science does not study all concepts. If a researcher or an individual is interested in studying a certain aspect, he will have to link suitable fields as per the requirements. For example, if someone wants to study the effect of pollution on groundwater, he will have to consider the fields of chemistry, biology, geology, etc.

In environmental science, it is known that all organisms have positive or negative effects on the environment. It is also true that the environment is capable of affecting organisms if living beings misuse resources. In society, some people recognize themselves as protectionists or conservationists. These are the people who protect the environment from any harmful activities and focus on sustainable resource utilisation. Environmental science helps create awareness among the public about the protection of natural resources and induces effective decision-making with respect to the use of natural or man-made resources.

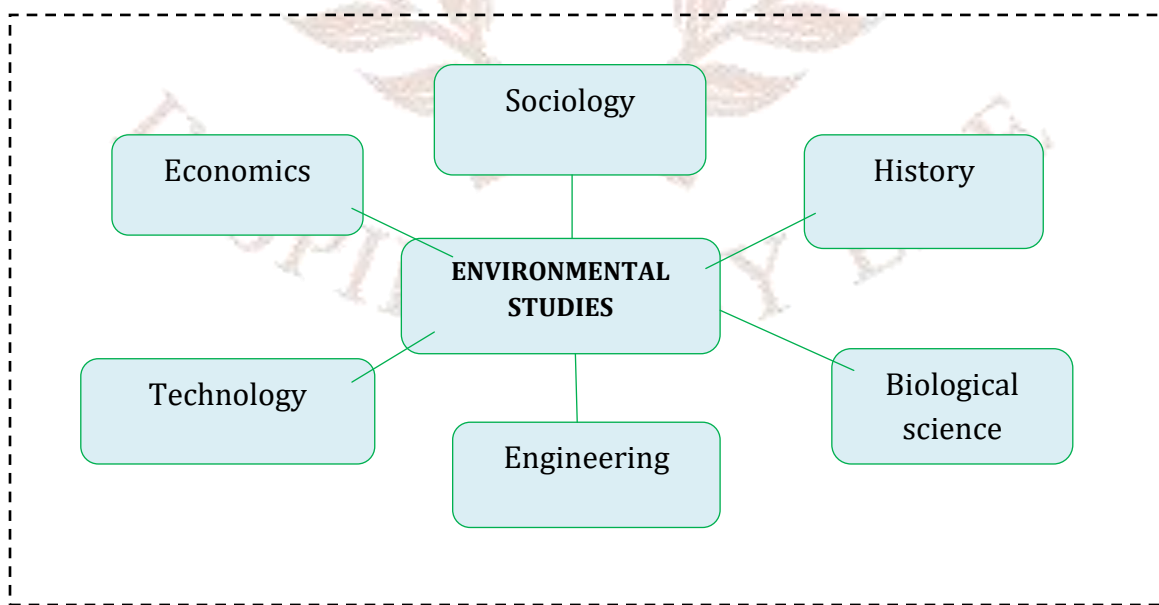


Fig 1: Branches of Environmental studies

2.3. Meaning Of Natural Resources and Types of Natural-Resources

Natural resources are available in the environment and do not involve any human efforts for production. Natural resources include land, air, water, and minerals, which are very scarce in our society. We have to protect them from exploitation as it is impossible to survive in an environment without natural resources, these being the basic necessity for living on the earth.

The industries that produce goods and services for fulfilling society's requirements are using natural resources at a large scale, and in the process, are also depleting them. There are two types of pollutants in the environment, i.e., biodegradable and non-biodegradable pollutants. Bio-degradable pollutants can be easily decomposed in nature, whereas non-biodegradable are those that cannot be decomposed in nature. The government of every country takes necessary actions against the individuals and organizations that participate in the depletion of natural resources. There are two categories of natural resources, as mentioned below:

A. Renewable natural resources: These are the resources that can be used and generated with other natural processes.

For example, a forest is a renewable resource, and if trees from the forest are cut down to produce goods, we can plant new trees and take measures to conserve the forest. Another example is solar energy, which can be produced with the help of solar panels, thus saving electricity costs. Solar energy is generated with the help of sun rays on solar panels.

B. Non-renewable natural resources: These resources cannot be regenerated, and their absence will affect the environment badly. Thus, it is very important for society to procure and safeguard non-renewable resources.

These resources are not available in large quantities as they are limited, and we cannot increase their availability. The two types of non-renewable resources are mentioned below.

1. Recyclable: Example: Aluminium from old cars, Aluminium cans
2. Non-recyclable: Example: Coal, Fossil fuels, Mineral Oil

For example, fossil fuels are available today, but if they get depleted in the future, it will not be possible to reconstitute them or make them available in the environment again.

Natural resources cannot be developed as they are available naturally under the surface, above the surface, and on the surface. Under-the-surface, resources like coal, minerals, etc., are further used to produce industrial equipment. Several countries have a large number of natural resources, i.e., they could be rich in any kind of natural resources such as petroleum, coal, etc.

SELF-ASSESSMENT QUESTIONS – 1

1. Which of the following branches does environmental studies not include?
 - a) Accounts
 - b) Geography
 - c) Technology
 - d) Economics
2. In Recyclable resources we include things which can be recycled such as Aluminum cans, etc. [True/False]
3. Non-renewable resources are available in huge quantity and can be consumed at large scale. [True/False]
4. How will you create awareness about the positive measures needed for the benefit of the environment?
 - a) By creating social groups in society
 - b) By spreading information on social media
 - c) By providing the right guidance to the people
 - d) All of the above
5. _____ is the energy that can be converted with the help of sun rays.
6. Pollutants that are easy to manage and decompose in nature are known as _____.

3. SCOPE OF ENVIRONMENTAL STUDIES

The scope of environmental studies encompasses the aspects related to environmental studies or sciences. Scope enables decision-making and helps in finding a suitable solution against environmental exploitation. We know that the urban and rural population and industries use the resources. However, most of the resources in huge quantities are consumed by the urban population, who are also the least food producers. Urban people generate a large amount of waste after the consumption of resources as compared to rural people. Understanding the scope is equally important for everyone in society. The study has a broad scope and helps identify the environment's cause-and-effect relationships.

- It explains the different types of resources available in the environment, i.e., non-renewable, and renewable resources.
- It also identifies alternative solutions to environmental problems.
- It applies theoretical knowledge to the practical world and helps in understanding multiple uses of the environment.
- It creates awareness among people for sustainable utilisation of the available resources.
- It focuses on developing eco-friendly technologies for the production processes in companies
- It emphasizes more on research and development in the area of resource utilization. Environmentalists focus on promoting a sustainable culture
- It gives information related to ecology and how to protect the species of animals and plants from danger
- It also guides and helps citizens who live and use natural resources from their surroundings to make good choices and appropriate decisions by knowing environmental facts, rules, regulations, etc., to protect the environment
- It also helps in guiding the people about 3 R's in environmental studies, i.e., reduce, reuse and recycle. Everyone needs to understand this for a safe and clean environment

There are four segments that we need to focus on, in order to understand the components of the earth. They are given as follows:

1. **Atmosphere:** It is composed of oxygen and nitrogen and saves all species, including humans, from the hostile environment of outer space. The atmosphere helps to sustain life on the earth. It absorbs the cosmic rays and radiation from the rays of the sun.
2. **Hydrosphere:** This segment consists of all types of water resources that are naturally available to the environment, such as rivers, oceans, lakes, glaciers, seas, etc. In the hydrosphere, 100 percent of the water resources are distributed in the following ways:
 - a. About 97 percent of the earth's supply of water is in the oceans.
 - b. Only 2 percent of water is available and locked in the glaciers.
 - c. 1 percent of water is available as fresh water in rivers, lakes, etc.
3. **Biosphere:** It consists of all the living organisms and their interaction with the surroundings, atmosphere, and other environmental components such as the hydrosphere and lithosphere.
4. **Lithosphere:** It consists of the natural resources available under and above the surface, such as minerals, and it is the outer layer of solid earth. It includes various types of rocks on the earth.

SELF-ASSESSMENT QUESTIONS - 2

7. Which of the following combinations is the best fit for the 3 R's that are concerned with the environment?
 - a) Reduce, reuse, recycle
 - b) Recall, register, read
 - c) Reuse, recall, rate
 - d) Restrict, recycle, read
8. Which of the element includes interaction between surroundings and other environmental components?
 - a. Lithosphere
 - b. Biosphere
 - c. Hydrosphere
 - d. Atmosphere
9. _____ Segment or Sphere of Earth is composed of nitrogen and oxygen.

4. IMPORTANCE OF ENVIRONMENTAL STUDIES

As the environment is very dynamic in nature, we should understand the importance of the environmental study based on the comprehensive view related to the environment system. The understanding helps make citizens to independently take decisions to protect the environment. The citizens who live in the present environment are responsible for positively and negatively impacting the environment. Listed below is the importance of environmental studies.

- As we know, the population is increasing day by day, so we need to control the utilization of resources
- In today's time, resources are exploited to the fullest, and it is important to take action to reserves resources for the future needs
- It is essential to provide education and create awareness about the environment, biodiversity, and ecology. There is a rise in pollution and other harmful activities in the environment, affecting the life of a human being
- The environmental study helps in maintaining the ecological balance between the environment and the organisms, as it makes people aware of our surroundings
- The environmental study also allows the solving of real-time problems by applying methodologies, techniques, and ethical practices in the businesses and by doing CSR activities
- As we know, pollution is a result of the decomposition of waste from various industries, which affects the rivers, air quality, etc. It is important to consider the economic activities or the activities to operate the business and earnings for survival. However, it is important to understand the environmental studies approach to take necessary actions for controlling pollution and preserving the environment from side effects.
- This multidisciplinary approach in environmental studies includes management and awareness that tries to identify how society can make profits without exploiting the environment. As the natural resources get exploited to a large extent for the growth of society's can balance our environmental condition only by using minimum natural resources and by avoiding methods or practices that lead to environmental pollution. So the concepts of geology, economics, and ecology are considered important in the study of the environment

- The environment provides the natural resources required by man for producing goods and services. These resources include land, air, water, food, timber, etc. We have studied that some resources are renewable, and some are non-renewable. Hence, it is important to preserve these resources and save them for future needs. Everyone is aware that most natural resources are scarce. Hence it becomes essential to adopt the optimum utilization of resources

SELF-ASSESSMENT QUESTIONS – 3

10. It is very important to create awareness among people, about the scarcity of resources and for better utilization of resources so that we can preserve resources for fulfilling future needs. [True/False]
11. To understand the concept of Environmental Studies, Geology, Ecology and Economics are not required. [True/False]

5. CONCEPT OF SUSTAINABILITY

Have you ever considered what if all the resources available today will get depleted and not available for use in the future? What if the coming generations would not be able to use these available resources? How will they survive and live their lives without these resources that are the necessities of life?

The population of the world is growing at a constantly increasing rate. The availability of resources is limited. How will upcoming generations be able to use these resources? To resolve this problem, here comes the concept of sustainability. The word sustainability has been derived from the Latin word '*sustinere*,' which means to hold or support. The basic meaning of the word sustainability is something that can be kept continued.

The concept of sustainability depicts that the available resources should be used in the most efficient ways to be saved to the maximum limit for future use. The existence of humankind cannot be possible without the sustainability concept. All the civilizations that existed in history have made use of sustainability and had spread awareness during their time so that the natural resources could be saved for the future to fulfill the upcoming civilizations' needs.

Every civilization is responsible for giving rise to another civilization in the form of human life that cannot survive without having the necessities of the life-like water, food, land, forests, etc. But such natural resources are limited and take a longer time to renew. They may soon get depleted too if not used judiciously. To preserve these resources that are normally an essential part of the people's daily lives for billions of years, these resources need to be taken care of, properly.

To sustain the environment, the environment's ecological balance should not be disturbed so that the resources can take their natural processes to get renewed over a specific time. This can be explained best through the example - If the rate of deforestation keeps on increasing at an alarming rate, there are more possibilities that there will be less rain or no rain in the upcoming years, that in turn will ruin the soil fertility, stop plant growth, dry up natural water bodies, drop the availability of the water on land, make the air drier and polluted, and increase the temperature. Infertile land will not grow crops well.

Evaporated water bodies will confine the accessible amount of water downpour which assists with absorbing the destructive particles in the air. The temperature will increase due to the unavailability of rain that can increase the effect of scorching heat. Such problems will give rise to many disasters like famine, floods, drought, earthquakes, etc., that can be a reason to end human life to a larger extent. This is why we need to study the concept of sustainability, as it helps in utilizing the available resources as per the individual's requirement and sustaining them for future requirements.

5.1. Principles of Sustainability

There are three aspects of sustainability, such as social aspects, environmental aspects, and economic aspects. These are also called pillars of sustainability, which depict that the process of sustainability cannot be implemented without satisfying these three principles. These three principles are interrelated to each other such that if any one of these aspects is not satisfied, the process of sustainability cannot be completed. Let us understand how these pillars affect sustainability.

1. **Social aspect:** The social parameter of sustainability depicts society and its adapted cultures and practices. It emphasizes spreading awareness in society to conserve

natural resources and adopt practices that will encourage others to conserve the environment and be attentive towards the efficient use of resources.

Many laws and legislation are being formulated and implemented to support these conservative practices, non-compliance, which can be considered a severely punishable offence.

This can be explained with an example. Water is required for performing some household chores like dishwashing. In order to conserve water, people are informed to practice water-saving techniques to complete household chores. People are recommended to use practices such as rainwater harvesting, installing hand pumps, etc., and in some countries, people are instructed by law to pay water tax for using water resources. The revenue from such tax collection is used to install water-saving techniques in the country.

2. **Economic aspect:** More often, business firms have access to natural resources for using them in business operations. Business firms make profits by establishing their operations by using these resources. So, to preserve the resources from being exploited by the businesses, certain eco-friendly operations are to be implemented. Businesses are directed to pay the cost so as to use these resources in the form of doing something for environmental conservation. For example, manufacturing businesses are instructed to treat the chemicals that are used in the production of their products before dumping them into the water bodies. Business firms are also directed to pay all the required taxes to use natural resources for commercial purposes in order to comply with all the established laws and legislation.
3. **Environmental aspect:** This aspect depicts the concept of reuse and recycle. The social and economic aspects consist of the environmental concept. Every entity, whether an individual person or a business firm, is formalized to take up eco-friendly practices that will not disturb the ecological balance. Every entity that is deriving any kind of benefit from the environment should contribute to the environment. This, in turn, will add balance to the environment and maintain the same for the longest period. People and business entities are required to use bio-degradable products that will not harm the biological elements such as wildlife and plants on the earth. People using banana leaves

as a plate for consuming food in the Indian state of Kerala can be considered as one of the best eco-friendly practices.

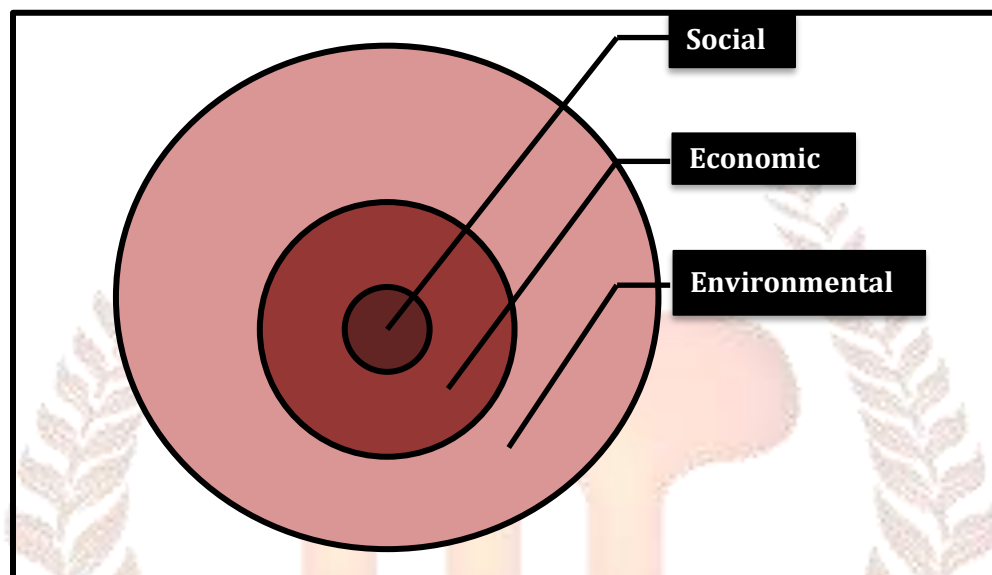


Fig 2: Three Pillars of Sustainability

You must now know how social aspect, economic aspect, and environmental aspects are interrelated to each other. Let's take another example to clearly understand the relationship between these three pillars and how they are interdependent on each other.

Farmers in Rajasthan are in the habit of practicing rainwater harvesting to conserve water in the rainy season so that it can be used for the entire year as Rajasthan receives less rainfall and often encounters a shortage of the water level in the state. So, farmers build multiple cement tanks in their farmyards or keep big drums in their houses that can be filled by rainwater in the rainy season. After the rainy season ends, these tanks and drums are covered for using the stored water over a period of one year. The water is used for household and irrigation purposes. The tanks and drums are fitted with pipelines to provide irrigation facilities to the crop fields for the whole year. Various farmers pool money to construct the tanks as it is costly, and when the cycle revives, they construct more tanks together with the money they receive after selling their crops.

Similarly, hand pumps are fitted in every house to use conserved groundwater in desert areas. Also, the groundwater conservation facilities are fitted with pipelines to use groundwater for irrigation purposes rather than household purposes.

A few more sustainability practices are mentioned below

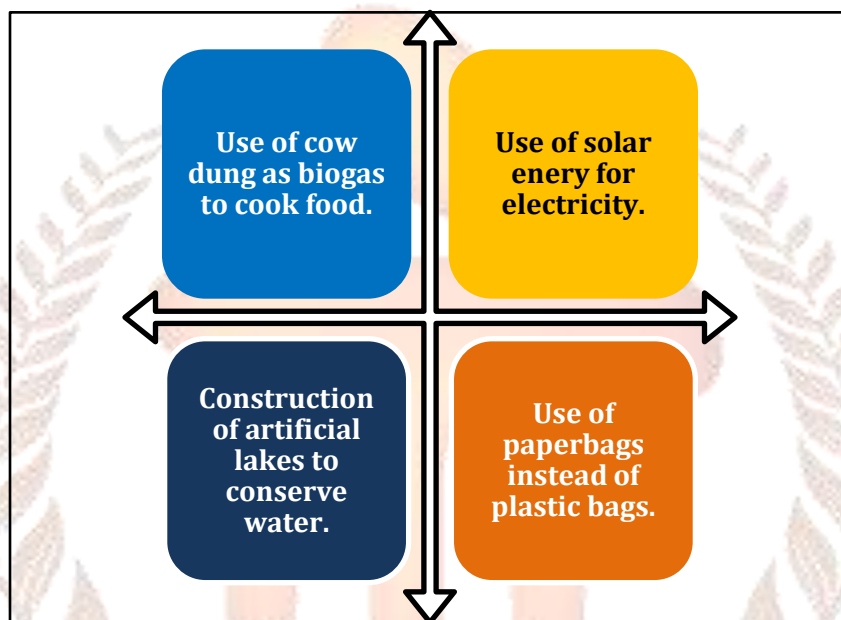


Fig 3: Sustainability in Practice in the Real World

You have studied how resources can be preserved through various traditional and eco-friendly practices rather than modern technology-oriented practices. You must have noted that there were many traditional practices in history, which were successful in using natural resources without any wastage.

The alarming rate of depletion of resources has made some countries strict and aware of environmental resources. Many countries have passed the laws and implemented them strictly to follow the sustainability concept mandatorily. Some of the laws are stated below.

- ❖ Environment Protection Act, 1986, India
- ❖ Environmental Impact Assessment Act, China
- ❖ Environmental Impact Assessment Act, 2000, Austria
- ❖ Air Pollution Control Law, Japan
- ❖ Water(Prevention and Control of Pollution) Act, 1974, India
- ❖ The Forests Act, Kenya

❖ Chemical Act of 2008, Ireland

These laws are meant to protect the various bio-diversities in the environment, non-compliance of which can be considered a severe offense and can attract heavy punishments and penalties. Multinational companies are specially administered to follow these laws and regulations on a mandatory basis. They are supposed to pay various taxes and complete legal procedures to use environmental and societal resources. These taxes and legal procedures ensure that any activity done for the purpose of business would not harm the environment and society. In fact, companies are often told to execute some welfare programs for the environment and society to exchange the resources used for business purposes.

5.2. Sustainable Development

You must have often watched around you the construction of certain heavy infrastructure such as dams, installation of windmills, construction of water tanks at height, the establishment of artificial lakes, installation of solar panels, etc. These infrastructures are built to preserve natural resources and use them for various personal or commercial purposes such as producing electricity and energy, providing irrigation facilities, and many more.

Thus, we can say that infrastructure is developed to assist the sustenance of natural resources. This is called sustainable development. Sustainable development refers to meeting the present human needs by keeping in mind the needs of future generation. Sustainable development ensures that the future generation needs will not be compromised for meeting the present needs. Let us now discuss some of the techniques of sustainable development to understand the concept more clearly.

1. **Use of wind energy:** Wind is considered as a renewable source of energy and is available permanently. The wind is used to produce electricity and power using windmills and wind turbines. These are the heavy pillars installed in the large open grounds where the flow of wind is uninterrupted.
2. **Use of solar energy:** Sun is an unlimited source of energy, and so, solar panels are installed to produce electricity and thermal energy to make it useful for various commercial and personal purposes.

3. **Crop rotation:** It is the practice of planting different crops in the same field to enhance or increase soil fertility for a longer period. Some of the crops are planted as they are good for improving the soil's quality so that other crops planted subsequently can receive the required nutrients and minerals.
4. **Green Space:** Green spaces are created or left for the wildlife and the plants to grow and prosper. There are green fields that are specially created for the flora and fauna so that they can grow and develop without any hindrances.
5. **Rainwater harvesting:** Rainwater harvesting is the practice of collecting and storing rainwater in cemented tanks or in an artificial reservoir so that it can be used later for various purposes. It is very important to implement rainwater harvesting practices as we know there is a shortage of water due to dried lakes, rivers, etc.
6. **Use of CNG gas:** Nowadays, motor vehicles are fitted with compressed natural gas rather than tanks of petrol and other fuels. CNG is also an environment-friendly gas that does not produce pollution, thus reducing air pollution.
7. **Use of biogas:** Earlier, rural areas used biogas for cooking with the help of cow dung and food waste, but now urban areas have also adopted the same practice, reducing the use of liquefied petroleum gas. Biogas is made using natural waste from the environment and is eco-friendly in nature. Today, many households in urban areas are fitted with biogas pipelines.

5.3. Importance of Sustainable Development

Let us now understand the importance of sustainable development:

1. **Eco- friendly:** The techniques used for sustainable development and conservation of nature are eco-friendly and do not introduce any harmful emissions and waste material into the environment.
2. **Limited use of scarce resources:** The concept of sustainable development makes it possible to use scarce natural resources in a limited and restricted form. This promotes the idea of using such resources in the most efficient ways.
3. **Promotion of efficient living:** Sustainable development focuses on implementing the techniques that will make the living standard of urban and rural people efficient enough. It mainly focuses on practicing the lifestyle that is based on saving up the resources and hence, satisfying all the basic needs of the people.

4. **Promotion of a healthy lifestyle:** Sustainable development lays emphasis on using renewable sources of energy for daily life activities. This prevents every person from getting exposed to harmful emissions and production of waste from man-made sources of energy that pollute the environment and hence, saving the people from suffering due to severe diseases.
5. **Maintaining ecological balance:** The use of eco-friendly practices and renewable sources of energy does not hamper the growth and balance of the environment and prevents the occurrence of any disaster. The environment and its elements can take their own natural time to prosper and renew.
6. **Depletion of limited or natural resources:** If we know that the natural resources are available in limited quantity in the environment, it can be substituted by unlimited resources such as human-made resources. Only those resources can be replaced that is possible to do. This will help to reduce the rate of depletion of natural resources. As well as, nature can preserve them for a more extended period.
7. **Up gradation of living standards:** As discussed above, sustainable development lays focus on efficient living and so, it allows human beings to develop through these efficient ways. It establishes the techniques to develop a standard of living and increase the quality of life.
8. **Clean energy and clean water:** The concept of sustainability provides us with clean energy to use, which will not harm us and the planet earth in any way. The methods of sustainability make it possible for us to use clean water that is preserved and purified.

SELF-ASSESSMENT QUESTIONS – 4

12. What refers to the development done for making the natural resources to sustain?
- a) Sustainable development
 - b) Sustaining development
 - c) Infrastructure development
 - d) None of the above
13. What is called the space created for the growth of flora and fauna?
- a) Wide space
 - b) Green space
 - c) White space
 - d) None of the above
14. What do you mean by sustainable development?
- a) Eco-friendly in nature
 - b) Helpful in attaining healthy lifestyle
 - c) Helpful in conserving the natural resources
 - d) All of the above
15. _____ is the process of cutting down of forests.
16. The _____ aspects of Principle of Sustainability talks about using eco-friendly business operations for manufacturing of goods and services and for earning profits in business.
17. Environmental Impact Assessment Act, China is considered as the law to protect biodiversities from exploitation. [True/False]
18. The process of collecting and storing rainwater in cemented tanks or in an artificial reservoir is known as Rainwater harvesting. [True/False]

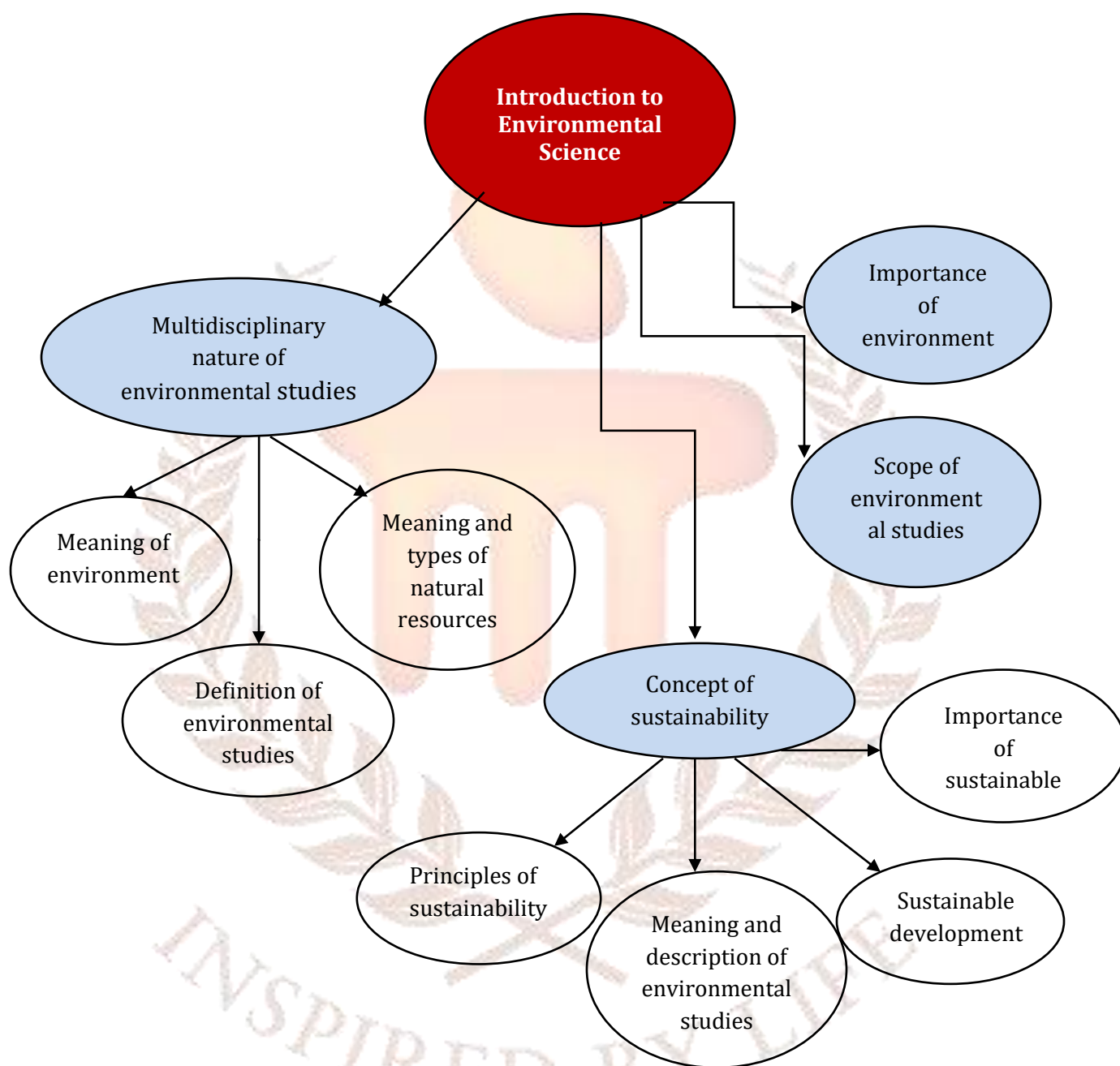


Fig 4: Conceptual Map

6. SUMMARY

- The environment consists of living and non-living things that are important for economic growth. Each and every species, human being, plants have a major role in economic growth.
- Various elements are present in the environment, contributing to the ecosystem and playing a vital role in economic development
- There are two types of resources, renewable and non-renewable, provided to us by nature. Renewable resources such as timber can replenish with natural regrowth, but non-renewable resources cannot replenish and are thus available in a limited quantity
- Various problems, such as over-exploitation, and scarcity of resources, industrialization, pollution, deforestation, etc., affect our environment
- There is a need to create public awareness to stop unethical, illegal, and hazardous activities that damage our environment
- The multidisciplinary approach involves the study of environmental science from the aspect of physics, sociology, geology, history, law, and chemistry
- In a multidisciplinary approach, there are three economic development concepts: environmental pollution and conservation of resources
- The scope of environmental studies focuses on how we can use technology for the purpose of environmental safety
- Sustainability is the process of using the available resources to meet the requirement of the present generation such that it does not compromise the needs of the future generation
- There are three pillars of sustainability—social, economic, and environmental.
- The governments of different countries have implemented various laws that help citizens comply with sustainability
- Multinational companies are specially instructed to follow the laws for the use of natural resources for business purposes
- Sustainable development involves such infrastructure development that the natural resources remain conserved for future use
- Sustainable development can be implemented in various ways, such as the use of wind and solar energy, rainwater harvesting, crop rotation, use of CNG and biogas

- The concept of sustainable development is important in various aspects such as social, economic and environmental
 - ❖ Social aspects talks about creating awareness among people for conserving natural resources, better living standards, etc.
 - ❖ Economic aspects talks about the generation of income, profits for livelihood. It includes activities related to employment, economic growth, research and development.
 - ❖ The environmental aspect talks about the scarcity of resources how we can preserve natural resources for fulfilling future needs. It includes resource management, policies related to the prevention of resources, etc.

7. CASE STUDY

COCA-COLA, INDIA

In 2009, Coca Cola faced severe criticism for exploiting the ground water and natural resources for its plant operations at various remote locations in India. The company allegedly overused the ground water and other natural resources for operating its plants and dumped the wastewater in the agricultural fields. It also dumped toxic wastes near its bottling facilities. All this affected the lives of the poor farmers and the local people by polluting the natural resources and preventing them from using those resources. The depletion of ground water caused water scarcity in the area which led to severe criticism.

To redress these criticisms and allegations, Coca-Cola took many initiatives for implementing sustainability development at the location of its plants. It undertook many practices such as rainwater harvesting and groundwater restoration. It adopted water-neutral methods to minimize the use of water in its operations.

Coca Cola also took initiatives such as recycling and reusing its bottles to prevent the excessive use of natural resources and conserve them. It also started many welfare programmes for the benefit of the local communities near its plants. Coca Cola took these initiatives on the global level to encourage green and eco-friendly strategies.

Source: Article from times of India newspaper, 27 November 2017

Discussion Questions:

1. Discuss all the concerns hampering the environment in the case of Coca Cola.
2. In reference to the Coca Cola case, explain why is it the need of the hour to change the industrial development strategy globally and focus on sustainable socio-economic development?
3. Discuss few other ways by which industries or businesses can be made more responsible for sustainable development of the environment.

8. TERMINAL QUESTIONS**Short Answer Questions**

- Q1. List some natural resources?
- Q2. Differentiate between renewable and non-renewable resources.
- Q3. What is meant by multidisciplinary nature?
- Q4. What is sustainable development?
- Q5. What are the three principles of Sustainability?

Long Answer Questions

- Q1. Explain the scope and importance of environmental studies.
- Q2. List three elements of Sustainability and explain them.
- Q3. Explain the three pillars of Sustainability in your own words.

9. ANSWERS

Self-Assessment Questions

1. A. Accounts
2. True
3. False
4. D. All of the above
5. Solar energy
6. Biodegradable pollutants
7. A. Reduce, reuse, recycle
8. B. Biosphere
9. Atmosphere
10. True
11. False
12. A. Sustainable development
13. B. Greenspace
14. D. All of the above
15. Deforestation
16. Economic
17. True
18. True

Terminal Questions

Short Answer Questions

Answer 1: Natural resources are generated from the environment, i.e., resources that are naturally and freely available such as air, sunlight, minerals, etc. Human beings cannot generate natural resources as we have only one source for natural resources, i.e., the environment. Natural resources are of two types, renewable resources, and non-renewable resources.

Answer 2: Renewable resources can be renewed over time from available resources. Ex: Electricity can be generated from solar panels. Renewable resources can be generated if they are not present in the environment with natural process help such as solar energy, wind

energy, wood, etc. At the same time, non-renewable resources cannot be renewed over time due to the scarcity of resources such as natural gas, oil, and coal. They cannot be generated with natural processes.

Answer 3: Multidisciplinary nature helps understand environmental studies from different perspectives such as philosophy, physics, geography, history, etc. With this help, we can understand the problems and solutions related to environmental problems from various aspects and make decisions accordingly.

Answer 4: Sustainable development is recognized as an organizing principle related to the economic growth that is continuously involved in maintaining the ecosystem and natural resources. It is considered as a development that focuses on current requirements without compromising on future needs.

Answer 5: There are three principles of sustainability, which are as follows:

1. **Social aspect:** The social aspect of sustainability encourages society's people to conserve natural resources. It includes environmental safety and health, use of marketing communication for creating awareness related to natural resources, etc.
2. **Economic aspect:** The economic aspect depicts how the business uses eco-friendly resources to operate its business to earn revenues. Economic elements talk about the financial position of the economy. Businesses should prevent our resources by utilizing them in a better way so that they will not cause any harm to society. In this, it includes giving better product and services to society, Employment opportunity, etc.
3. **Environmental aspect:** The environmental aspect helps use eco-friendly products for business operations, that can be easily recycled and reused. It includes green initiatives, resource management, etc.

Long Answer Questions

Answer 1: The scope and importance of environmental studies are very important to understand, as listed below:

- It helps in creating public awareness among citizens and industries
- As we know, resources are scarce in nature, and their preservation is very important
- It is important to develop eco-friendly products and technologies for safety purposes.

- As the population is increasing, the utilization of resources is also increasing, and we should use resources as per our requirement only and do not waste them.

Answer 2: Sustainability is described as an ability to endure continually. It concentrates on the needs or requirements of the present generation without disturbing the future generation's needs. The three elements of sustainability are:

- **Environmental:** the environmental element is also informally recognized as the planet. The environmental element introduces the concept of recycling and reuse. It assists businesses or individuals to utilize the eco-friendly practices that later recycle and reuse.
- **Social:** The social element is also recognized as people. It involves spreading information in society regarding how to preserve natural resources.
- **Economic:** Economic element of sustainability is also known as profit. It emphasizes the business to use eco-friendly procedures and resources which help in protecting the environment. It also helps cut-off extra expenses and keeps nature happy.

Answer 3: The three pillars of sustainability are social, economic, and environmental

1. **Social Pillar:** Social pillar of sustainability involves spreading awareness among the society and culture to prevent natural resources. It also encourages people to use the natural resources attentively that do not harm the environment or surrounding. To protect the environment, many laws, rules, and regulations are imposed by the government.
2. **Economic Pillar:** Many companies or businesses utilize natural resources in their production and operations in today's era. The economic pillar suggests that the industries use eco-friendly resources, so they are later recycled or reused, reducing production costs and protecting the environment.
3. **Environmental Pillar:** The environmental pillar introduces the theory of recycling and reuse. Environmental pillars highlight the use of eco-friendly resources to reduce the expense and emphasizes other people to adopt eco-friendly practices.

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BACHELOR OF COMPUTER APPLICATIONS

SEMESTER 1

ENVIRONMENTAL SCIENCE

Unit 2

Ecosystems

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1. INTRODUCTION

In the present times, due to progress, our world is facing various problems such as loss of natural resources, ozone layer depletion, global warming, etc. In the multidisciplinary approach, we study the different concepts that help to understand the concept in-depth, which means we study smaller units under the ecosystem from various other points or disciplines to understand the ecosystem from a broader aspect. Environmental studies help understand various environmental phenomena that occur naturally within the ecosystem.

As we know, our day-to-day activities are directly associated with the surroundings and the organisms in the ecosystem. It is our responsibility to respect each organism in the ecosystem and take care of the environment. Environmental study is basic and applied science. It mostly talks about the social sciences. In this chapter, we will study various concepts related to the structure and functioning of ecosystems, human life processes, food chain, etc. There are many organisms in the environment that interact with each other, which comprises the ecosystem. Humans should make a conscious effort to ensure not to disturb the rhythm of the ecosystem due to our development. Therefore, it is important to know that there should be proper sustainable development to maintain the balance of the ecosystem.

Ecology is a word that is widely used in environmental concepts; the word is derived from the Greek word "oekologue", which means the study of the surroundings. Ecology is a concept that studies organisms, animals, plants, and their relationships.

In environmental studies, we study the ecosystem as a whole including nature's functional units such as living organisms that interact with the surrounding physical environment and with each other. The ecosystem is very wide and is spread across a large space. The multidisciplinary approach gives a broad view of the ecosystem, that is, the role that each and every plant, animal or any other living organism plays in the environment. This also includes the flow of energy, such as in the food chain, nutrient cycling, etc. This ultimately results in the harmony of the energy. The whole cycle works in a circular flow.

In this unit, let's look at some factors that cause the degradation in land resources. We will also see some of the powerful renewable energies, which are:

- **Hydropower:** This energy is generated by converting the mechanical energy of flowing water into electrical energy.
- **Geothermal energy:** This energy comes from the heat generated in the deep core of Earth.
- **Wind energy:** The turning of wind turbines generates wind energy.
- **Solar energy:** Sun's radiation can be converted to electricity using photovoltaic cells.

1.1 Learning Objectives

After studying this chapter, you will be able to:

- ❖ *Explain the meaning of the ecosystem.*
- ❖ *Explain the structure and function of ecosystem.*
- ❖ *Describe the concept of food chains and food webs.*
- ❖ *Explain the importance of food chains and food webs in the environment.*
- ❖ *Describe how deforestation and other human activities have caused degradation in our environment*

2. ECOSYSTEM

We will study the environment and environmental studies through a multidisciplinary approach. The environment comprises of different organisms that are present in our surroundings. It helps to establish a relation between many organisms. We will study the structure and function of the ecosystem and the different interactions occurring within them in further sections. Survival of any organism is impossible in isolation. It is essential to understand the importance of interdependence in the environment and identify the problems that may arise related to the ecosystem's disturbance in the near future. A multidisciplinary approach is a very wide concept of environmental studies consisting of living and non-living species.

STUDY NOTE

The incentives to promote a greener environment are proving to be a success. In 2018, as per EIA (U.S energy information administration), renewable energy produced was close to 11.5 quadrillion Btu (British thermal units).

We have to study the ecosystem in every aspect of environmental studies. When we talk about the multidisciplinary nature, based on our awareness, we make decisions towards protecting our ecosystem from external environmental changes and disturbance. The ecosystem has a different lifecycle as it works with the help of biogeochemical cycles and mechanisms of energy transfer. It consists of abiotic features in an ecosystem, which includes non-living organisms such as air, climate, water, etc. It also includes biotic features or living organisms such as plants and animals.

The term "ecosystem" refers to an individual, corporation, or other institution recognised as part of a larger structure of which each small component interacts and is interdependent to survive.

2.1. Meaning & Definition of Ecosystem

The natural process occurring in the environment helps in maintaining the lifecycles of human beings, animals, and plants. There are various factors that influence and affect the growth of people, animals, and plants in the environment. The environment also explains the interrelationships among different organisms, factors, etc.

STUDY NOTE

Ecosystem includes understanding the organisms and their interactions that affect the environment.

For example, in the environment, man is the geological force that helps to maintain balance in the environment, reshapes the landscape, and favours some organisms. Humans help to protect the natural species in the environment necessary for survival and growth on the earth.

A selection of species and their physical space is referred to as an ecosystem. The term "ecosystem" refers to the different ways in which an organism communicates with and is affected by its surroundings. The environment combines the "food chain" and "food web" principles, allowing for a deeper understanding of interdependence and consumption relationships. Plants, for example, provide animals with not only food but also shelter, shade, moisture, and other advantages.

While organisms in an ecosystem can compete or prey on one another, the ecosystem principle emphasizes interdependence — one organism's reliance on another or the ecosystem as a whole.

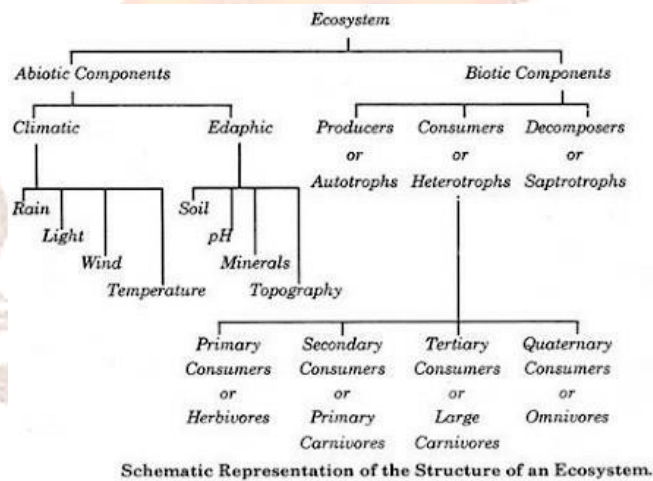
Ecosystems have historically been thought of as economic and social structures. Suppliers, partners, and other users, as well as the market's function and behavior, infrastructure, and social background, make up an "ecosystem". Suppliers and other partners are creating an environment that encourages relationship-building by forming partnerships with other firms that may be our rivals, or any other firm existing in the market. The partnership is formed to take extra advantages in society in terms of economic interactions, growth, and development. There are a plethora of possible economic relationships, just as there are a plethora of possible biological ecosystem relationships.

"ecosystem" is interchangeable with "machine," which refers to social actions. The term "ecosystem" is used to describe the shared internal dependencies of a larger structure, particularly when they apply to a specific component. To understand B's dependency on A in the wider sense, one might claim, "A is part of B's ecosystem." As a result, "ecosystem" is almost a synonym for "climate," but it emphasizes the nature of different facets of the environment rather than seeing it as a whole.

2.2. Meaning of Ecosystem

Ecosystem is a unit made up of all the species (biological factors) in a given region interacting with the physical factors in the ecosystem to create a well-defined trophic (nutrient-requiring) structure, biotic diversity, and material cycles as a result of a flow of energy (i.e., exchange of materials between living and non-living sectors).

The abiotic elements of an ecosystem include rocks, vegetation, air, water, sunshine, and all other non-living elements, while the biotic elements include all of the ecosystem's living members. Two main factors link these elements: the flow of energy within the ecosystem and the cycling of nutrients within the ecosystem. The Sun is the source of sunlight in ecosystems. The sun's energy is passed on to the ecosystem's autotrophic, or self-sustaining species, which use photosynthesis to generate energy in the form of carbohydrates.



Source- unese.campusquotient.org

Fig 1: Ecosystem

2.3. Definition of Ecosystem

The ecosystem is a functional and structural unit of ecology in which living organisms interact with one another and with their physical surroundings. To put it another way, an ecosystem is the product of a series of encounters between the living and non-living organisms and their surroundings.

A.G. Tansley, an English botanist, was the first to coin the word "ecosystem" in 1935. A specified physical area made up of two inseparable components is referred to as an ecosystem.

- **Abiotic:** A living world that has specific physical properties, such as temperature, wind, and humidity.
- **Biotic:** A group of living beings that is constantly in contacts with one another, such as plants, animals, or even small microorganisms and is, therefore, in a situation of mutual dependence.

SELF-ASSESSMENT QUESTIONS – 1

1. Which of the above is not an ecosystem component?
 - a) Micro-organism
 - b) Animals
 - c) Plants
 - d) None of these
2. The physical environment is referred as biotic component of the ecosystem. [True/False]
3. A group of living organisms who are independent on each other and interact for survival in the environment refers to _____ components.
4. The _____ component comprises of factors like humidity, light which has only physical properties.

3. STRUCTURE AND FUNCTION OF ECOSYSTEM

The organisms that make up an ecosystem communicate with one another as well as the physical environments in their respective ecosystems. Clarke says, "The animals and physical characteristics of the landscape shape an ecosystem" (1954).

A.G. Tansley was the first to propose the environment idea (1935). The ecosystem is the most important ecological entity. It is both structurally and functionally sound. The arrangement is designed to represent the species' diversity.

According to E.P. Odum, an ecosystem is the fundamental functional unit of an organism and its environment interacting with one another. The ecosystem's core functions are energy transfer, decomposition, nitrogen cycling, and major biomes.

3.1. Structure of Ecosystem

The structure of an ecosystem comprises two essential components.

1. Abiotic component
2. Biotic component

1. Abiotic components: These include non-living factors which impact the distribution of organisms. This consists of the following:

- (i) Inorganic substances which occur in the form of compounds dissolved in soil or in a free state. For example, carbon, oxygen, nitrogen, calcium, phosphorus, sulphur and micro-nutrients, etc.
- (ii) Organic compounds are the compounds on which all living beings are dependent for their survival. For examples, lipid, proteins, carbohydrate, nucleic acids, vitamins etc.
- (iii) These natural elements, like temperature, light, water, gases, wind, humidity, soil, etc., are included in the abiotic component. They are all physical properties.

2. Biotic components: Biotic components, or biotic factors, can be categorized as any living component which has an effect on another organism or shapes the ecosystem. The animals that consume other organisms present in the ecosystem and the organism consumed for deriving nutrition are included in this component.

(a) Producers: The organisms which produce their own food are autotrophic. They consist of chlorophyll which helps them to perform photosynthesis and make their food. The examples of producers are green plants, fruit, phytoplankton, small shrubs

and algae. There are two categories which are photo-autotrophs and Chemoautotrophs.

- **Photoautotrophs:** These are green plants, Euglena, green sulphur bacteria etc., These are the organisms which perform photosynthesis and prepare food. They control the terrestrial ecosystem. They are being used by all animals and human beings.
- **Chemoautotrophs:** This includes iron, sulphur bacteria which give out energy during a chemical reaction and make the organic food by the process, which is called chemosynthesis. Producers use CO₂ and give away O₂. Hence, they help in maintaining CO₂—O₂ balance in nature. They are also called transducers as they also convert the radiant energy of the sun into chemical energy.

(b) Consumers: Consumers are the organisms which derive energy by eating other organisms. They are divided into Primary, secondary, tertiary and quaternary organisms. They are heterotrophs and cannot produce food. These are of three types.

- **Primary Consumers:** These are also called herbivores and feed on plants and their products, e.g., cattle, rabbit, rat, grasshopper (terrestrial ecosystem) and snails, tadpole, tortoise (aquatic ecosystem). etc. Elton is the key term also known as herbivorous animals which is used for industry animals for primary consumers as they convert plant material into animal material.
- **Secondary Consumers:** Consumer is the organism that derives energy by eating other animal flesh. They are also called carnivores. They mostly eat herbivores. Foxes, cats are called transducers and are examples of the terrestrial ecosystem. The aquatic ecosystem includes water bugs, frogs etc.
- **Tertiary consumers:** They are the animals that eat primary and secondary animals to obtain nutrients. The example of this type of consumer in the ecosystem is large fishes like dolphins, jellyfish, sea lions, etc. Tertiary consumers are also known as apex predators because they are not predated by any other animals.
- **Quaternary:** They are the largest carnivores e.g., lions/tigers which exist on land and whales which exist in water and hence, a part of the aquatic ecosystem. These are also called top carnivores.

(c) Decomposers (Saprotrophs): They are also called micro consumers (reducers) due to their small size. They help in adding inorganic materials to the environment and help in mineralization. They eat the dead bodies and materials of animals and plants

such as animal carcasses, leaf litter, etc. To receive nutrients, it also degrades complex organic substances into simpler compounds. Decomposers are important for preserving the ecosystem's complex existence. They break down bigger items, such as dead animals or trees, into smaller ones.

E.coli, Tapeworm, etc., are called parasites or consumers and get their food from them. Animals like the frog are herbivore when at a tadpole stage and secondary consumers when at an adult stage while, man, ants, etc., are omnivores.

Activity 1

Make a team of 5-6 students in which some are ecologists and some are industrialists. Invite them for a discussion on the climate action project. Let the discussion revolve around encouraging sustainable development education in the schools to make the future generations more considerate about environment. Further, design a speech which will help you to deliver the main agenda behind how to develop ourself but not at the cost of the environment.

3.2. Function of Ecosystem

The functional and life-sustaining environmental structure, which comprises both abiotic and biotic elements, is referred to as an ecosystem. We've also looked at the biotic components, such as living cells, as well as the abiotic components, such as organic matter and energy. In the ecosystem, there are three functional elements.

1. Inorganic constituents
2. Organisms
3. Energy input

To create an environmental structure, the cell, inorganic constituents, and energy input interact with one another. The primary producers in our atmosphere use the sun's energy to turn inorganic constituents into organic components by the process of photosynthesis. The farmers provide energy to the herbivores, and the herbivores provide energy to the carnivores. Animals of various kinds accumulate organic matter in their bodies that is used

as food and are referred to as secondary producers. With the aid of bacteria and fungi, decomposers feed on the decaying organic matter of plants and animals and then complex molecules are further broken down, releasing inorganic components. In this step, some energy is released in the form of heat. As a result, we should classify the following functions as ecosystem functions.

- Production of food by autotrophs, chemoautotrophs
- Utilizing solar energy by green plants
- Recycling minerals
- Ensuring CO₂/O₂ balance ratio in the atmosphere
- Controlling the flow of energy

SELF-ASSESSMENT QUESTIONS – 2

5. The industry animals who convert plant material into animal material are called _____.
6. _____ helps in maintaining the dynamic nature of the ecosystem.
7. The process by which sulfur bacteria makes the organic food is called as _____.
8. The _____ type of consumers in the ecosystem eat the dead bodies of animals and plants.
 - a) Primary
 - b) Secondary
 - c) Quaternary
 - d) Tertiary
9. In ecosystem, for smooth functioning and life-sustaining environment structure, which factors are included?
 - a) Abiotic
 - b) Primary
 - c) Biotic
 - d) Both A and C

4. FLOW OF ENERGY

The capacity to do work is called energy. Potential energy is the energy generated while you are at rest, and kinetic energy is the energy produced when you are moving. Both living beings need a supply of energy, which is produced by the chemical energy of food. The chemical energy is obtained from the sun's radiant energy. In a healthy environment, energy can flow in one direction only. The flow of energy is determined by the law of thermodynamics. The radiant energy produced from the thermonuclear reactions in the sun reaches earth through sunlight (and other forms of radiation) and provides energy to all living organisms.

- Approx 29 percent of the solar energy is available at the topmost or uppermost layer of the atmosphere, which is reflected back to space due to different particles present in the atmosphere. This energy is of no use and has no role in changing climatic condition of Earth's system.
- Approximately 23 percent of solar energy that reaches the earth's atmosphere is absorbed by the dust, water vapor, etc.
- The remaining 48 percent of the solar energy goes through the atmosphere and is consumed by the surface.
- A very small fraction drives the wind and waves.

However, the total solar energy reaching on earth per year is $1,73,000 \times 10^{12}$ watts. From this, only 40×10^{11} watts is utilised in photosynthesis. It is to be noted that both earth's surface and the atmosphere reflect a significant fraction of the solar radiation received from the sun.

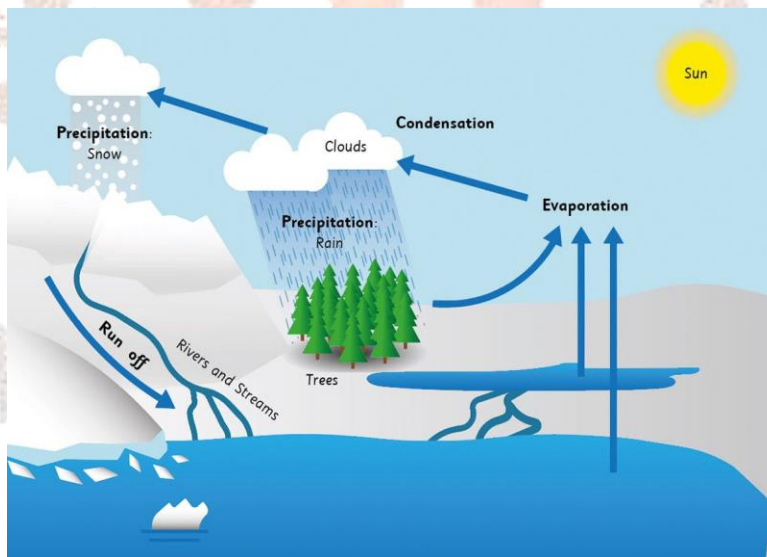
Autotrophs use the energy trapped by the sun for preparing the food through photosynthesis which is then taken by herbivores, and in turn by carnivores and top carnivores.

A large amount of energy is lost during this transfer at each trophic level. Just 10% of energy is moved from one trophic stage to the next. Heat or respiration energy cannot be passed to another organism. Also, the decomposition of dead plants and animals releases chemical energy. In a shorter food chain, more energy is available than in a long food chain. The energy always flows unidirectional, i.e., from autotrophs to herbivores and herbivores to carnivores.



4.1. Water Cycle

When it rains, rainwater runs, sinks, and spills into rivers, or it falls straight into the sea. Any rainwater that falls on land and is collected by the ground, where it is processed for the rest of the year. The roots of the plants suck up the water from the ground along with the nutrients from the soil. This water is evaporated from the leaves and added to the atmosphere as water vapour. Water vapour rises and forms clouds when the air gets light, and when the clouds rise higher, the condensation occurs, the water changes into droplets, which results in rainfall back to the land. However, pollution created by humans is altering rainfall patterns is impacting this endless cycle. Long-term droughts in countries like Africa, on the other hand, will cause flooding in countries like the United States.



Source- encounteredu.com

Fig 2: Water Cycle

4.2. Carbon Cycle

Both the biotic and abiotic aspects of the world contain carbon, which is contained in organic compounds. Both animal and plant tissues use it as a building stone. In the presence of sunshine, plants absorb carbon dioxide from the soil by their leaves. Water is drained by the roots from the soil and mixed with carbon dioxide by the plants. They develop carbohydrates that have carbon in the presence of sunlight, which is called photosynthesis.

Plants grow and develop through this complex process of photosynthesis. Photosynthesis releases oxygen into the atmosphere and is used by animals to breathe. As a result, plants have control over the percentage of oxygen and carbon dioxide in the atmosphere. As a result, all of humanity is dependent on the oxygen created by this cycle. It also assists in keeping consistent CO₂ levels. Herbivorous species get their energy from seeds. Plants and animals both emit carbon dioxide during respiration. They return fixed carbon to the soil in their waste as they excrete. When plants and animals die, they return their fuel to the earth, completing the carbon cycle.

4.3. Oxygen Cycle

During respiration, animals and plants absorb oxygen from the air. Photosynthesis is the process by which plants return oxygen to the atmosphere. Deforestation decreases the amount of oxygen in the atmosphere. As a result, plant life plays an important role in our lives that we sometimes forget. Participation in forestation projects is important for this cause.

4.4. Nitrogen Cycle

Carnivorous species eat herbivorous (plant-eating) animals. When the animals defecate, worms and insects, mainly ants and beetles, decompose the waste. The tiny 'soil animals' break down the waste material into smaller fragments, which are then broken down further into nutrients by microscopic bacteria, which can then be consumed by plants and used for their growth. As a result, nutrients are returned from animals to plants.

In addition, the bodies of dead animals are broken down into nutrients and used by plants for development. As a result, the nitrogen cycle, on which life depends, is complete. Some nitrogen-fixing bacteria and fungi live in the soil, providing this essential nutrient to plants,

which is then consumed as nitrates. They're a part of the plant's metabolism, and they help to make new plant proteins. Herbivore species make use of this.

Different animals get nitrogen through different modes. e.g., when carnivores feed herbivore animals, they get nitrogen from them; likewise when herbivore animals feed on plants, they get nitrogen in the form of protein from plants. As a result, soil species, bacteria, and even fungi play an important role in human survival. When we think about food chains, we also think of huge mammals and other large forms of life. However, we must recognise that the invisible small mammals, trees, and microscopic modes of life are critical to the ecosystem's functioning.

4.5. Energy Cycle

The energy cycle is based on the energy flux within the ecosystem. Plants use the energy from sunlight to produce new plant material, which includes leaves, fruits, bulbs, branches, and roots. They are known as habitat creators because they can expand by directly converting the sun's energy into their tissues. Herbivorous species use the plants as a source of nutrition.

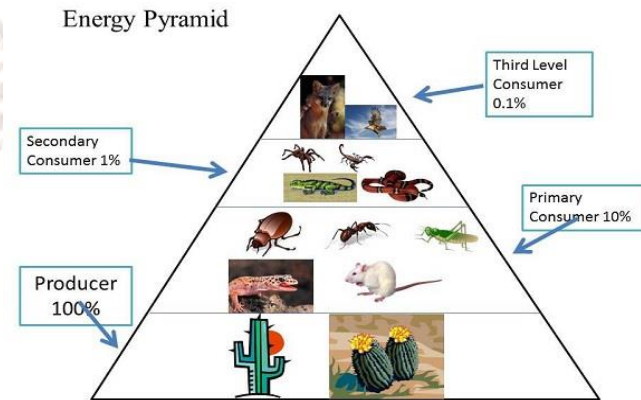
The everyday activities of these species, such as food absorption, breathing, promoting tissue growth, and sustaining body temperature and blood, absorb a significant portion of this energy. Energy is also used for subsistence tasks such as seeking food, finding shelter, breeding, and raising children. Herbivores provide food for carnivores. As a result, food chains connect various plant and animal species.

However, since each animal or plant can be linked to a large number of other animals or plants by multiple linkages, these interconnected chains can be depicted as a complex food network. As a consequence, it is known as the 'web of life,' which means that there are interrelationships in the environment. A food pyramid or an energy pyramid may be used to depict the energy in the ecosystem. The pyramid's base is made up of a vast variety of plants known as growers.

The middle segment of the pyramid, which represents the number and biomass of herbivorous animals known as 'first-order consumers,' becomes smaller. The limited

biomass of carnivorous animals known as 'second-order consumers' is represented at the apex. Humans are one of the creatures at the very top of the pyramid.

To sustain civilization, there should be a broad base of herbivorous animals and a much greater amount of plant material. When animals and plants die, decomposers such as beetles, bacteria, and fungi break down this material into simpler substances, which are then returned to the soil so that plants can consume nutrients from their roots.



Source- tutorialspoint.com

Fig 3: Energy Pyramid

SELF-ASSESSMENT QUESTIONS – 3

10. The energy trapped by sun is used by _____.
11. _____ can be found as organic matter in both biotic and abiotic ecosystem components.
12. To depict the energy present in the eco system we use _____, whose base is made up of growers.
13. The willingness to work is referred to as _____.
14. _____ determines the energy balance within the environment.
15. _____ is also called web of life.
16. What does middle segment of energy pyramid represent?
 - a) Amount of plant material
 - b) Number and biomass of herbivorous animals
 - c) Decomposers
 - d) None of the above

5. FOOD CHAIN

Many of the biotic factors in an ecosystem are linked by food. Herbivores, for example, consume the products created by the farmers. Carnivores consume herbivores as carnivores are the ones who eat only meat, and herbivores eat plants. A food chain is a group of species that transmit food resources via a sequence of repetitive eating and being consumed behaviours.

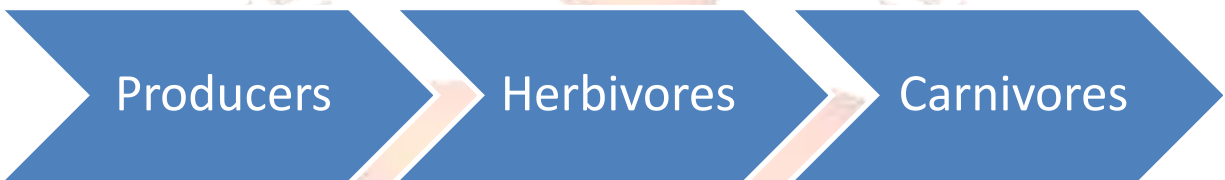


Fig 4: Food Chain

Each level of the food chain is called the trophic level.

There is a continuous flow of energy through the food chain. Energy flow is the transition of energy from one trophic stage to the next. The light, which is the ultimate source of energy, is still at the top of the food chain. It provides the energy needed to keep all biotic life alive. Plants use this energy to carry out the photosynthesis process, which produces their food.

In this biological process, light energy is converted into chemical energy and carried on through successive stages of the food chain. The food chain is the transfer of energy from a plant to a recipient and then to a detritivore or apex predator.

Furthermore, scavengers break down the dead and rotting matter, and organic waste into its constituents, which are then consumed by the reducers. The reducers release molecules into the atmosphere after gaining electricity, which the producers then consume.

Grass Land

Grass land refers to a piece of land that is generally flat and full of grass where small plants and insects like clovers, mouse, snake and hawk live and complete their food chain.

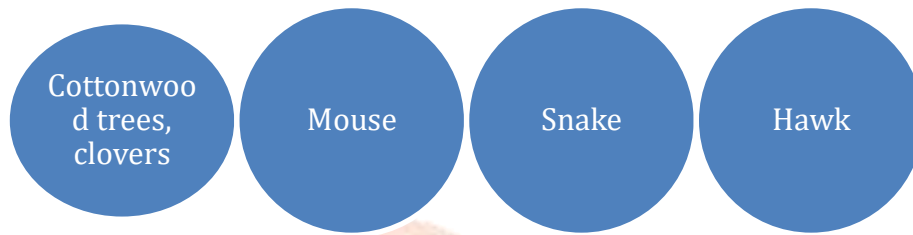


Fig 5: Animals and Plants in Grass Land Examples

Forest

Forest is the area that is generally dominated by trees and wild animal, and it is considered as the predominant terrestrial ecosystem, and the food chain in forest starts from shrubs, ferns, and other trees, which are later consumed by animals like goats and the food chain is completed by big animals like lion.



Fig 6: Animals and Plants in Forest Examples

The most famous types in nature are land-based food chains. The decomposer, which is ubiquitous, is a curious split in the food chain. The decomposers, which include bacteria and maggots as well as the noble cockroach, feed on the dead and break them down into nutrients that serve to sustain the food chain. Through excreting, decomposers send the nutrients of dead species to the soil, which then helps to nourish the plants that restart the chain.

Aquatic food chains are often dynamic, which intrigues land dwellers. But even the aquatic food chains follow a specific pattern. The most famous example is chemosynthesis. It is the process by which food is made by bacteria using chemicals that are utilized as the energy source.

Decomposers play an important role in aquatic decomposers. They disperse nutrients to the soil and the water column, where they feed the plankton that is the foundation of all marine food chains.

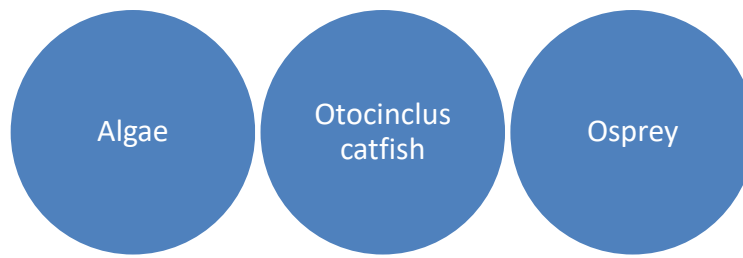


Fig 8: Aquatic Food Chain

5.1. Types of Food Chain

There are two types of food chains which are as follows:

1. Grazing food chain
2. Detritus food chain

1. Grazing Food Chain: Green plants are at the top of this food chain, and carnivores are at the end. This food chain is highly reliant on autotrophs, which derive their nutrition from sunlight. The following diagram depicts the pattern of food chains in the terrestrial ecosystem:

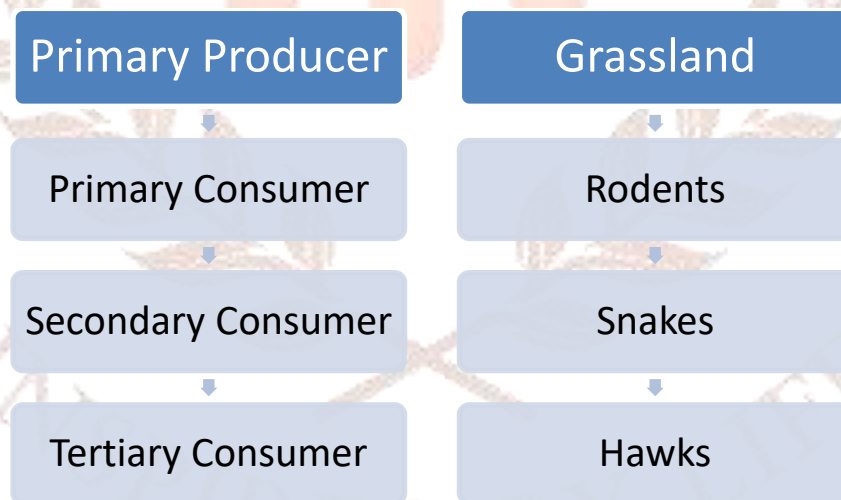


Fig 9: Grazing Food Chain

2. Detritus Food Chain: Detritivores are species that prey on the dead bodies of plants and animals. Bacteria, fungi, protozoa, and other microorganisms are included. The dead organic materials are ingested and digested by these species, which then turn them to water and CO₂.

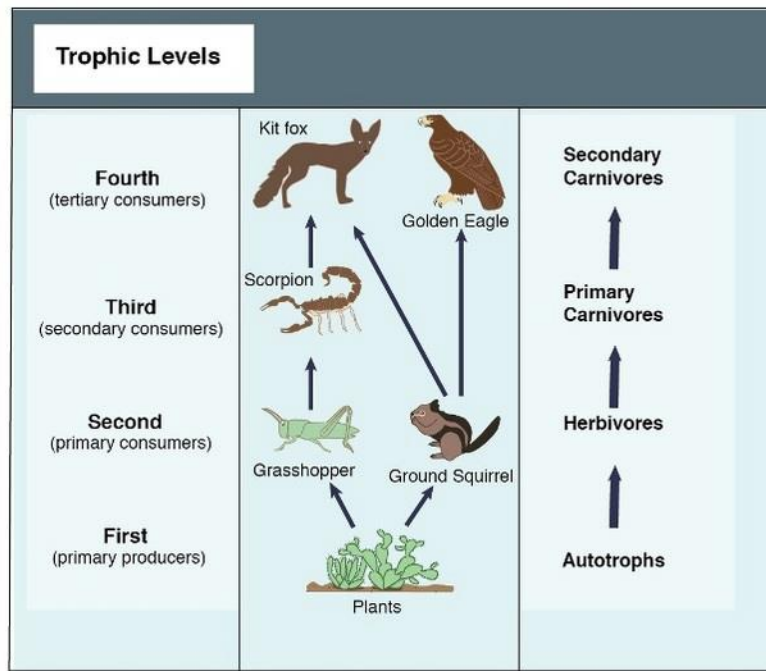
SELF-ASSESSMENT QUESTIONS – 4

17. Each level of food chain is called as _____.
18. The organisms that feed on the dead bodies of plants and animals are called _____.
19. In _____ food chain, carnivores are present at the lower or end stage where as plants are at the top.
20. _____ animals eat only meat and _____ animals eat the products produced by the farmers or eat plants.
 - a) Carnivores, herbivores
 - b) Primary, carnivores
 - c) Herbivores, carnivores
 - d) Herbivores, Primary

6. FOOD WEB AND ECOLOGICAL SUCCESSION

Food webs are a network of intertwined food chains that exist at various trophic stages. Since most species consume more than one form of plant or animal, food chains and food webs are interconnected. Autotrophs, such as trees, are the main food source, as they generate food using solar energy by the mechanism of photosynthesis. In the predatory chain, a herbivore species is eaten by a carnivore animal. A smaller organism eats a fraction of a larger host and may itself be parasitized by smaller organisms in a parasite chain. Microorganisms feed on dead organic matter in a saprophytic chain. The chains do not normally encompass more than four or five trophic levels as the energy is lost at each trophic level as heat. Humans can increase the food supply by eliminating one step in the food chain rather than consuming herbivore, or the people consume the grains themselves. In that case, the total energy available to the final consumers is increased because the food chain is made shorter.

- The vegetation (producers) and fine organic debris (decomposers) constitute the base of a food web.
- Carnivores (secondary consumers) and Herbivores (primary consumers) constitute the higher levels.
- The intermediate level in the food web includes omnivores.
- Food webs are often complicated as different species feed at various levels.



Source- nature.com

Fig 10: A Food Web

Succession is the inevitable mechanism by which a culture is colonised by another group of individuals. Succession is the orderly and gradual substitution of one group by another before a healthy community (known as the climax community) conquers its territories. The various stages of a plant succession that occurs in a certain ecosystem are referred to as sere. Pioneer plants are plants that thrive on the bare ground. After several modifications, the ecosystem becomes dominated by the most tolerant species, forming a climax population.

6.1. Hydrosere

In an aquatic climate, this is a form of succession. In a virgin pond, hydrosere starts with phytoplankton colonisation and progresses to a climax forest level. The numerous stages of succession are depicted below.

- **Phytoplankton stage:** Phytoplanktons and zooplanktons are the pioneer colonizers which begin the succession. These species aid in the addition of a considerable volume of organic matter and nutrients to the bottom layer.
- **Rooted submerged stage:** There is a development of mud which develops at the bottom of the pond after the death and decomposition of phytoplanktons which becomes a new

habitat for rooted hydrophytes like *Myriophyllum*, *Elodea*, *Hydrilla*, *Potamogeton*, *Vallisneria*, *Utricularia*, etc.

- Floating stage: The underwater and floating plants flourish together at first, but the submerged plants are eventually replaced entirely. Physically and environmentally, the habitat can change. Plants that have died have sunk to the ground. In a vertical direction, the substratum will rise much further. *Limnanthemum*, *Nelumbium*, *Trapa*, *Pistia*, and *Nymphaea* are essential plants at this point.
- Reed-swamp stage: This period is also known as the 'amphibious' stage because the plants in this group, *Scirpus*, *Typha*, *Sagittaria*, and *Phragmites*, are rooted but the majority of their shoots are exposed to the air. This plants' rhizomes form dense foliage.
- Sedge meadow stage: The substratum is affected by a further drop in water depth. *Carex*, *Eleocharis*, *Juncus*, and *Cyperus* are some of the *Cyperaceae* and *Gramineae* species that colonise the region to form a swamp or marsh. As a result of the lack of moisture in the region, marshy vegetation eventually fades.
- Woodland stage: The formation of woodland begins with a few shrubs and then grows to include medium-sized trees. The trees like *Butea*, *Acacia*, *Cassia*, *Terminalia*, *Salix*, *Cephalanthus*, and other species of *Butea* dominate the woodland community..
- Forest stage: This is the last succession stage of hydrosere. In this stage, the climax vegetation develops and forms a community, where all plants meet together. Like herbs, shrubs and other plants depict their separate community.

6.2. Xerosere

Xerosere is the plant succession stage that has limited water availability. There are different succession stages of xerosere. The Xerosere begins their life on unweathered rock surfaces. Lichens colonise the primitive form of substratum. In Xerosere, all animals and plants go through a sequence of transformations. The different stages are described below.

- Crustose- lichen stage: There isn't enough soil to reach the roots. The pioneer plants are lichen and blue-green algae. In cooler climates, crustose lichens such as *Rhizocarpon*, *Lecanora*, and *Rinodina* are typical pioneers. They produce acids that cause rock weathering. Dead organic matter from lichen and algae combines with small rock fragments to create a thin layer of wet soil on the rocks.

- Foliose-lichen stage: They survive on a substratum that is partly made up of crustose lichens. It's made up of Parmelia species called Dermatocarpon, which looks like big leaf (foliose: leaflike shape) which is made up of a lichen. The accumulation of a fine thin soil layer on the rock surface as a result of the weathering of rocks with humus results in a change in the habitat.
- Moss-stage: Many xerophytic mosses, such as Grimmia, Tortula, Bryum, Polytrichum, Barbula, and Funaria, benefit from the formation of a thin humus-rich soil layer on the rock surface.
- Herb-stage: The growth of mosses causes more soil to settle. The addition of minerals by leaching has resulted in the growth of herbaceous plants such as Aristida, Festuca, Justicia, Tridax, and others.
- Shrub-stage: Owing to the deposition of dirt, the ecosystem becomes suitable for shrubs. This stage is led by Rhus, Phytocarpus, Zizyphus, and Capparis species. The shrubs also overshadowed the herbaceous plants, resulting in the development of more organic matter.
- Forest-stage: Forest stage represents the final group in Xerosere. The weathering of rocks creates a thin layer of soil that is ideal for small trees such as Acacia, Prosopis, and Boswellia. To get to the climax level, plants need more rainfall. Dense climax forest occurs in humid and rainy climates, as well as in temperate climates.

SELF-ASSESSMENT QUESTIONS – 5

21. _____ and _____ are the pioneer species of crustose-lichen stage.
22. _____ stage witnesses the development of mud under hydrosere.
23. _____ constitutes the base of the food web.
24. _____ is an inevitable phenomenon that occurs when a community is colonised by another group of people.

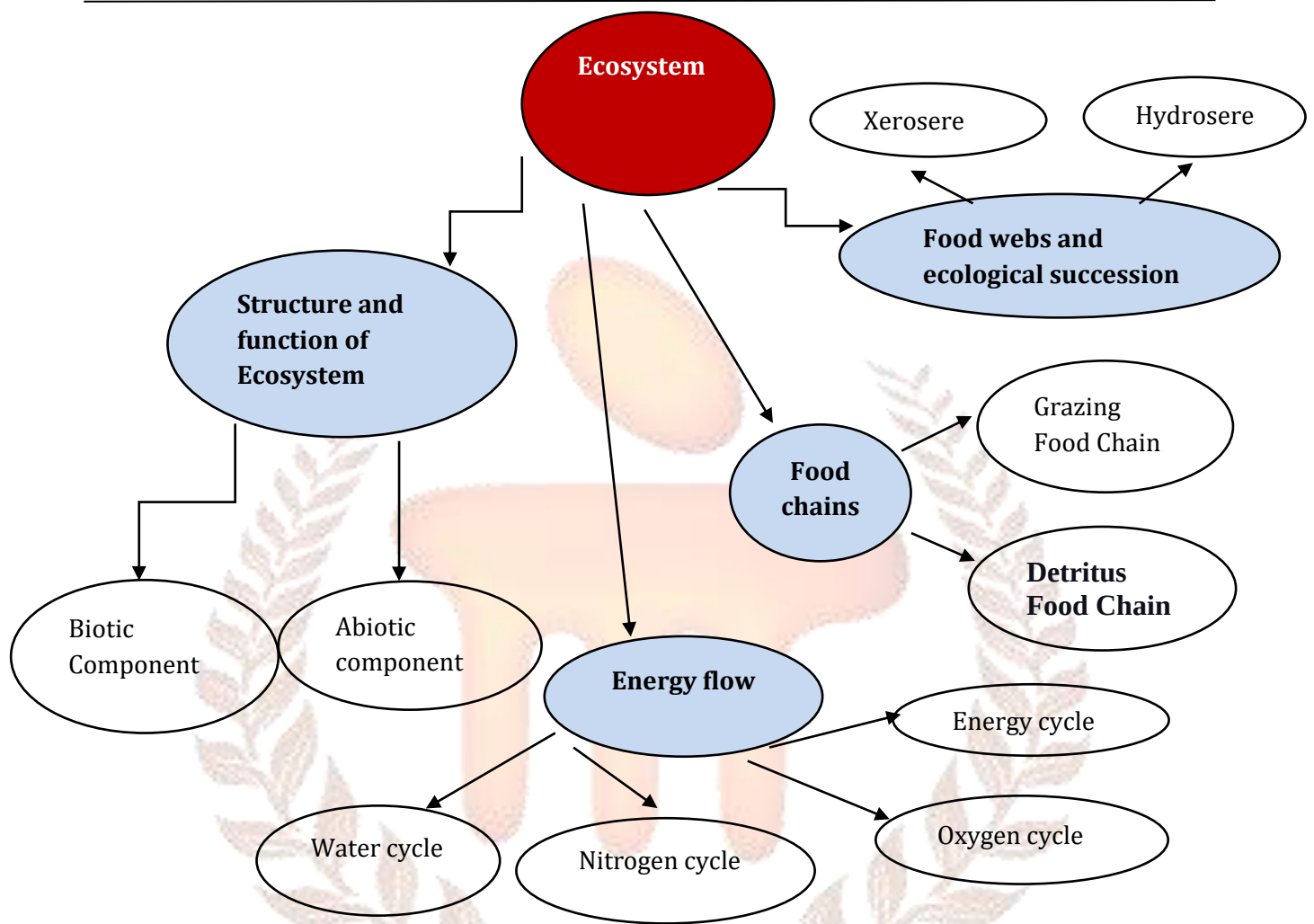


Fig 10: Conceptual Map

7. LAND RESOURCES AND LAND USE CHANGE

The maintenance of all terrestrial ecosystems depends largely on land and its usage by humans. Land as a resource is essential for the survival of humans as well as their prosperity. All the features and processes of land that are used to fulfil various human needs are termed land resources.

To define the land, we can look at its characteristics outlined below:

- The land is an area on the Earth's terrestrial surface that encompasses all the parts of the biosphere immediately above or below it.
- The land also comprises near-surface climate, terrain forms, soil, surface hydrogeology (this includes lakes, rivers, shallow lakes, swamps, and marshes).
- Other resources which come under the category of land resources are near-surface sedimentary layers, associated groundwater, plant and animal population, geohydrological reserves, human settlement patterns.
- The physical results of past and present human activity like terracing, drainage structure, building, roads, and water storage.

Over the years, people have become experts in exploiting land resources and the human demands of land have become an issue since the land resources are finite. The increased pressure on land demand has contributed to the decline in crop production, more competition for land, and a quality and quantity degradation of land resources.

7.1 Functions of Land

Environmental economists define land as a renewable stock resource. Land resources do not fit exactly into the category of renewable or non-renewable resources as they are slowly renewable. But the rate of degradation of land far exceeds its natural regeneration. This means that if a portion of land is lost to degradation, it cannot be regenerated in a human time frame, resulting in the loss of opportunities for future generations. The various functions of land which support human and various other terrestrial ecosystems are summarized below:

STUDY NOTE

Currently, there is a total of 4.4 billion hectares of total world land area, which is available for cropping. The total land area currently being used for cultivation is 1.6 billion hectares out of which 20% of cultivation is happening on marginally suited lands.

- It is a store of wealth for communities, groups, or individuals.
- It is a storehouse of raw materials and minerals for human needs.
- It produces food, fuel, fiber, and many other biotic materials which humans use.
- It regulates the flow and storage of surface water and groundwater.
- It is a physical space where settlements, recreation, and industrial activities take place.
- It stores and protects many historic and prehistoric records (fossils, archaeological remains, evidence of climates in the past, etc.)
- It enables (or hampers) the movement of people, animals, and plants across different areas.

7.2 Land Use

A land cover is the physical land type on the ground surface: vegetation, forest, water, bare soils, etc. Land use is defined by the purpose that a certain land cover fulfils such as recreation, sports, agriculture, etc. In other words, land use is the characterization of a given land based on what can be built on it, or what the land can be used for.

Activities, arrangements, and inputs give the characteristics of land use by humans to produce, alter, or maintain a specific type of land cover. The terms “Land use” and “zoning”, are used differently. Where land use means the way humans adapt the land to suit their needs, and zoning is about the way the government regulates the land. A land buyer needs to understand land use for two main reasons:

- It helps prevent buyers from spending too much time and money on land when they know what can be built on it. For instance, if someone has plans for building a farm, they would not buy a residential property once they know the purpose of that land. If a buyer knows the purpose of certain land, they are better positioned to decide what to buy that will meet their needs.

STUDY NOTE

Since humans will always rely on livestock and crops as a source of their food, it is safe to assume that a large portion of land would be associated with agricultural activities.

STUDY NOTE

A landform can get created in just a few years or might take millions of years. Landforms can be changed in a very short period of time by mainly two natural methods: strong earthquakes and volcanic eruptions.

- On a larger scale, if one understands land use, they can understand the patterns that associate land and urbanisation. If you understand how the land was used in the past, you can judge how that land would be used in the future.

7.3 Types of Land Use

Land use has many different categories; the most primary ones are transport, recreational, agricultural, commercial, and residential. When one plans to invest in a property, understanding the specifics of each of these types can be extremely helpful.

- **Recreational:** If a property's land use is defined as recreational, then it denotes it is to be used for people's enjoyment. Many lands fall under this category, such as parks, playgrounds, open spaces, athletic fields, swimming pools, etc. Every city has a recreational land space in its blueprint since it is important to have areas in a city where people can go and have some leisure or fun time. Recreational lands do not contribute to a community's development, but they attract visitors and permanent residents.
- **Agriculture:** Agricultural land includes ranches, pastures, farms, etc. They are used for agricultural purposes to grow and harvest crops and livestock.
- **Transport:** Transport land's purpose is to help people travel from one destination point to another. Examples of transport lands are airports, roads, subway stations, railways stations, etc.
- **Residential:** Residential lands are used to build homes of different types like mobile homes, apartment complexes, or single-family homes. This type of land use has certain restrictions such as permission of only certain kinds of pets on the property, minimum lot size and the permitted distance between lots in your building, etc. When building a residence on any land, one must consider many factors like accessibility to local amenities, crime rate, proximity to school, etc.
- **Commercial:** This type of land use refers to infrastructures related to commerce like warehouses, businesses, shops, etc. The most common use of such lands is to build restaurants, shopping malls, office buildings, and so on.

STUDY NOTE

In 2006, the average number of cultivated land areas that were required to feed one person was 0.22 hectares.

SELF-ASSESSMENT QUESTIONS – 6

25. What does the term “green funds” mean?
- a) A fund that is used in generating renewable energy.
 - b) A fund that invests in generating non-renewable energy.
 - c) A fund that invests only in companies that produce goods with more responsibility towards environmental health.
 - d) A fund that supports companies that use fossil fuels.
26. Which one out of these is not a renewable source of energy?
- a) Hydropower
 - b) Geothermal energy
 - c) Solar energy
 - d) Fossil fuels
27. The two natural processes which can change landforms very quickly are:
- a) Strong earthquakes and deforestation.
 - b) Volcanic eruptions and soil erosion
 - c) Strong earthquakes and volcanic eruptions.
 - d) Volcanic eruptions and deforestation.
28. Non-renewable resources are replenished faster than they are consumed. [True/False].
29. Which of the following categories of land uses do athletic fields belong to?
- a) Transport
 - b) Agriculture
 - c) Recreational
 - d) Commercial

8. LAND DEGRADATION

With the degradation of lands, their soils lose or deteriorate in their productive capacity for the present and future. This issue has become a global challenge. Everyone on the planet is affected by it as it poses food insecurity, higher food prices, adverse changes in climate, loss of biodiversity, degradation of ecosystem services, and many other environmental hazards.

The rate at which land degradation is happening in today's age is quite alarming. Due to this, we are witnessing a dramatic decline in the productivity of rangeland and croplands across the globe. There is a degradation of 16% of arable land and 25% of total land area globally, and the numbers are only rising. The traditional systems used to manage land are no longer effective, or these systems are breaking down. The technology for better management of land is not always available.

STUDY NOTE

It is projected that the world population would grow by 35% in 2050 to 9.7 billion and there will be rising demands for food, fiber, fuel, and other agricultural products.

8.1 Reasons for Land Degradation

The primary reason for the land degradation is the unprecedented rate at which the population is growing globally. This explosion of the population has put high demands on the land. The other factors contributing to increasing pressure on land resources are:

- There is an uneven distribution of land resources between countries as well as within countries. This difference in access to land as per the population needs makes it difficult to meet the population demands.
- Changes in climate and natural (or human-induced) disasters also degrade the quality of the land. Due to climate change, there is an exacerbation in agricultural yields and income which threatens the stability of agroecosystems and food production systems.
- The advent of many socio-economic changes like trade liberalization, privatization, decentralization of decision making, and globalization of markets has made it difficult for humans to manage resources efficiently.
- Application of unscientific agricultural practices for cropping. For example, in some portions of the Western Ghats, the forest

STUDY NOTE

In cyclone-prone areas, the rise in sea level along with cyclones can cause land degradation.

is cleared to sow commercial crops like potatoes and ginger on the slopes. This results in loss of nutrients, soil erosion, waterlogging, etc.

- **Water logging:** The water table gets saturated due to various reasons like over-irrigation, inadequate drainage, seepages from canals, etc. The land lying under waterlogged areas is unusable either for settlements or agriculture.
- **Political reasons:** Short-term political gains are being made at the cost of long-term damage to the land. Decision-makers deals with difficult decisions on how to increase production to eradicate poverty. Such decisions have to be balanced with resource conservation, but often, decision-makers forfeit these long-term benefits for fulfilling people's immediate needs.

8.2 Effects of Land Degradation

When there is land degradation, soil releases carbon and nitrous oxide into the atmosphere, which causes climatic changes. Some of the areas aggravated by land degradation are mentioned below:

- **Soil erosion:** One of the biggest effects of land degradation is the detachment of the top layer of soil from the land. This soil gets washed away by ice, water, or sea waves, or it could be blown away by the wind.
- **Salinity or alkalinity:** In areas with high temperatures and temporary water surplus, the moisture sinks down, dissolving the underground salts along with it. During a dry period in these heavy rainfall regions, this saline solution comes on the surface through capillary action. The heat evaporates the water, forming a crust of magnesium, sodium, and calcium salts. This salt wreaks havoc by rendering vast stretches of soil infertile.
- **Desertification:** Desertification happens when sand from the deserts advances to adjoining regions. It affects the fertility of the soil by forming a cover on it.
- **Food shortage for consumption:** If less fertile land is available for agricultural purposes, it cascades into less yield and shortage of food.
- **Land conflicts:** With land becoming scarce, less land is available for biomass production or conservation purposes. That is why huge competition for land for different uses leads to complex conflicts.
- **Vulnerability to natural disasters:** Land degradation can cause floods, droughts, and other natural calamities.

- **Water shortage:** Due to land degradation, there is a loss of biodiversity in drylands, resulting in low precipitation and longer spells of dry seasons. This leads to water scarcity.
- **Fuel wood shortage:** Loss of forests causes widespread fuelwood shortages.
- **Lack of adequate housing:** As per UN (United Nations) estimates, close to 100 million people are sleeping without shelter in public spaces like buses, railway stations, etc. If one counts the number of people in temporary shelters which are of poor quality and do not provide enough protection, then that number rises to more than a billion globally. This problem is prevalent not only in developing countries. 12 countries of the European Union have 1.8 million people sleeping on the roads.

STUDY NOTE

There are 3.2 billion people who are affected badly by land degradation and most of them belong to rural communities.



Source: fao.org

Fig 11: Symptoms of the issues due to pressure on land resources

8.3 Measures for Preventing Land Degradation

Intervention is required to prevent further land degradation to restore and prevent deterioration of land's essential functions. The most pressing need right now is to break the current downward spiral of land resources. The landowners and users need to be well informed and taught all the technologies that are needed to make conscious decisions on land usage.

- The UNCCD (United Nations Convention to Combat Desertification) has to serve the benefits of the environment globally. They have recognized LDN (Land Degradation Neutrality) as a concept that will help boost food production without degrading land further. LDN is geared towards managing land in a more sustainable manner, reducing degradation and increasing the land restoration rate.
- The partnership of IAEA and FAO: International Atomic Energy Agency (IAEA) has come up with a technical cooperation program using nuclear technologies. IAEA has partnered with the Food and Agricultural Organization (FAO) to assist scientists and farmers residing in 60 countries in controlling and measuring soil erosion. This will help them measure the rate of soil erosion and track hot land degradation spots.

STUDY NOTE

Soil erosion is a very big reason for land degradation and leads to a loss of 75 billion tons of soil every year.

SELF-ASSESSMENT QUESTIONS – 7

30. What is the name of the condition where the water table gets saturated due to over-irrigation?
- a) Waterlogging
 - b) Desertification
 - c) Salinity
 - d) Alkalinity
31. Which of the following is a political reason for land degradation?
- a) Application of unscientific agricultural practices for cropping
 - b) Trade liberalization
 - c) Deforestation
 - d) Soil erosion
32. _____ occurs when sand from the deserts advances to adjoining regions.
33. Salinity occurs when the top layer of soil gets detached from the land and gets washed away by the sea, water, or ice. [True/False]

9. DEFORESTATION

Forest is a biotic (living) renewable resource that provides us clean air, water, wood products, stable soil, habitat for wildlife, a source of recreation, and a beautiful environment that is rich in biodiversity. Forests are integral to any nation's economy as they provide a wide range of environmental goods, production inputs, fuel, medicines, and other raw materials of industrial processing,

Different kinds of forests all around the world vary in diversity and composition. Some of the major benefits that human society derives from forests are:

- In the absence of services provided by forest resources, there will be more capital expenditure and a decline in human well-being. For example, forests can store water, regulate its flows, cleanse impurities, and protect channels. By these activities, forests form a structure for hydrological services, similar to transportation and communication.
- Forests are also significant in maintaining political and economic relations amongst different nations. The conditions of forests in a country affect that nation's dependence on international trade. Forests supply materials for many domestic industries, export commodities like wood and processed products, and a range of specialty products like ornaments and medicines.
- A forest ecosystem provides many commercial and scientific services like living space, climate regulation, and genetic resources.
- Forests are used for recreational purposes and are great for a country's tourism industry.
- Forests serve as a laboratory for the genetic resources of plants and animals. Tropical Forests are deemed to have more than 50% of the entire living species on the earth.
- Forests supply cheap forms of energy in the form of woods, and wood is the primary fuel in most tropical nations. Wood is also seen as a replacement for agricultural biomass and fossil fuels, bringing down the cost of these resources.

STUDY NOTE

As per FAO estimates, the wage employment provided by the forestry sector is equivalent to 60 million work years globally. 80% of this wage employment is in developing countries. Most of this work is in charcoal or fuelwood activities.

- Forests also influence the environment's ability to cope with natural calamities by functioning as barriers to intense rains, landslides, and windstorms. They aid in the prevention of soil erosion, landslides, and so on.

With so many global benefits derived from forest resources, forests are gaining value rapidly. Many institutions are coming forward to protect forests and develop strategies to translate these forest resources into tradable forms.

Forests are being cut down to make room for something other than forests to meet different needs. Deforestation happens when forests are removed permanently. The various reasons for deforestation could be to clear the land for agricultural activities or grazing, urbanization, mining activities, etc. Since 1960, deforestation has accelerated rapidly and harms our ecosystems greatly. UN FAO estimates that 1.3 million km² of forests are being wiped out every decade.

Most of the deforestation is happening in the tropics, with inaccessible areas in the past being within reach due to new roads being constructed by clearing dense forests. Forest is a mature resource, which means that even though it is renewable, it takes a long time to build a forest.

STUDY NOTE

Human-lit fires in forests skyrocketed in Brazil in the year 2019. More than 80,000 fires were burnt in the Amazon forest which was 80% more in number from 2018.

9.1 Causes of Deforestation

Multiple factors can cause deforestation; they could be of natural origin or due to human actions. Some of the natural factors are diseases caused by parasites or natural forest fires, leading to deforestation. Despite natural factors, the biggest cause of global deforestation is human activities. FAO has identified that 80% of global deforestation has been caused due to agricultural purposes. Other factors include the construction of infrastructure like dams, mines, roads, etc.

- **Agriculture is the numero uno reason for deforestation:** Out of all the agricultural activities that cause deforestation, around 33% of deforestation is a consequence of subsistence farming which is agriculture by local peasants in developing countries for their own use. The other 40% of forest loss is caused by agriculture for industrial or

commercial purposes. Forests are being frequently cut down in the search to grow fibers, food, raise livestock, and biofuel.

- **15% of deforestation is due to new constructions:**

Deforestation is being done to serve the current lifestyle of humans. Energy generation, transportation, etc. are some of the reasons for cutting down trees. Roads, railway tracks, and airports are being built to move goods (cereals, fruits, spices, etc.) and people.

STUDY NOTE

Scientists at the University of Maryland came out with a report in 2017 according to which the tropics lost 61,000 square miles of forest area which is almost the size of Bangladesh.

- **Building dams:** Dams are another major cause of forest loss, either directly or indirectly. People have often seen these large hydroelectric dams as a sign of development, but little do they know that 40,000 large dams (measuring more than 15 meters in height) are obstructing the world's rivers. The reservoirs of these rivers are 400,000 km² of land which is inundated with millions of hectares of forests. This is especially happening in the tropics, where many trees were not even logged but left to just rot slowly. The indirect way dam building has caused deforestation is that farmers get displaced due to building dams. These farmers then relocate and clear forests to build their homes and grow crops.
- **Mining:** Mining, in general, is extremely destructive for our environment. It is considered one of the main causes of deforestation. Trees and vegetation must be burned and cleared to begin mining. Large-scale mining begins by making the ground completely bare and then using bulldozers to extract minerals and metals from the soils. Once the extractions are ready, they need to be amalgamated (clustered) using chemicals such as mercury, cyanide, or methylmercury. The chemicals go through pipes (tailings) and get discharged into rivers, streams, oceans, and bays. This contaminates all the living bodies within the water.
- **Commercial logging:** This involves cutting down trees to sell them as pulp or timber. This timber is used to build many things such as, furniture, homes, etc. Paper and paper products are made using the pulp of the trees.
- **Natural fires:** Natural fires are rare but very intense in tropical forests.

STUDY NOTE

As per WWF (World Wildlife), the tropical forests hold more than 210 gigatons of carbon in them.

9.2 Effects of Deforestation

Deforestation is a major threat to the resilience of our ecosystem and the planet. Deforestation directly affects all living organisms and damages the environment deeply. Forests are home to more than 80% of diverse trees, animals, plants, and microbes. The effect of deforestation reaches far and wide, which is summed up below:

- **Deforestation affects biodiversity:** Forests are the most variable hub for biodiversity. Many rare and fragile species, including plants, animals, mammals, birds, amphibians, and insects, have their homes in forests. If these forests are destroyed, it creates a natural imbalance and threatens life. The complex and interconnected natural world with interdependencies provides many functions to our environment. Apart from many other functions, trees provide shade so that certain plants and animals who cannot survive in direct sunlight can survive in the shade. Trees also provide food to animals and birds.
- **It hampers the livelihood of the local or tribal population:** Globally, 1.6 billion people depend on healthy forests for their livelihoods. Amongst them, around 1 billion are the poorest in the world. They cannot continue their small-scale agricultural programs by hunting and gathering raw materials from forests. Big businesses often take away these forests for their other means and disrupt the lives of a huge tribal population. The locals of these forests have to resort to only two available options. The first one is to abandon their homes and face the challenges of a new life elsewhere. The second one is to work for these big companies under inhuman conditions and extremely low wages.
- **Future Food insecurity:** Almost 52% of land used for food production is either moderately or severely impacted by soil erosion. If this progresses at the same rate, there is a high chance of low yield in the future due to the poor quality of the soil. This will lead to less food production giving rise to food insecurities.
- **Reduced rainfall:** In tropical areas, deforestation can alter the way water vapour produces over the canopy, reducing rainfall, as if trees will not be there, then it will be very difficult to cope with climate changes. Deforestation removes vegetation that is crucial for absorbing carbon dioxide from the atmosphere and causes greenhouse gas emissions.

- **Climatic changes due to greenhouse gases:** Trees absorb carbon dioxide, which helps mitigate greenhouse gas emissions. Tropical trees are responsible for mitigating 23% of climatic changes. Trees can capture and store excess carbon dioxide, helping in carbon sequestration. Just the act of clearing forests also emits greenhouse gases into the atmosphere.
- **Soil erosion:** The weakening and degradation of soil is another massive downside of deforestation. Forested soils have rich organic matter and have high resistance to erosion. Soils surrounded by forests can withstand bad and extreme weather too. This is because the roots of the trees help soil fix on the ground and the shades of the trees promote slow drying of soils. Deforestation leaves more fragile soils that can get washed away, leading to landslides and floods.

SELF-ASSESSMENT QUESTIONS – 8

34. Which of the following is not a forest resource?
- a) Plants
 - b) Animals
 - c) Microorganisms
 - d) Sand
35. Forests can be built in a short period of time. [True/False]
36. Reduced rainfall is caused when deforestation affects the way _____ is formed on canopies.
37. Which of the following best fits the meaning of deforestation?
- a) Erosion of soil
 - b) Removal of forests permanently
 - c) Emission of greenhouse gases
 - d) Climatic changes
38. Which one of the following is not a cause of deforestation?
- a) Agricultural activities
 - b) Mining
 - c) Building dams
 - d) Soil erosion

39. Which one from the following list is not a result of deforestation?

- a) Soil erosion
- b) Damage to the livelihood of local people of forests
- c) Climatic changes due to greenhouse gas emissions
- d) Development projects and new constructions

10. SUMMARY

- The environment consists of many different types of organisms. Different organisms contribute to the environment and affect the other organisms.
- There is a biotic and abiotic component of the ecosystem. Abiotic components are those which contain only physical properties, whereas biotic components consist of living organisms.
- Various problems caused by human activities like pollution and deforestation for development affect the balance of the ecosystem.
- There is a need to create public awareness to think about sustainable development.
- The scope of environmental studies focuses on how we can use technology for the purpose of environmental safety.
- There is a constant dependence of many organisms within the ecosystem leading to the concept of the food chain.
- Many different organisms linked to each other at different levels is referred to as the food web.
- There is a unique composition of flow of energy which is flowing in a way that sustains all life kinds.
- Humans should stop doing activities which creates imbalance in the environment.
- A food chain is a type of cycle where energy flows from one organism to another.
- Each level of the food chain is called trophic level, which helps to connect with different organisms.
- Succession refers to the inevitable mechanism by which other classes of individuals colonise a culture.
- The concept of sustainable development is important in various aspects to respect every being and share the planet.

11. GLOSSARY

Chemosynthesis: It is the process by which food is made by bacteria using chemicals which are utilized as the energy source.

Detritivores: Detritivores are species that prey on the dead bodies of plants and animals.

Decomposer: Decomposers are the organisms which feed on the dead and decay matter.

Food chain: The progressive chain where the all living organisms are dependent on the other one for energy is called a food chain.

Producers: Producers refer to the plants which produce energy with the help of radiant energy from the sun.

12. CASE STUDY

The Age of Extinction

According to a recent study of the diversity of life on earth, human overconsumption, intensive agriculture and population growth are largely affecting the population of wildlife. According to the WWF and the Zoological Society of London (ZSLbiennial)'s Living Planet Survey 2020, global numbers of birds, rodents, fish, reptiles, and amphibians decreased by 68 percent on average from 1970 to 2016.

This study, which 134 scientists from around the world put together, is considered one of the most detailed analyses of global biodiversity possible. Nature is being used and manipulated by humans to a scale never seen before, from the rainforests of Central America to the Pacific Ocean.

The study gathered data from 20,811 populations of 4,392 vertebrate species all over the world. This includes well-known endangered species like polar bears and pandas, as well as lesser-known fish and amphibians. The data indicate that vertebrate wildlife populations are dwindling in almost every area of the world, with an estimated loss of more than two-thirds since 1970.

Robin Freeman, who researched at ZSL, said: "It looks like that we've spent 10 to 20 years discussing about these declines and not actually managed to take any actions about it. It

upsets me. We just sit at our desks and see these statistics but they do have real-life serious implications. It's really disappointing to communicate how drastic some of these declines are actually."

Mammals, amphibians, birds, fish, and reptiles have all declined significantly in Africa and the Asia Pacific, with declines of 65 percent and 45 percent, respectively. The most troubling decline is in Latin America and the Caribbean, where vertebrate animal numbers have dropped by 94 percent on average. Overexploitation of habitats, habitat fragmentation, and disease has had the largest effects on reptiles, amphibians, and fish in the area.

Source: *Article from Economic Times Newspaper, January 05 2021*

Discussion Questions:

1. Discuss all the concerns hampering the population of vertebrate population.
2. With reference to the research shown, explain why is it the need of the hour to change the industrial development strategy globally and focus on sustainable socio-economic development?
3. Discuss few other ways by which industries or individuals can be made more responsible for sustainable development of the environment.

2. Use of Biomass for Electricity in Kasai Village, Madhya Pradesh

Kasai is a tribal (Adivasi) village in a remote area of Madhya Pradesh. It has a population of 392 people spread in 55 households. The village is not connected to a grid but there is abundant biomass resource here in the form of woods (from farmlands and forests), cattle dung, crop residues, and oilseeds. The government has been supporting a project in the village since 2005.

The project is to generate electricity from a small biomass plant of 10 KW capacities. The ministry of Renewable Energy (MNRE) is funding 11 such projects in M.P. Government funded 100% of the capital costs for this project which is now managed solely by the local community. The local community gets technical support from the local forest department. The villagers gather the biomass for the plant, and they charge INR 120 for each household to take care of the plant's costs (operating and maintenance).

Apart from the maintenance fee, the households also need to pay a usage charge on the amount of electricity they consume. The whole operation is overseen by an elected group of (11 members out of which 5 are women).

All the lighting needs of households of the village are met by this plant. It comprises street lighting, schools, music systems and T.V installations in the village for entertainment, electricity for the flour mill, power to the milk-chilling unit, and water pumping.

The project has helped the villagers in many ways as outlined below:

- Migration from the area has been curbed.
- Due to water available for irrigation, it has promoted some amount of agricultural production.
- It prevented milk from being spoiled due to heat and made it available to markets. It is also deemed to promote a village dairy system which would further improve the livelihood of villagers.
- People are able to process wheat and rice in the flour mill and sell at profitable rates in the market.
- The project has helped a household piping system.

Currently, villagers use traditional biomass (cow dung, wood, charcoal, etc.) for their cooking and heating needs. With some modifications to the existing system, it is probable that in future, gas is made available to the village households for cooking purposes.

Source- Article from Times of India newspaper

Discussion Questions:

- 1) Discuss the various ways that biomass has helped villagers lead a better life.
- 2) Discuss what future modifications are possible to bring gas to the people of Kasai village.

13. TERMINAL QUESTIONS

SHORT ANSWER QUESTIONS

- Q1. What is the ultimate source of energy?
- Q2. Differentiate between food web and food chain.
- Q3. State the functions of ecosystem.
- Q4. What is the role of decomposers in food chain?
- Q5. What are some of the political reasons for land degradation?

LONG ANSWER QUESTIONS

- Q1. Explain the importance of food chain with example.
- Q2. Explain the structure of ecosystem.
- Q3. Explain the types of food chain.
- Q4. Explain how deforestation affects biodiversity

14. ANSWERS

SELF ASSESSMENT QUESTIONS

1. D. None of the above
2. B. False
3. Biotic
4. Abiotic
5. Herbivorous animals
6. Decomposers
7. Chemoautotrophs
8. D. Tertiary
9. D. Both A and C
10. Autotrophic plants
11. Carbon
12. Food pyramid/energy pyramid
13. Energy
14. Law of thermodynamics
15. Food web
16. B. Number and biomass of herbivorous animals
17. Trophic levels
18. Detritivores
19. Grazing
20. A. Carnivores, herbivores
21. Lichen and blue-green
22. Rooted submerged stage
23. Producers
24. Succession
25. C. A fund which invests only in companies that produce goods with more responsibility towards environmental health.
26. D. Fossil Fuels
27. C. Strong earthquakes and Volcanic eruptions.
28. False

- 29. C. Recreational
- 30. A. Waterlogging
- 31. B. Trade liberalization
- 32. Desertification
- 33. False
- 34. D. Sand
- 35. False
- 36. Water vapor
- 37. B. Removal of forests permanently
- 38. D. Soil erosion
- 39. D. Development projects and new constructions.

TERMINAL QUESTIONS:**SHORT ANSWER QUESTIONS:**

Answer 1: The major source of energy is sunlight. The sunlight is utilized by the plants who are the producers for making their own food. Other species depend on the plant directly or indirectly for energy.

Answer 2: A food chain is a continuous chain that depicts the transfer of energy from one entity to the next. A food web is multiple food chains that are interconnected at many trophic levels. Furthermore, a food web accurately represents all the various food chains that exist in an ecosystem. Example of a food chain is:

Producer: Grass

Primary consumer: Grasshopper

Secondary consumer: Frog

Tertiary consumer: Snake

Final consumer: Eagle



Answer 3: The following are the few functions of the ecosystem:

- (i) Production of food by autotrophs, chemoautotrophs
- (ii) Utilizing solar energy by green plants
- (iii) Recycling minerals.
- (iv) Ensuring CO₂/O₂ ratio balance in the atmosphere

Answer 4: Through excreting, decomposers send the nutrients of dead species to the soil, which then helps to nourish the plants that restart the chain. They assist with the clean-up of the planet. Without decomposers, dead leaves, insects, and animals would pile up everywhere.

Answer 5: Short-term political gains are being done at the cost of long-term damage to the land. Decision-makers deals with difficult decisions on how to increase production to eradicate poverty. Such decisions have to be balanced with resource conservation, but often, decision-makers forfeit these long-term benefits for fulfilling people's immediate needs.

LONG ANSWER QUESTIONS:

Answer 1: A food chain can be called the collection of organisms where they transfer food energy with the assistance of a sequence of consuming and being eaten.

- ❖ It helps in understanding the food interactions and relationship shared between various organisms in an ecosystem.
- ❖ It helps in understanding the basic mechanism of transfer of energy and food, nutrients through various components of nature.
- ❖ It assists in the understanding of hazardous material movement in an ecosystem.

Answer 2: The structure of ecosystem comprises two essential components.

1. Biotic component: Biotic components, or biotic factors, can be categorized as any living component which has an effect on another organism or shapes the ecosystem. The animals that consume other organisms present in the ecosystem and the organism consumed for deriving nutrition are included in this component.
2. Abiotic component: These include non-living factors which impact the distribution of organisms. This consists of the following:

- (i) Inorganic substances which occur in the form of compounds dissolved in soil or in a free state. For example, carbon, oxygen, nitrogen, calcium, phosphorus, sulphur and micro-nutrients, etc.
- (ii) Organic compounds are the compounds on which all living beings are dependent for their survival. For examples, lipid, proteins, carbohydrate, nucleic acids, vitamins, etc.
- (iii) Natural elements, like temperature, light, water, gases, wind, humidity, soil, animals, etc., are also included in the abiotic component.

Answer 3: There are two types of food chains which are as follows:

1. Grazing food chain.
2. Detritus food chain.

1. **Grazing Food Chain:** Green plants are at the top of this food chain, and carnivores are at the end. This food chain is highly reliant on autotrophs, which derive their nutrition from sunlight. The pattern of food chains in the terrestrial ecosystem may be shown as:



Fig 11: Grazing Food Chain

2. **Detritus Food Chain:** Detritivores are species that prey on the dead bodies of plants and animals. Bacteria, fungi, protozoa, and other microorganisms are included. The dead organic materials are ingested and digested by these species, which then turn them to water and CO₂.

Answer 4: Forests are the most variable hub for biodiversity. Many rare and fragile species, including plants, animals, mammals, birds, amphibians, and insects, have their homes in forests. The complex and interconnected natural world with interdependencies provides many functions to our environment. Plants and trees also provide food to animals and birds

11. SUGGESTED BOOKS AND E-REFERENCES

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